How Do Biologists Assemble Genomic Puzzles from Millions of Pieces?

Graph Algorithms

Phillip Compeau and Pavel Pevzner

Bioinformatics Algorithms: an Active Learning Approach

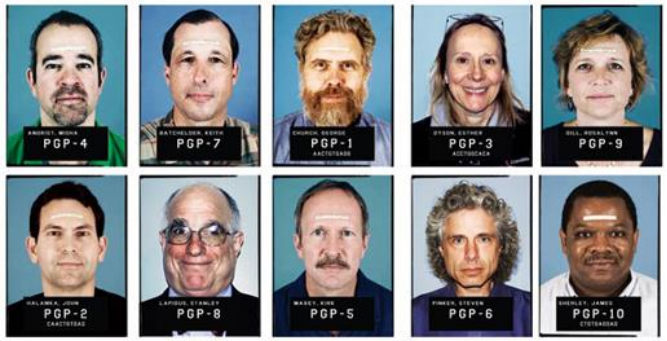
©2018 by Compeau and Pevzner. All rights reserved

Outline

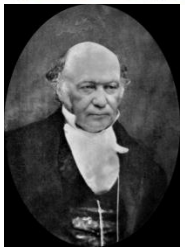
- **What Is Genome Sequencing?**
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Who Are These People?









Euler
1707-1783

Hamilton
1805-1865

De Bruijn
1918-2012

The human genome is a three billion nucleotide long "book" written in A, C, G, T alphabet.

Some genomes are 100 X larger than the human genome:

Amoeba dubia





Paris japonica

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Why Do We Sequence 1000s of Species?















- Applications in **medicine** (genomes of fungi-producing bacteria), **agriculture** (oil palm genome), **biotechnology** (genomes of energy-producing cyanobacteria), etc., etc., etc.

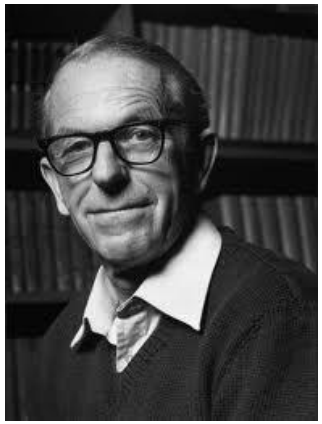
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Brief History of Genome Sequencing

- **1977**: Walter Gilbert and Frederick Sanger develop independent DNA sequencing methods.
- **1980**: They share the Nobel Prize.
- Still, their sequencing methods were too expensive ($3 billion to sequence the human genome).



Walter Gilbert



Frederick Sanger

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Race to Sequence the Human Genome

- **1990**: The public Human Genome Project, headed by Francis Collins, aims to sequence the human genome by 2005.



Francis Collins

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

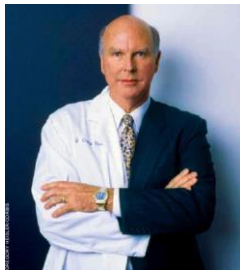
The Race to Sequence the Human Genome

- **1990**: The public Human Genome Project, headed by Francis Collins, aims to sequence the human genome by 2005.



Francis Collins

- **1997**: Craig Venter founds Celera Genomics, a private firm, with the same goal.



Craig Venter

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

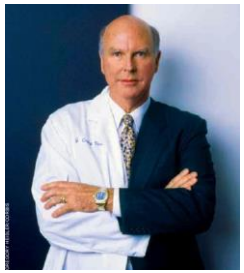
The Race to Sequence the Human Genome

- **1990**: The public Human Genome Project, headed by Francis Collins, aims to sequence the human genome by 2005.



Francis Collins

- **1997**: Craig Venter founds Celera Genomics, a private firm, with the same goal.



Craig Venter

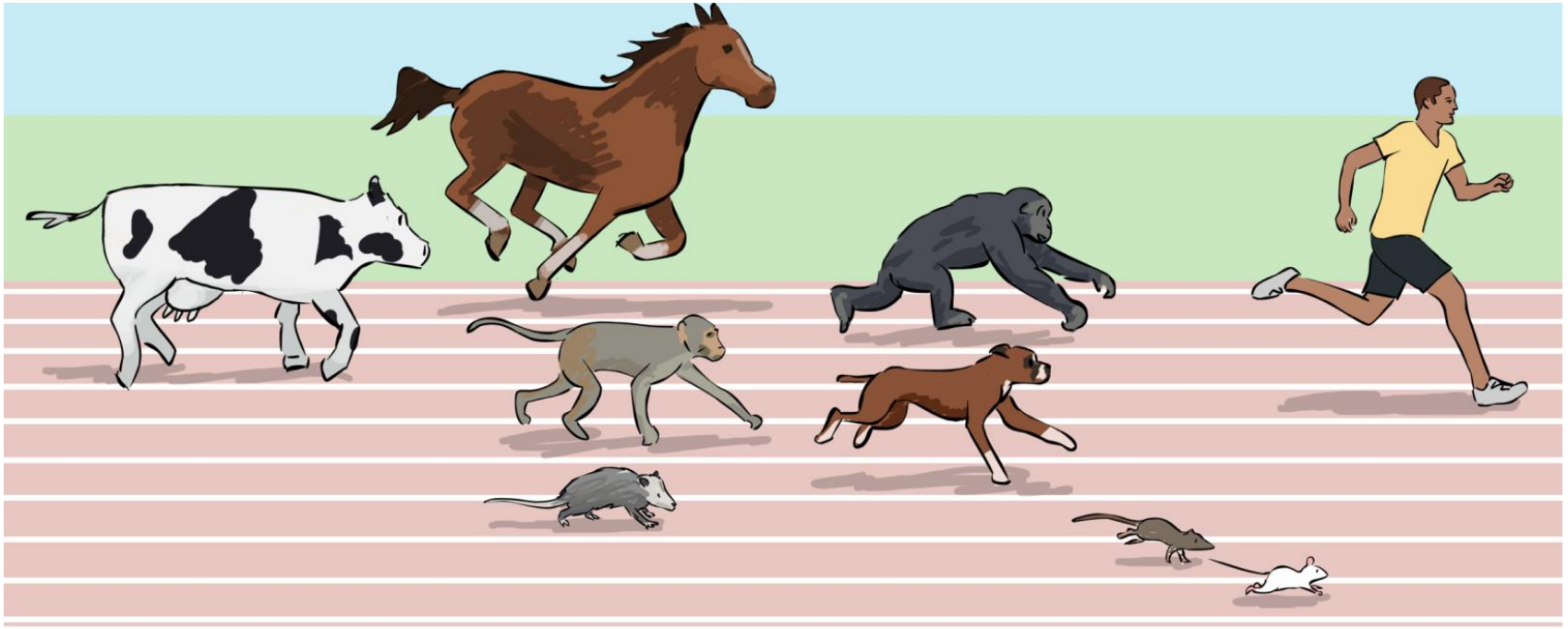
- **2000:**



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2013 Compeau and Pevzner.

From Human to Mouse to Rat to …

Early 2000s: Many more mammalian genomes are sequenced using the same Sanger sequencing method, but it is clear that new technology is needed for further progress.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Next Generation Sequencing Technologies



- **Late 2000s**: The market for new sequencing machines takes off.
  - Illumina reduces the cost of sequencing a human genome from $3 billion to $10,000.
  - Complete Genomics builds a genomic factory in Silicon Valley that sequences hundreds of genomes per month.
  - Beijing Genome Institute orders hundreds of sequencing machines, becoming the world's largest sequencing center.

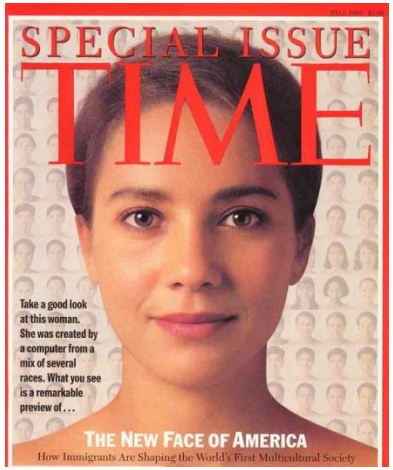
illumina

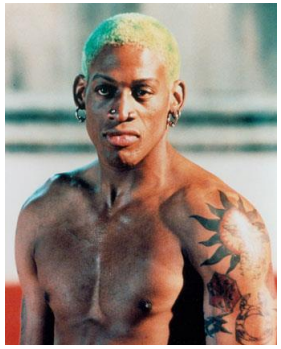
Complete genomics

华大基因 BGI

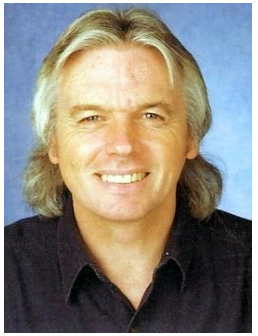
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Personal Genome Sequencing









Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Few Mutations Can Make a Big Difference…

- Different people have slightly different genomes: on average, roughly 1 mutation in 1000 nucleotides.

- The 1 in 1000 nucleotides difference accounts for height, high cholesterol susceptibility, and 1000s of genetic diseases.



CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGA
TCAGCTACCACATCGTAGCTACGATGCATTAGCAAGCTAT
CGATCGATCGATCGATTATCTACGATCGATCGATCGATCA
CTATACGAGCTACTACGTACGTACGATCGCGGGACTATTA
TCGACTACAGATAAAACATGCTAGTACAACAGTATACATA
GCTGCGGGATACGATTAGCTAATAGCTGACGATATCCGAT

CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGA
TCAGCTACAACATCGTAGCTACGATGCATTAGCAAGCTAT
CGATCGATCGATCGATTATCTACGATCGATCGATCGATCA
CTATACGAGCTACTACGTACGTACGATCGCGTGACTATTA
TCGACTACAGATGAAACATGCTAGTACAACAGTATACATA
GCTGCGGGATACGATTAGCTAATAGCTGACGATATCCGAT



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Why Do We Sequence Personal Genomes?

- **2010:** Nicholas Volker became first human being to be saved by genome sequencing.

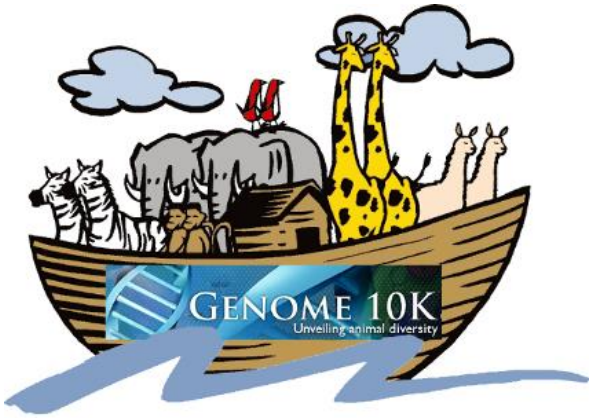


  - Doctors could not diagnose his condition; he went through dozens of surgeries.
  - Sequencing revealed a rare mutation in a *XIAP* gene linked to a defect in his immune system.
  - This led doctors to use immunotherapy, which saved the child.

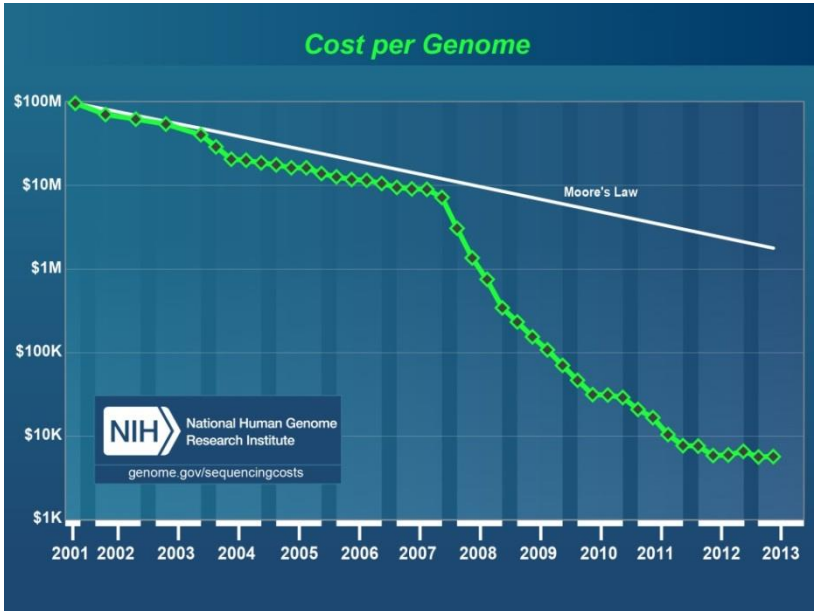
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

10,000 Genomes and Beyond

- **2010**: Scientists launch a project to sequence 10,000 vertebrate genomes.



- **Now**: Human genome sequencing costs just a few thousand dollars and under $1,000 human genomes may arrive any day now.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- **Exploding Newspapers**
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Newspaper Problem



stack of NY Times, June 27, 2000

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Newspaper Problem



stack of NY Times, June 27, 2000



stack of NY Times, June 27, 2000
on a pile of dynamite

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

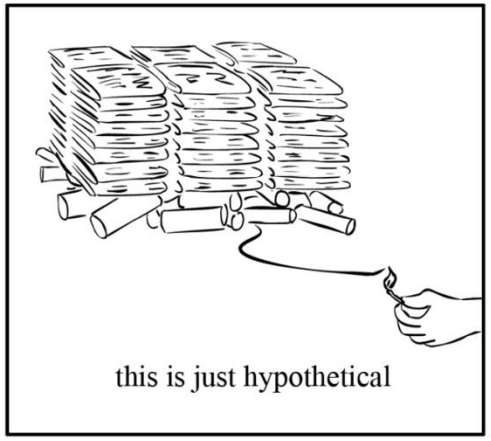
The Newspaper Problem



stack of NY Times, June 27, 2000



stack of NY Times, June 27, 2000
on a pile of dynamite



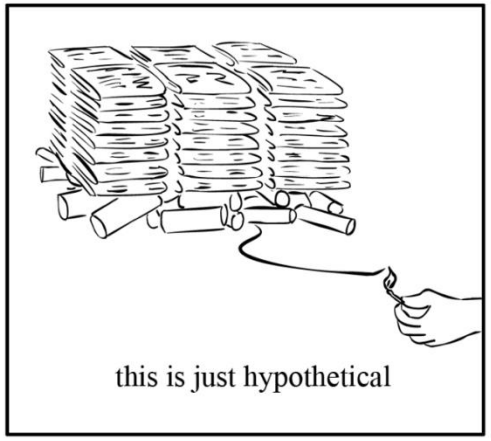
this is just hypothetical

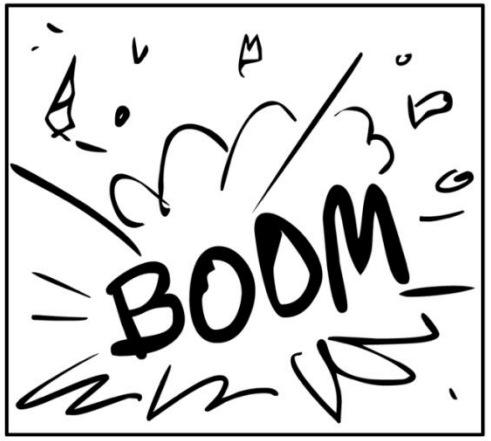
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Newspaper Problem









Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

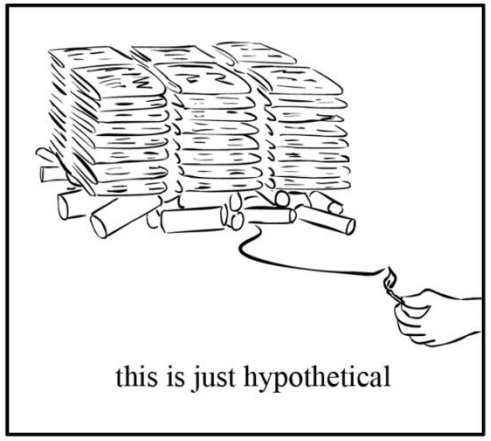
The Newspaper Problem



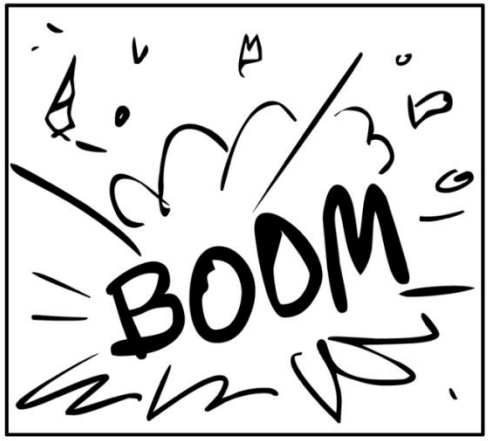
stack of NY Times, June 27, 2000

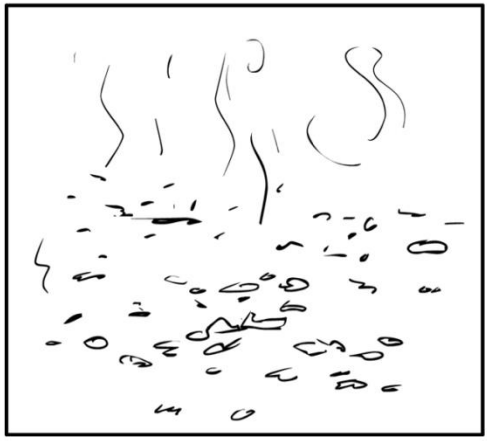


stack of NY Times, June 27, 2000
on a pile of dynamite



this is just hypothetical



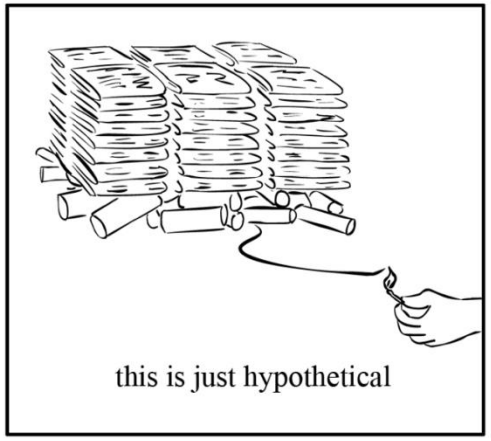


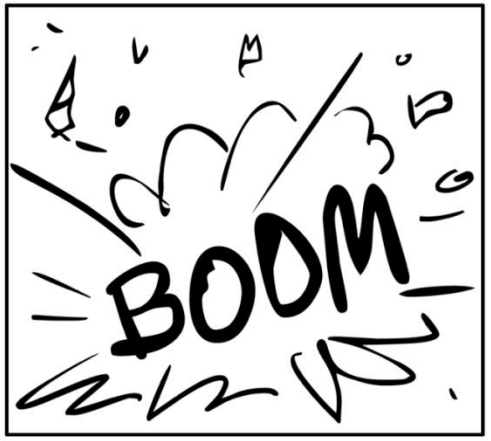
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

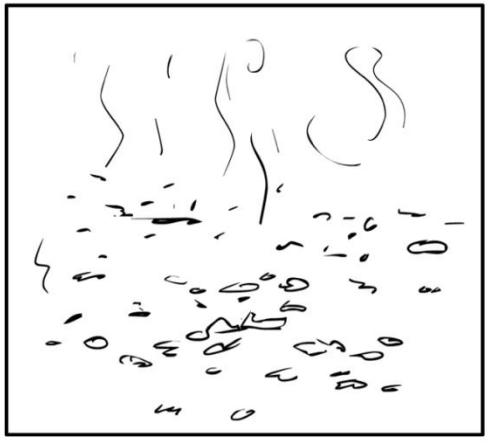
The Newspaper Problem

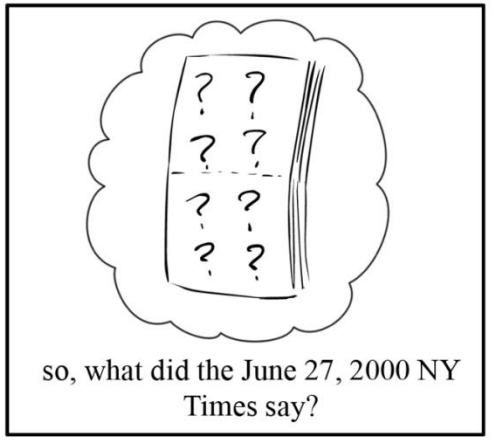






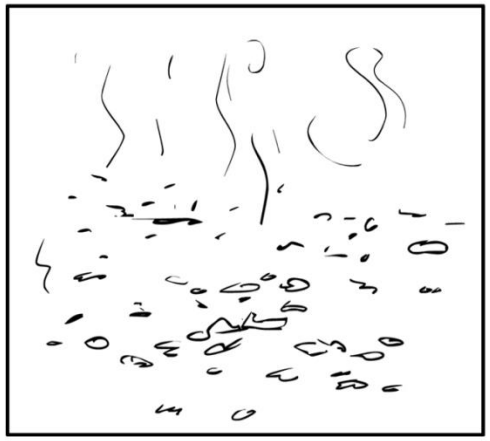


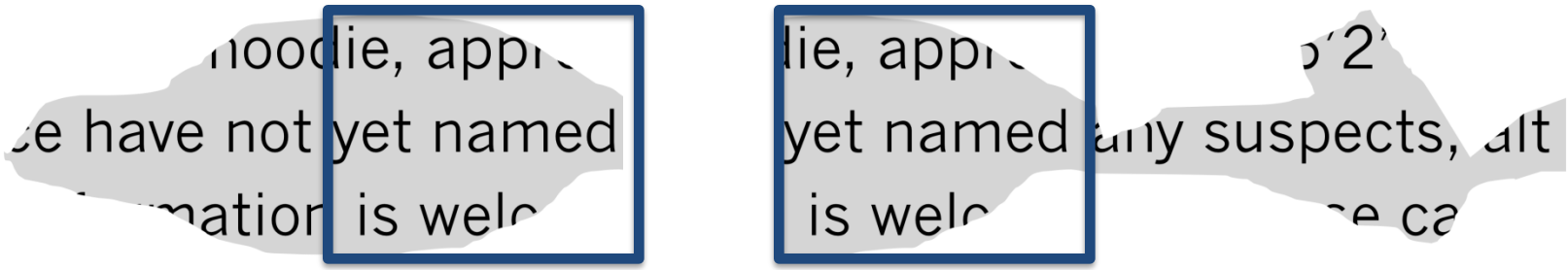




Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Newspaper Problem as an Overlapping Puzzle





Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Newspaper Problem as an Overlapping Puzzle


The New York Times

Genetic Code of Human Life Is Cracked by Scientists

JUSTICES REAFFIRM MIRANDA RULE, 7-2; A PART OF 'CULTURE'

A SHARED SUCCESS

A Pearl and a Hodgepodge: Human DNA

Clinton Raises Estimate of Surplus And the Stakes on How to Use It

Sharansky in Eyes of Israelis: A Hero or Betrayer of Peace?

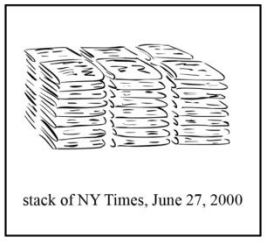
Buildings' Savior Now a Troubled Landlord

Vote in Zimbabwe Shows Opposition Making Big Gains

hoodie, appr
e have not yet name
mation is welc
'2'
spects, alt
e ca

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Multiple Copies of a Genome (Millions of them)



stack of NY Times, June 27, 2000

CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGATCAGCTACCACATCGTAGCTACGATGCATTAGCAAGCTATCGGATCAGCTACCACATCGTAGC

CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGATCAGCTACCACATCGTAGCTACGATGCATTAGCAAGCTATCGGATCAGCTACCACATCGTAGC

CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGATCAGCTACCACATCGTAGCTACGATGCATTAGCAAGCTATCGGATCAGCTACCACATCGTAGC

CTGATGATGGACTACGCTACTACTGCTAGCTGTATTACGATCAGCTACCACATCGTAGCTACGATGCATTAGCAAGCTATCGGATCAGCTACCACATCGTAGC

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Breaking the Genomes at Random Positions





Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Generating “Reads”

CTGATGA TGGACTACGCTAC TACTGCTAG CTGTATTACG ATCAGCTACCACA TCGTAGCTACG ATGCATTAGCAA GCTATCGGA TCAGCTACCA CATCGTAGC

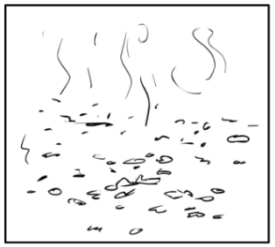
CTGATGATG GACTACGCT ACTACTGCTA GCTGTATTACG ATCAGCTACC ACATCGTAGCT ACGATGCATTA GCAAGCTATC GGATCAGCTAC CACATCGTAGC

CTGATGATGG ACTACGCTAC TACTGCTAGCT GTATTACGATC AGCTACCAC ATCGTAGCTACG ATGCATTAGCA AGCTATCGG A TCAGCTACCA CATCGTAGC

CTGATGATGGACT ACGCTACTACT GCTAGCTGTAT TACGATCAGC TACCACATCGT AGCTACGATGCA TTAGCAAGCT ATCGGATCA GCTACCACATC GTAGC

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

“Burning” Some Reads



CTGATGA TGGACTACGCTAC TACTGCTAG CTGTATTACG ATCAGCTACCACA TCGTAGCTACG ATGCATTAGCAA GCTATCGGA TCAGCTACCA CATCGTAGC

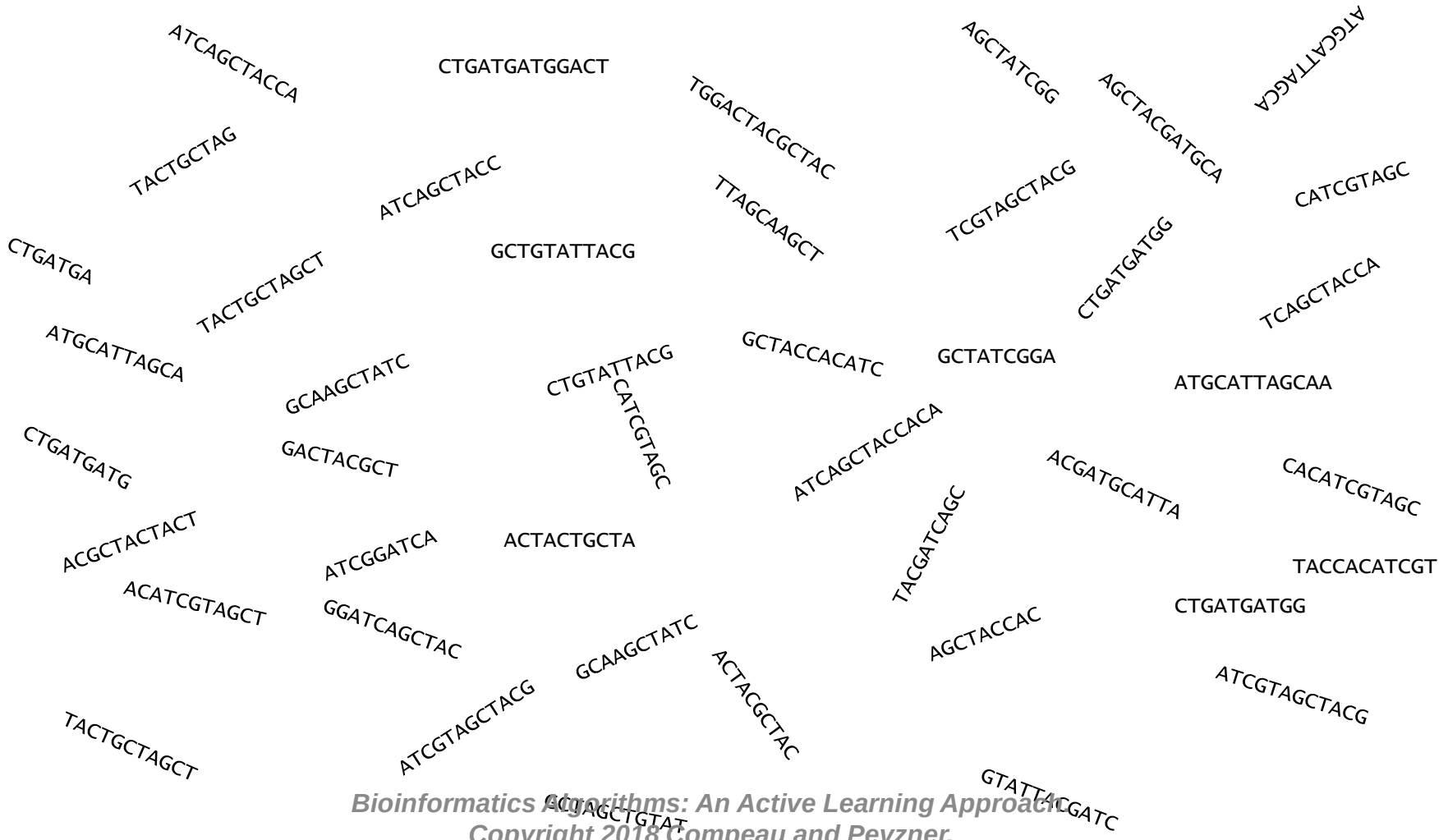
CTGATGATG GACTACGCT ACTACTGCTA GCTGTATTACG ATCAGCTACC ACATCGTAGCT ACGATGCATTA GCAAGCTATC GGATCAGCTAC CACATCGTAGC

CTGATGATGG ACTACGCTAC TACTGCTAGCT GTATTACGATC AGCTACCAC ATCGTAGCTACG ATGCATTAGCA AGCTATCGG A TCAGCTACCA CATCGTAGC

CTGATGATGGACT ACGCTACTACT GCTAGCTGTAT TACGATCAGC TACCACATCGT AGCTACGATGCA TTAGCAAGCT ATCGGATCA GCTACCACATC GTAGC

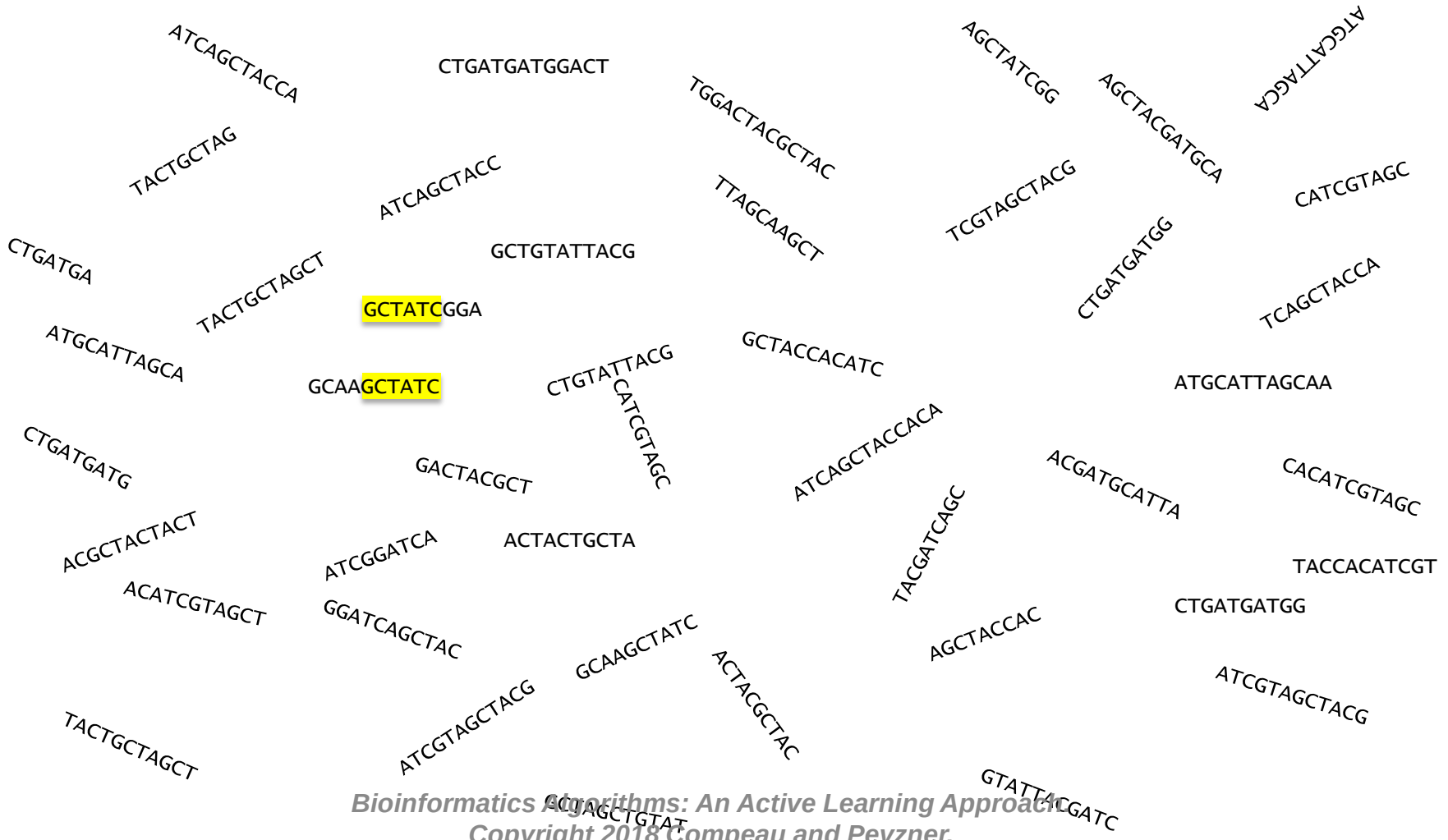
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

No Idea What Position Every Read Comes From



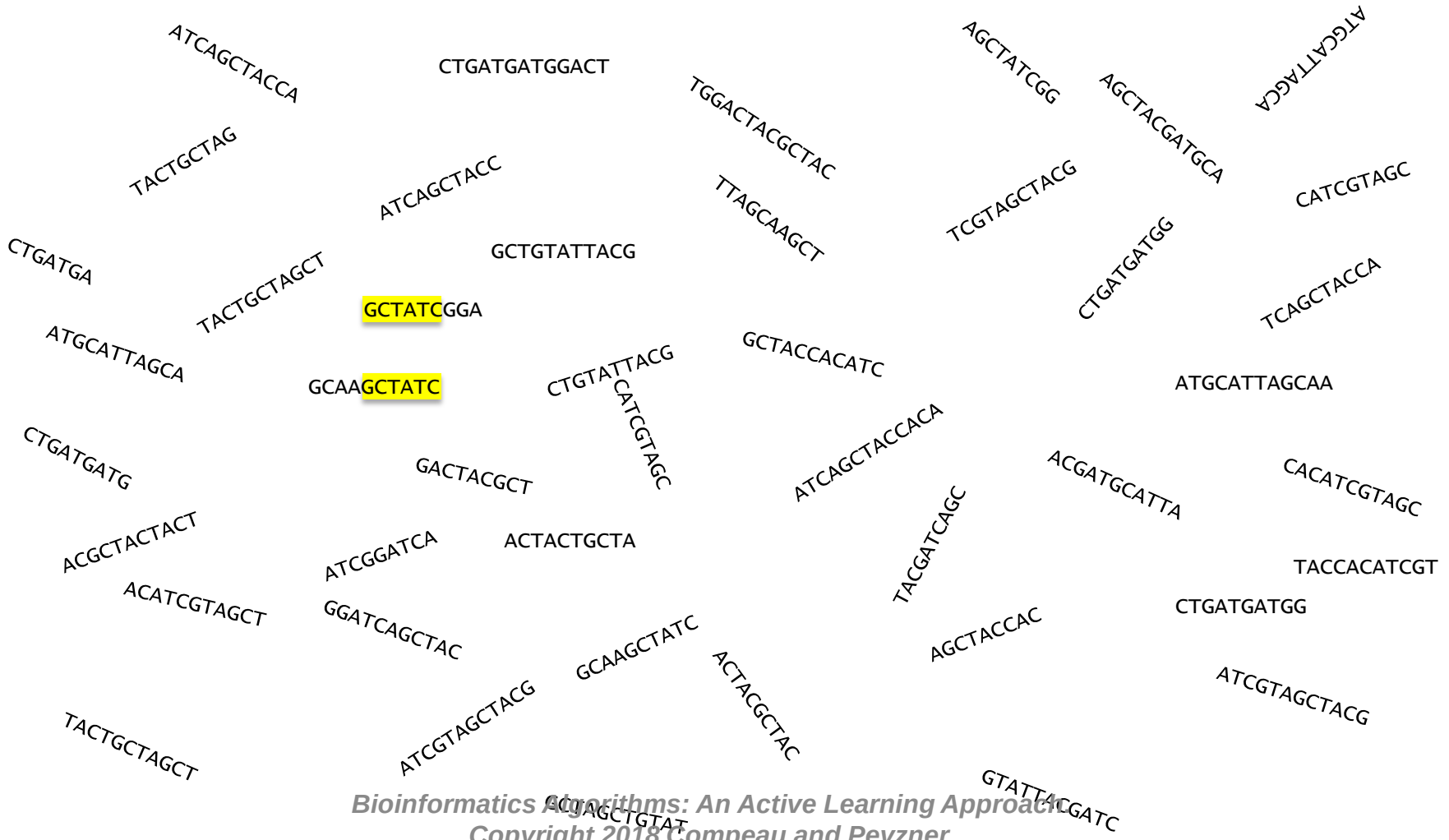
Bioinformatics Algorithms: An Active Learning Approach
Copyright 2018 Compeau and Pevzner.

No Idea What Position Every Read Comes From



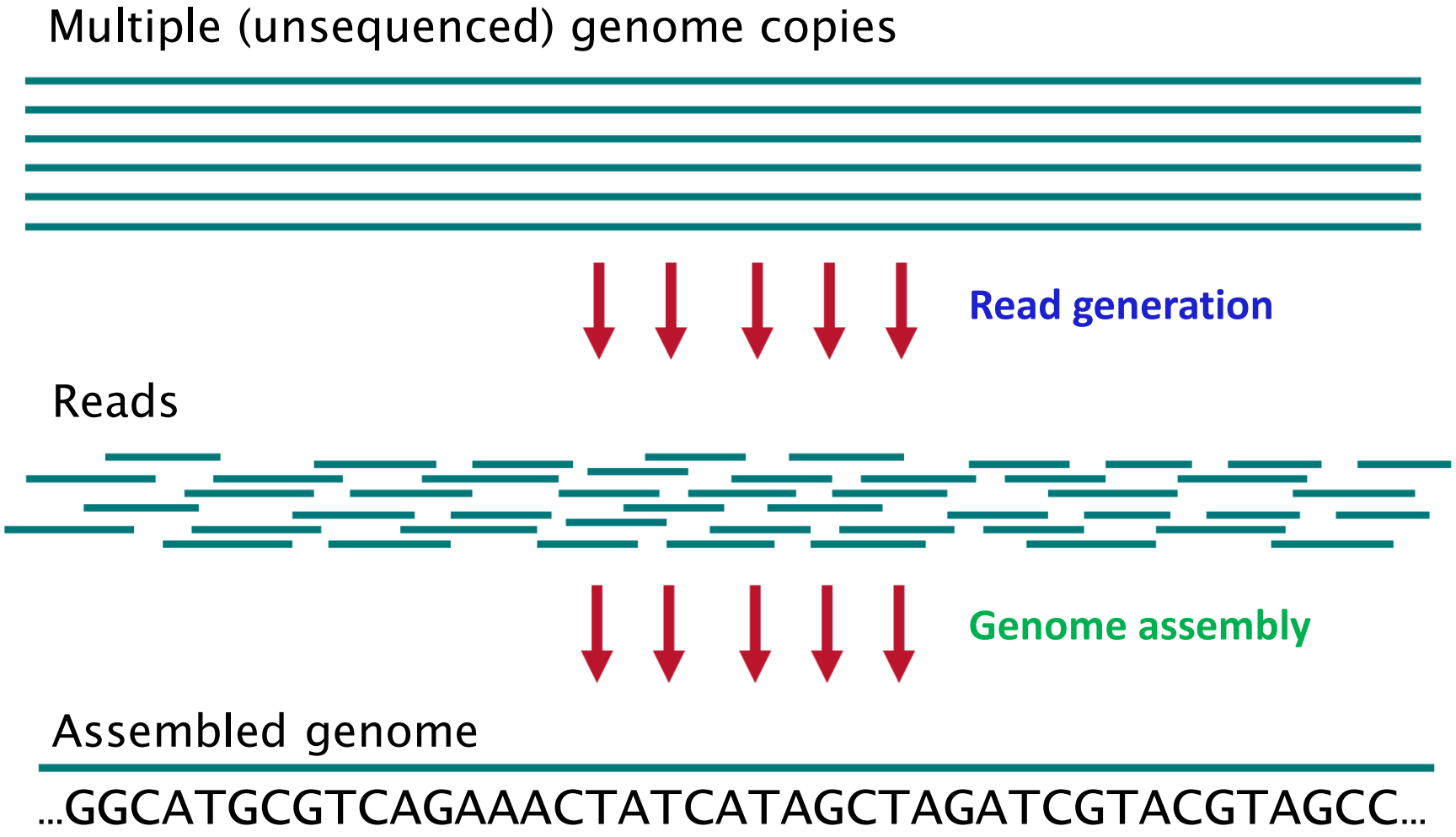
Bioinformatics Algorithms: An Active Learning Approach
Copyright 2018 Compeau and Pevzner.

No Idea What Position Every Read Comes From



Bioinformatics Algorithms: An Active Learning Approach
Copyright 2018 Compeau and Pevzner.

From Experimental to Computational Challenges



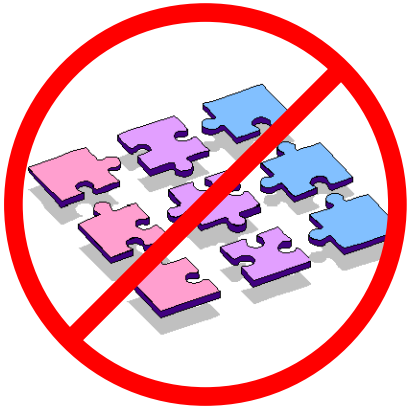
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What Makes Genome Sequencing Difficult?

- Modern sequencing machines cannot read an entire genome one nucleotide at a time from beginning to end (like we read a book)



- They can only shred the genome and generate short **reads**.
- The genome assembly is not the same as a jigsaw puzzle: we must use *overlapping* reads to reconstruct the genome, a giant **overlap puzzle**!



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- **The String Reconstruction Problem**
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Genome Sequencing Problem

Genome Sequencing Problem. Reconstruct a genome from reads.

- **Input**. A collection of strings *Reads*.
- **Output**. A string *Genome* reconstructed from *Reads*.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What Is *k*-mer Composition?

$Composition_3$(TAATGCCATGGGATGTT) =

TAA
AAT
ATG
TGC
GCC
CCA
CAT
ATG
TGG
GGG
GGA
GAT
ATG
TGT
GTT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

k-mer Composition

$Composition_3$(TAATGCCATGGGATGTT) =

TAA AAT ATG TGC GCC CCA CAT ATG TGG GGG GGA GAT ATG TGT GTT

=

AAT ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TAA TGC TGG TGT

e.g., lexicographic order (like in a dictionary)

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Reconstructing a String from its Composition

String Reconstruction Problem. Reconstruct a string from its *k*-mer composition.

- **Input**. A collection of *k*-mers.
- **Output**. A *Genome* such that $Composition_k(Genome)$ is equal to the collection of *k*-mers.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

AAT ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TAA TGC TGG TGT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

AAT ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

AAT ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA
AAT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA
AAT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA

AAT

ATG

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG CAT CCA GAT GCC GGA GGG GTT





TAA
AAT
ATG

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG TGT

TAA
AAT
ATG

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG

TAA
AAT
ATG
TGT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Naive String Reconstruction Approach

ATG ATG CAT CCA GAT GCC GGA GGG GTT TGC TGG

TAA
AAT
ATG
TGT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What's Next?

ATG ATG CAT CCA GAT GCC GGA GGG TGC TGG

TAA
AAT
ATG
TGT
GTT



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- **String Reconstruction as a Hamiltonian Path Problem**
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Representing a Genome as a Path

$Composition_3$(TAATGCCATGGGATGTT)=

TAA AAT ATG TGC GCC CCA CAT ATG TGG GGG GGA GAT ATG TGT GTT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Representing a Genome as a Path

$Composition_3$(TAATGCCATGGGATGTT)=



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Representing a Genome as a Path

$Composition_3$(TAATGCCATGGGATGTT)=

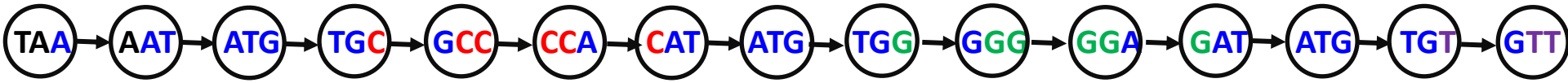


Can we construct this genome path without knowing the genome **TAATGCCATGGGATGTT**, only from its composition?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Representing a Genome as a Path

Composition$_3$**(TAATGCCATGGGATGTT)=**



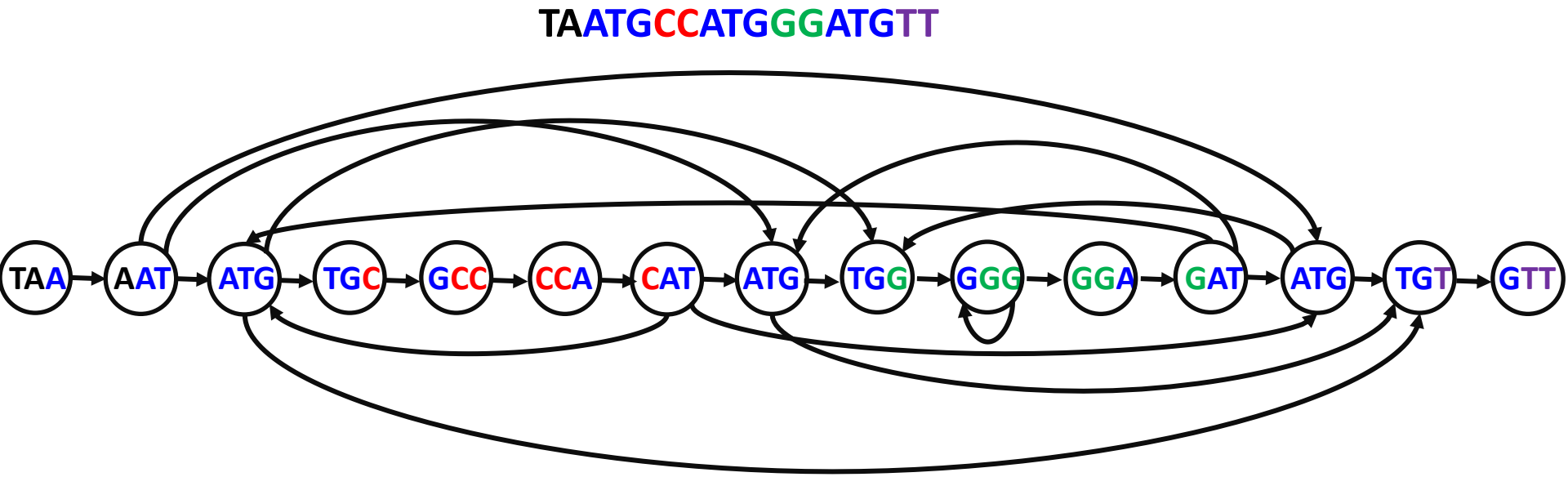
Can we construct this genome path without knowing the genome **TAATGCCATGGGATGTT**, only from its composition?

Yes. We simply need to connect *k*-mer$_1$ with *k*-mer$_2$ if ***suffix***(*k*-mer$_1$)=***prefix***(*k*-mer$_2$).

E.g. **TAA** → **AAT**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Path Turns into a Graph

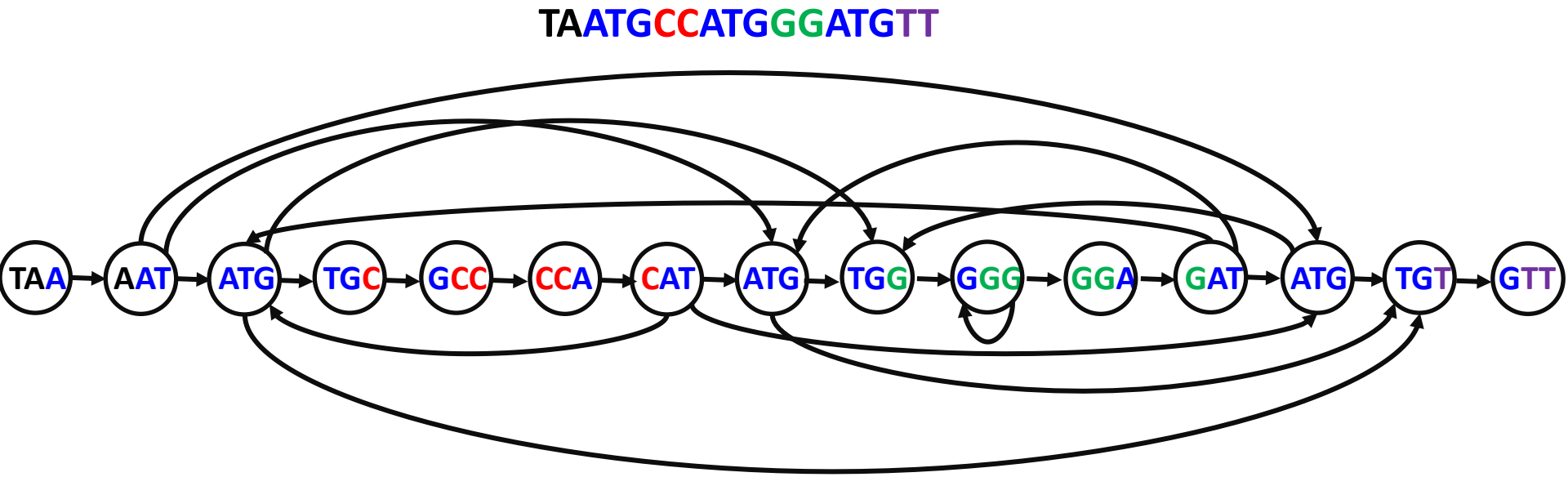


Yes. We simply need to connect k-mer$_1$ with k-mer$_2$ if ***suffix***(k-mer$_1$)=***prefix***(k-mer$_2$).

E.g. **TA**A → **A**AT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

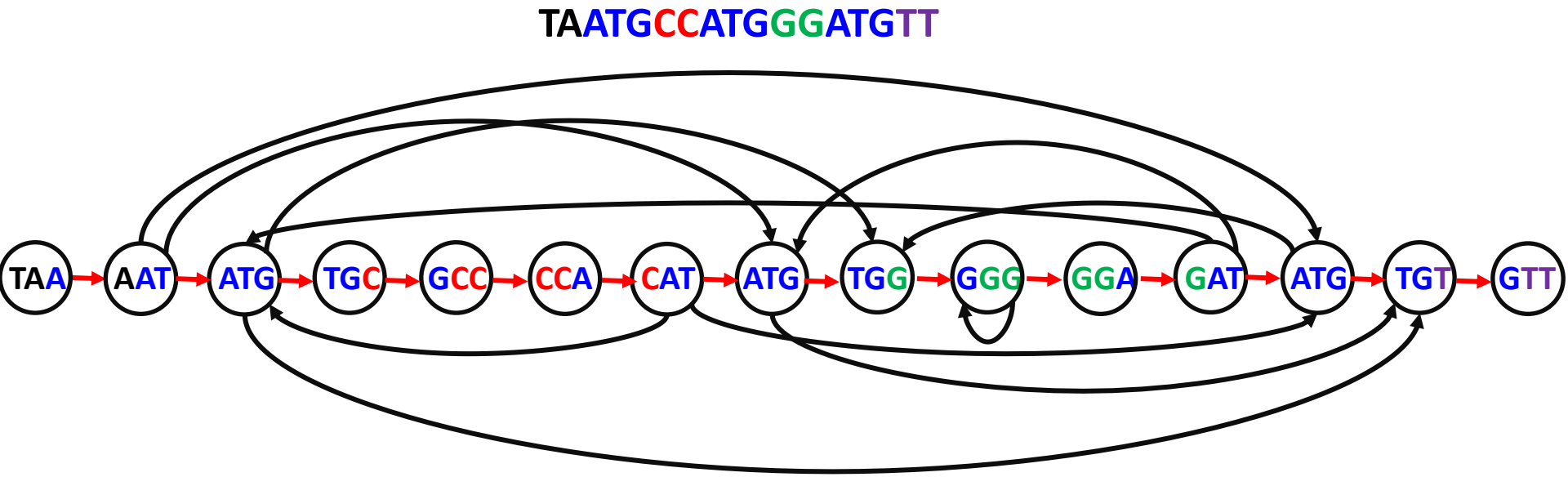
A Path Turns into a Graph



Can we still find the genome path in this graph?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Path Turns into a Graph

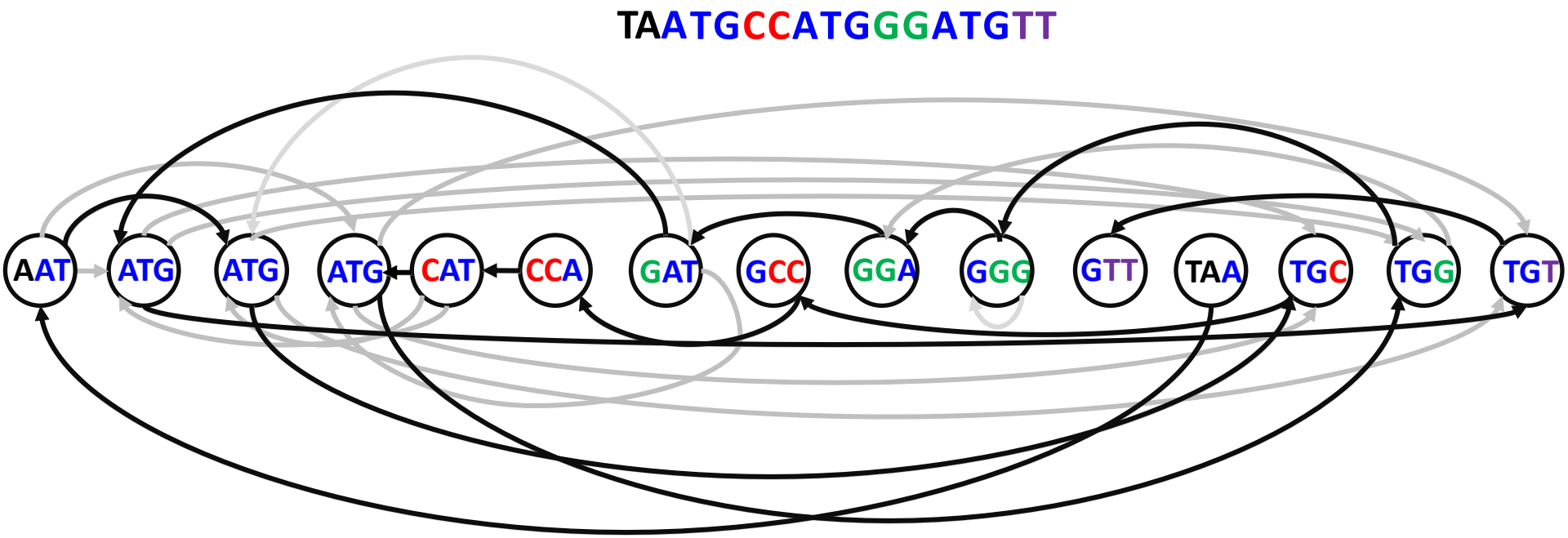


Can we still find the genome path in this graph?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Where Is the Genomic Path?

A **Hamiltonian path:** a path that visits each node in a graph exactly once.



What are we trying to find in this graph?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Hamiltonian Path Problem

Hamiltonian Path Problem. Find a Hamiltonian path in a graph.

- **Input.** A graph.
- **Output.** A path visiting every **node** in the graph exactly once.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Does This Graph Have a Hamiltonian Path?

Hamiltonian Path Problem. Find a Hamiltonian path in a graph.
Input. A graph.
Output. A path visiting every **node** in the graph exactly once.



Icosian game (1857)



William Hamilton

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Does This Graph Have a Hamiltonian Path?

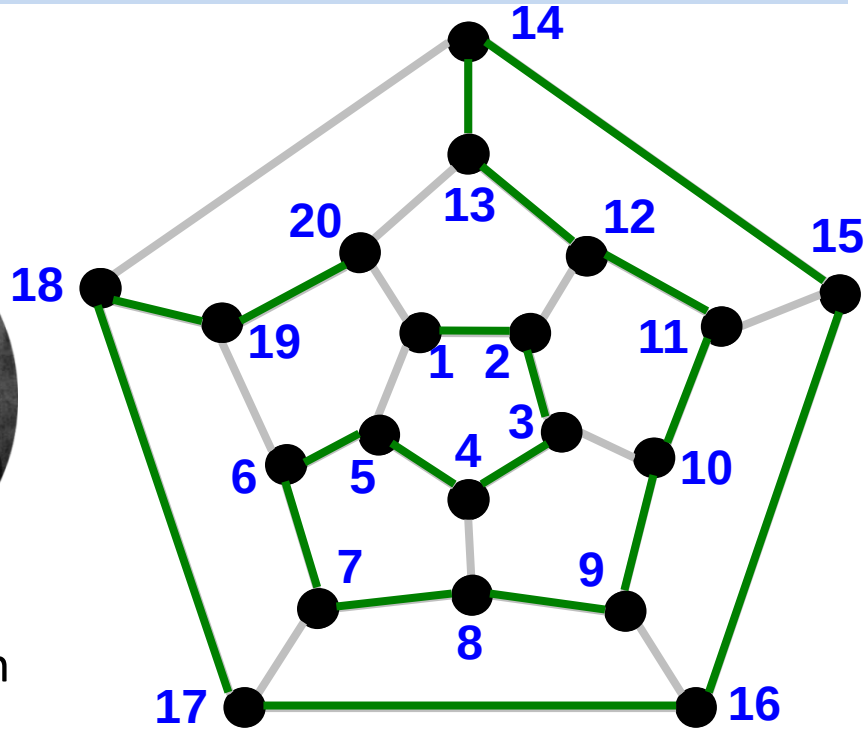
Hamiltonian Path Problem. Find a Hamiltonian path in a graph.
Input. A graph.
Output. A path visiting every **node** in the graph exactly once.



Icosian game (1857)



William Hamilton



Undirected graph

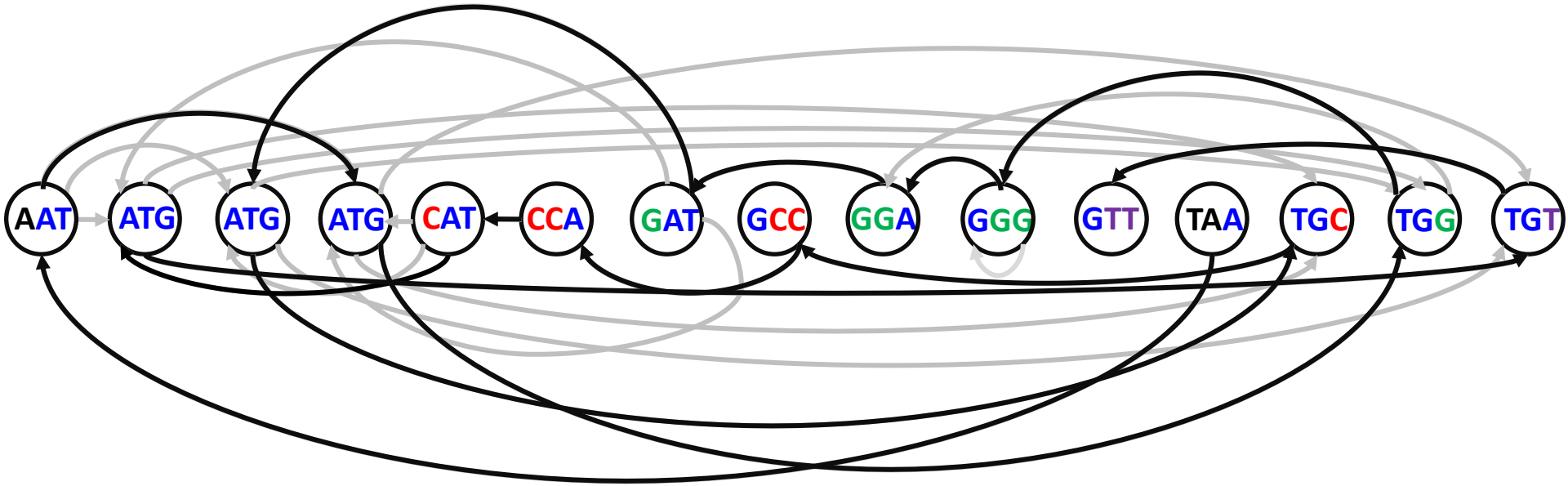
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

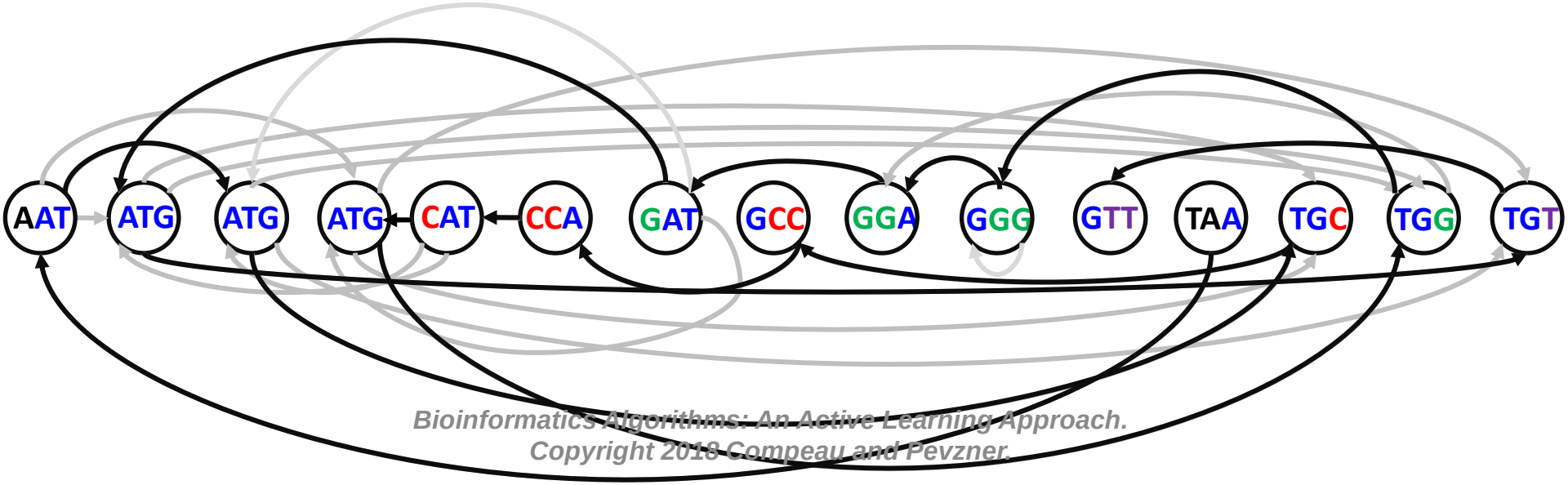
- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- **String Reconstruction as an Eulerian Path Problem**
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

TAATGGGATGCCATGTT



TAATGCCATGGGATGTT



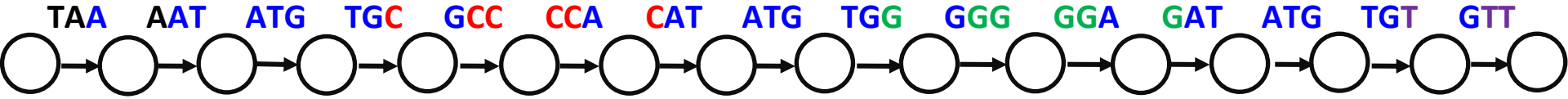
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Slightly Different Path

TAATGCCATGGGATGTT

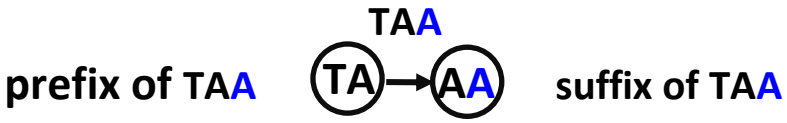


3-mers as **nodes**



3-mers as **edges**

How do we label the starting and ending nodes of an edge?



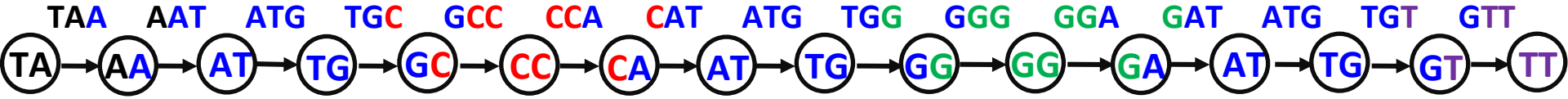
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Labeling Nodes in the New Path

TAATGCCATGGGATGTT



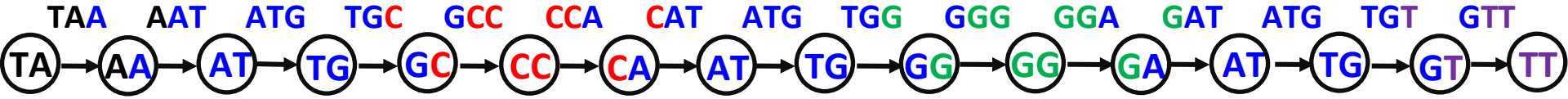
3-mers as **nodes**



3-mers as **edges** and 2-mers as **nodes**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

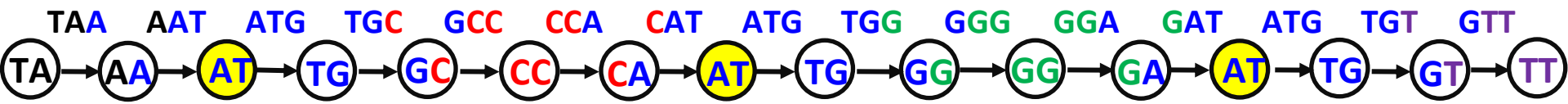
Labeling Nodes in the New Path

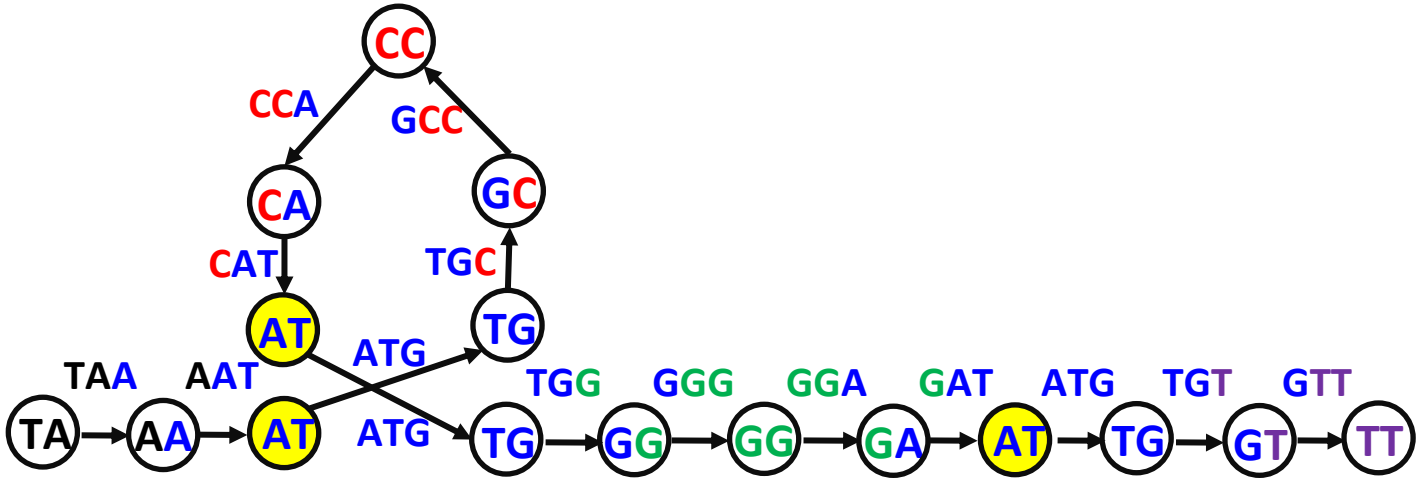


3-mers as **edges** and 2-mers as **nodes**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

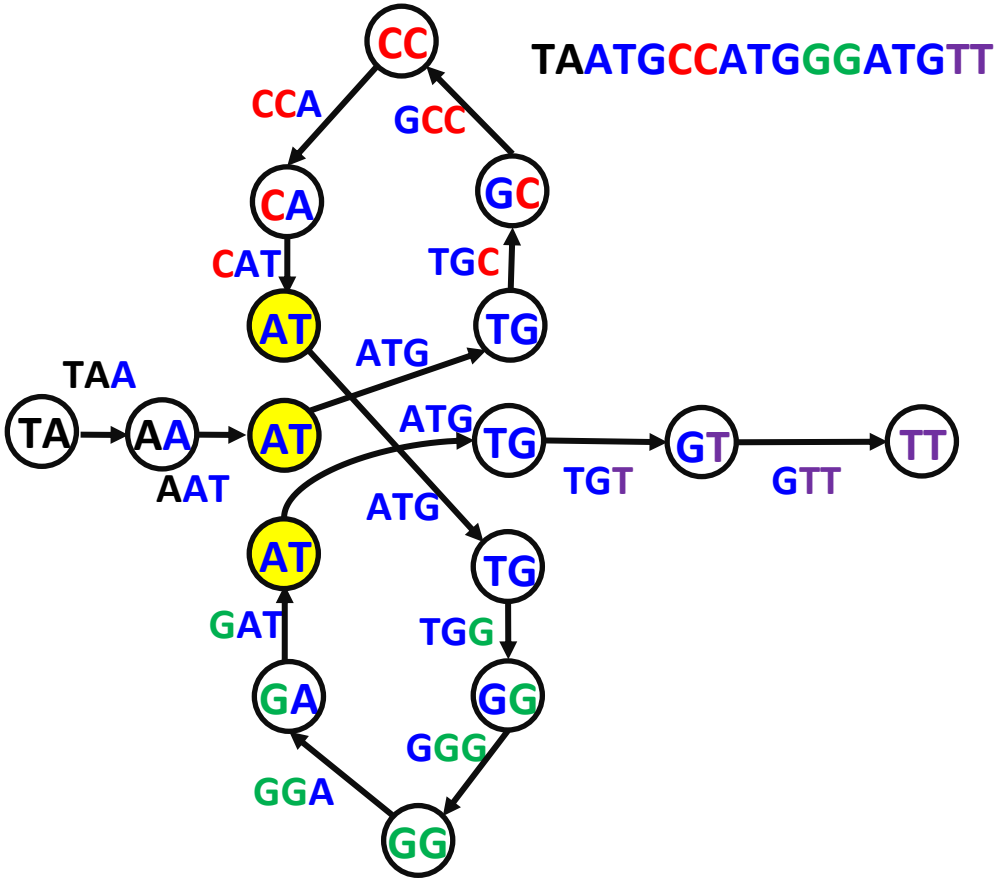
Gluing Identically Labeled Nodes





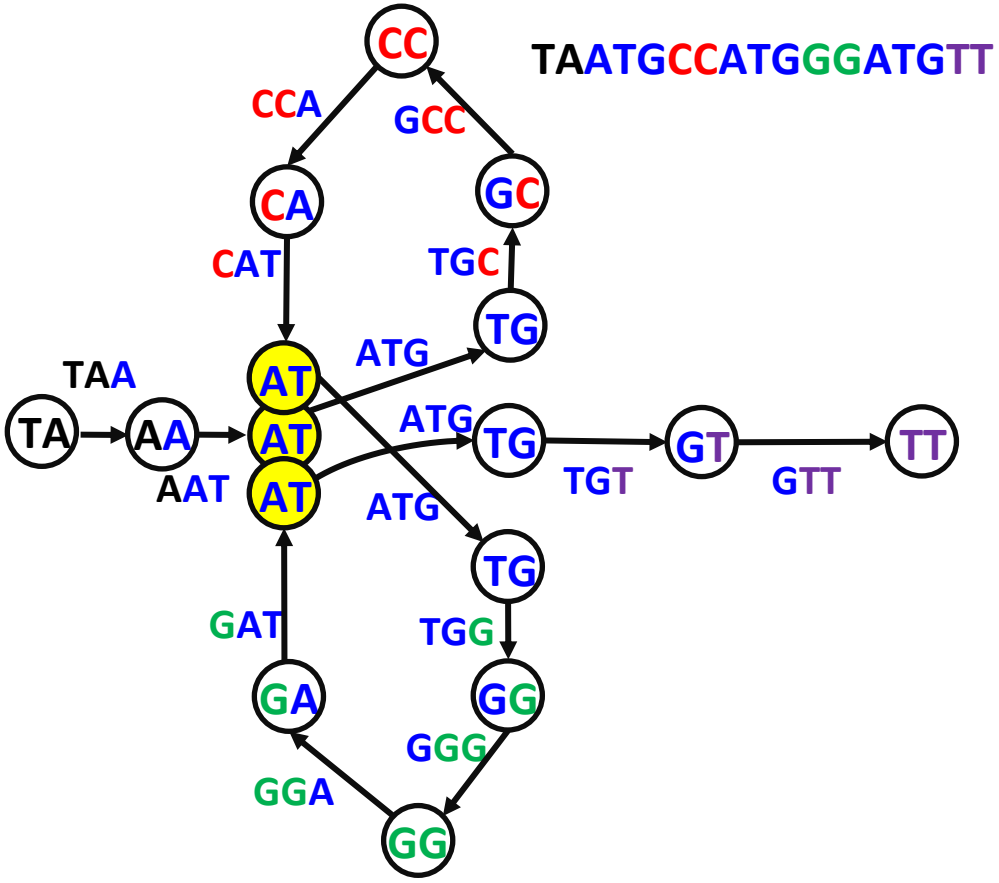
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



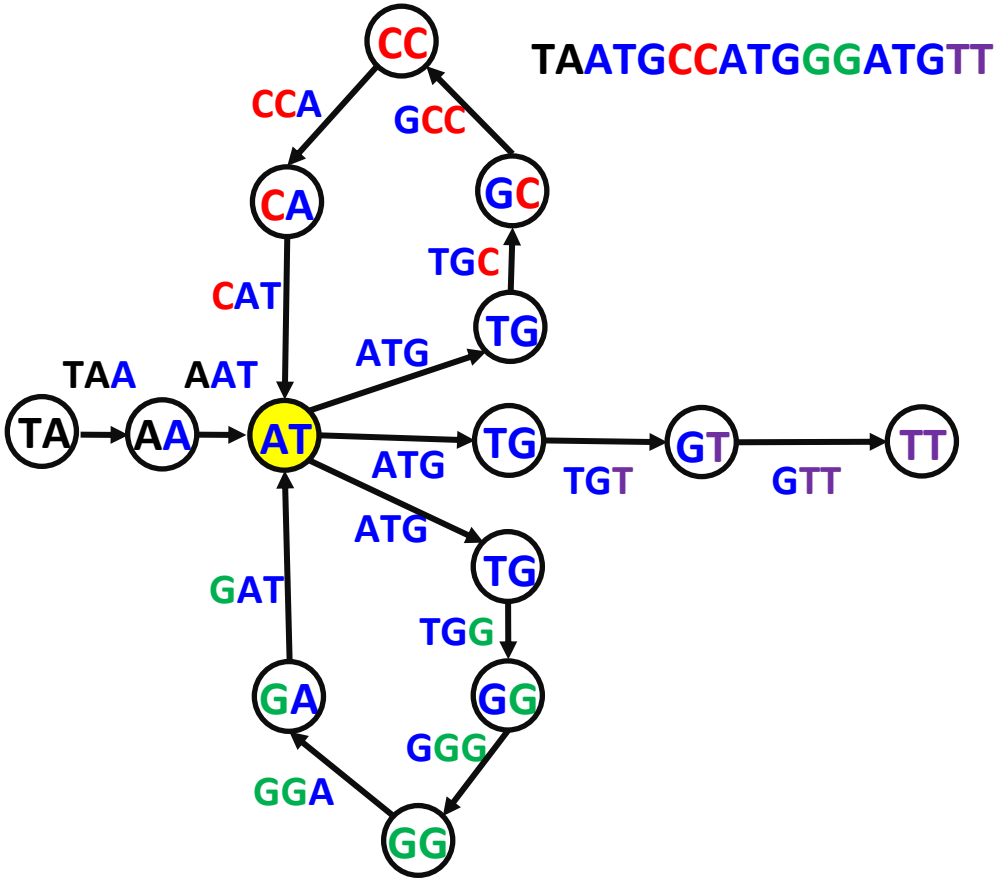
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



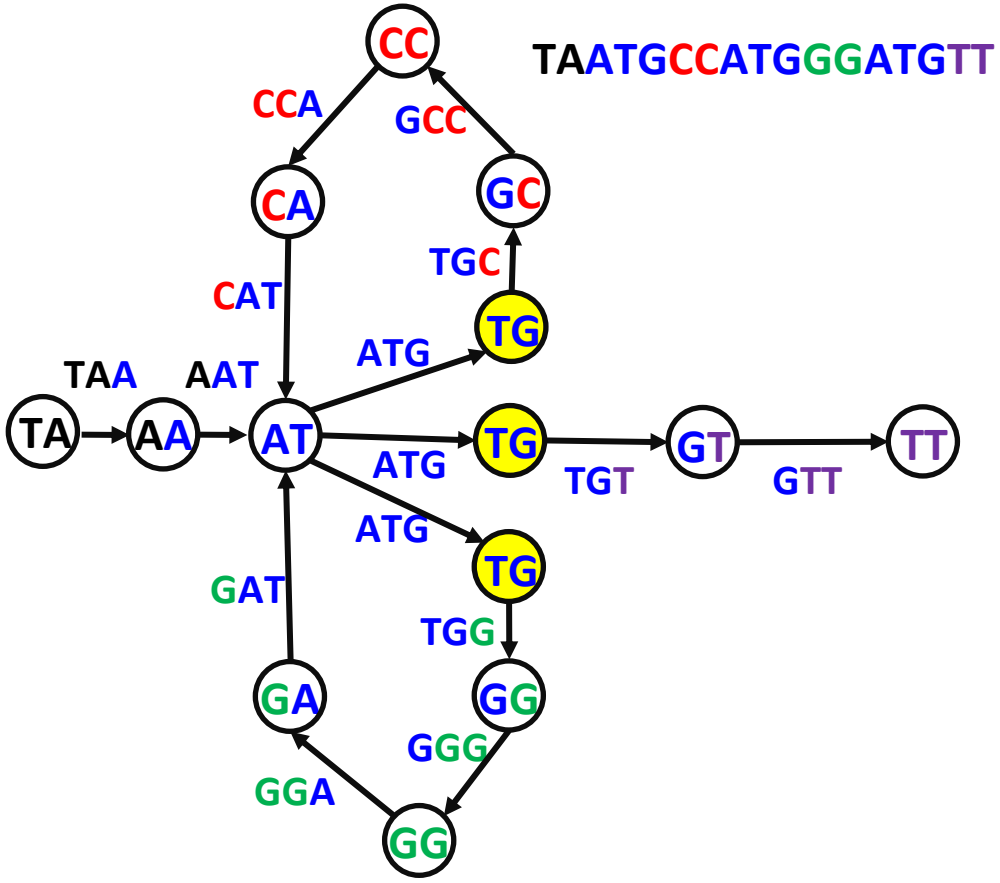
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



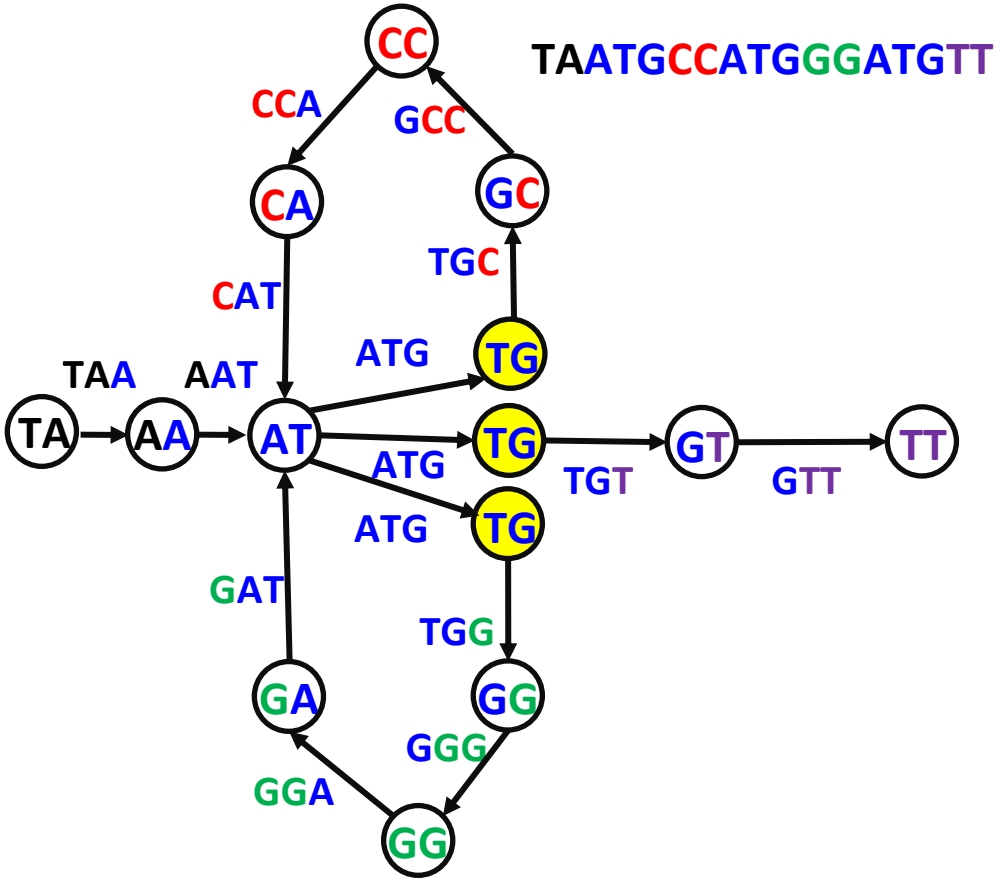
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



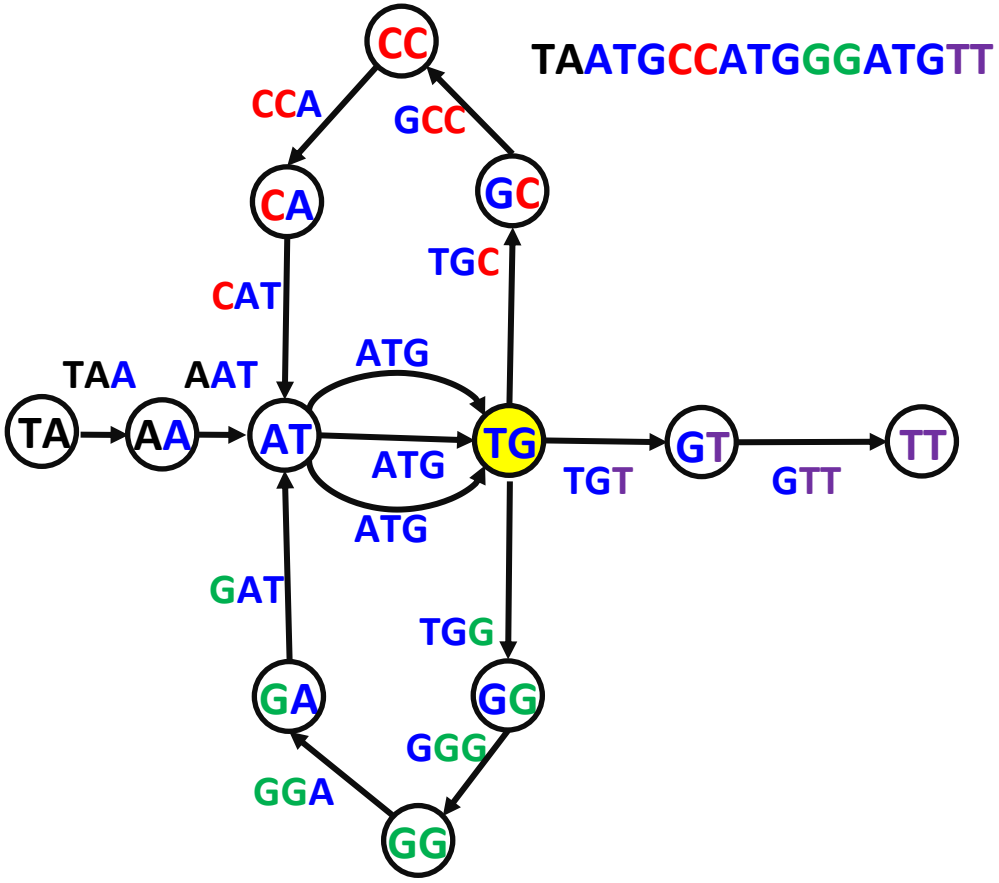
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



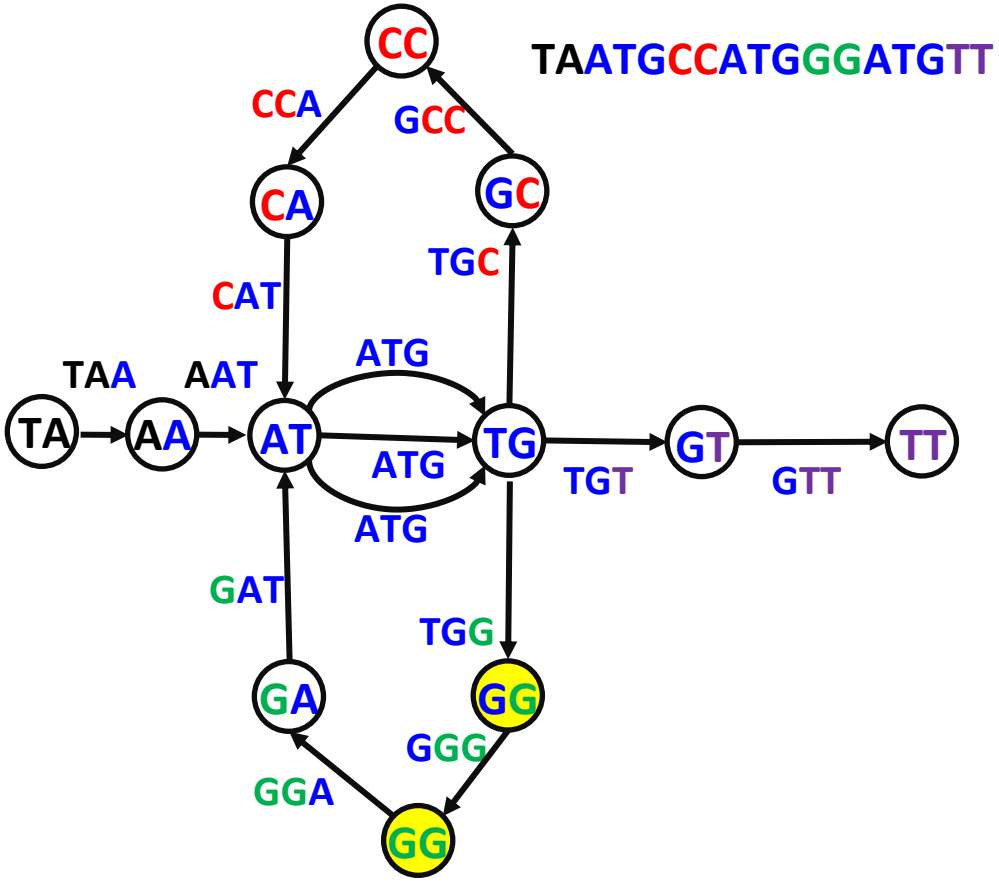
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



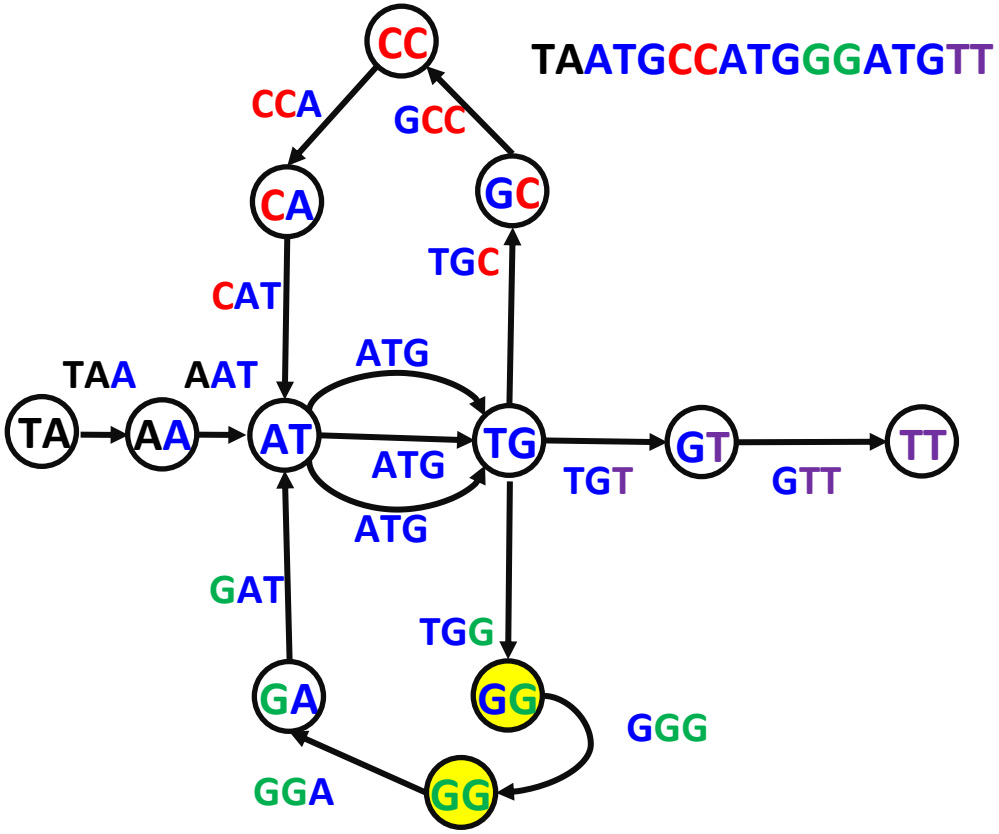
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



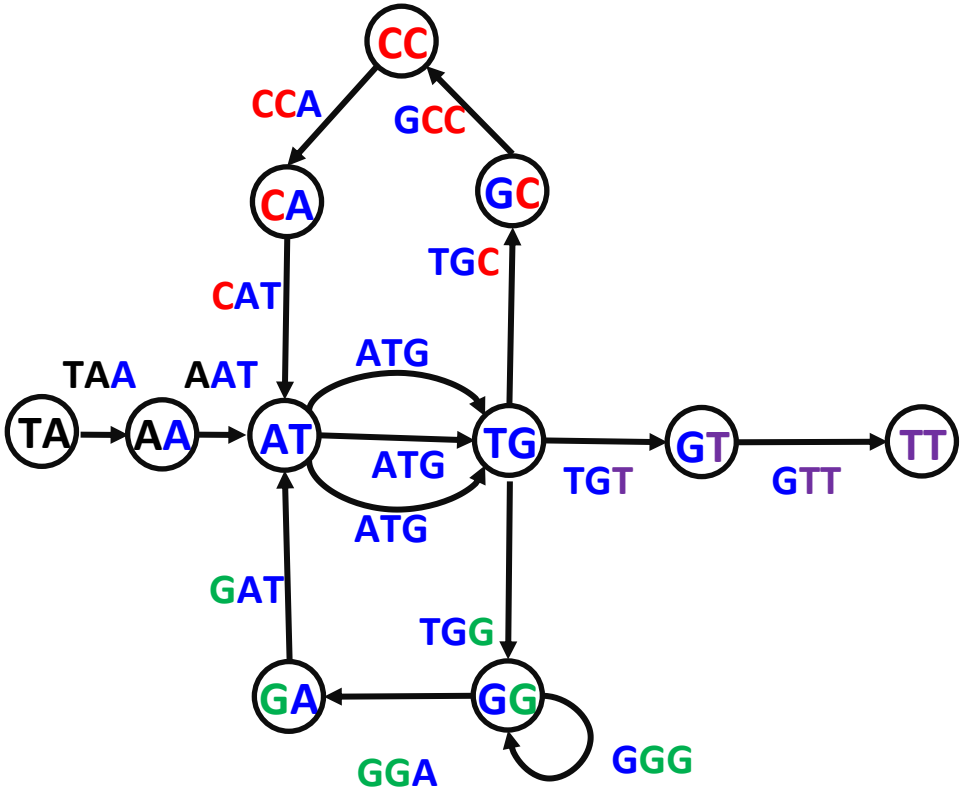
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

De Bruijn Graph of **TA**ATGCCATGGGATGTT

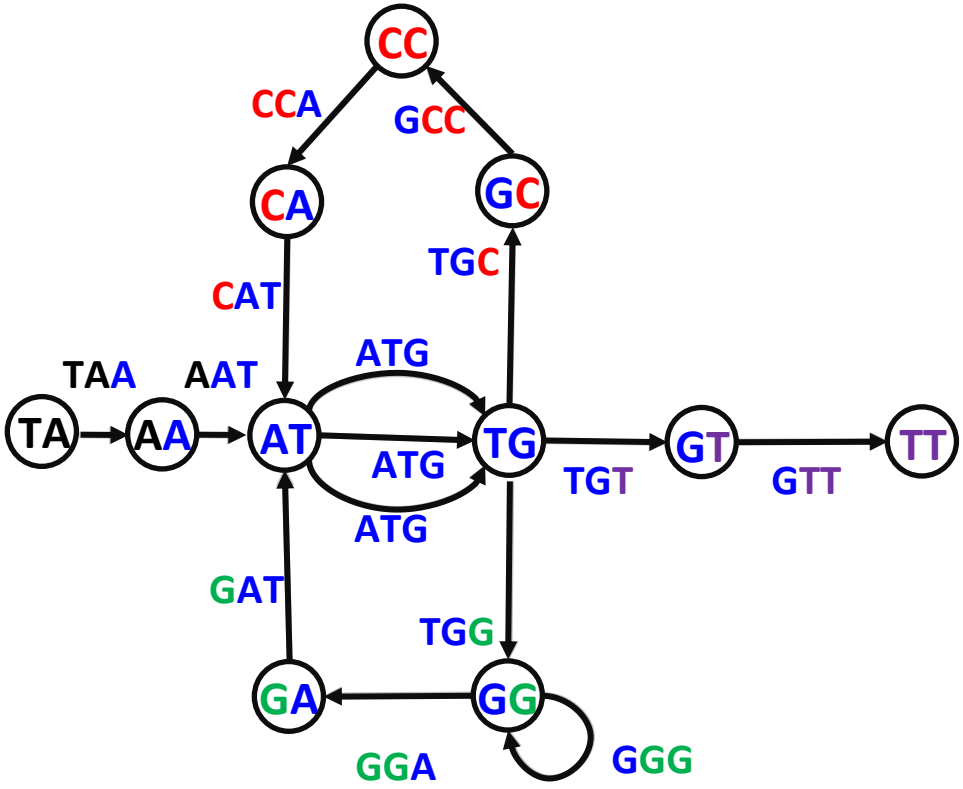


Where is the *Genome* hiding in this graph?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

It Was Always There!

TAATGCCATGGGATGTT



An **Eulerian path** in a graph is a path that visits each **edge** exactly once.

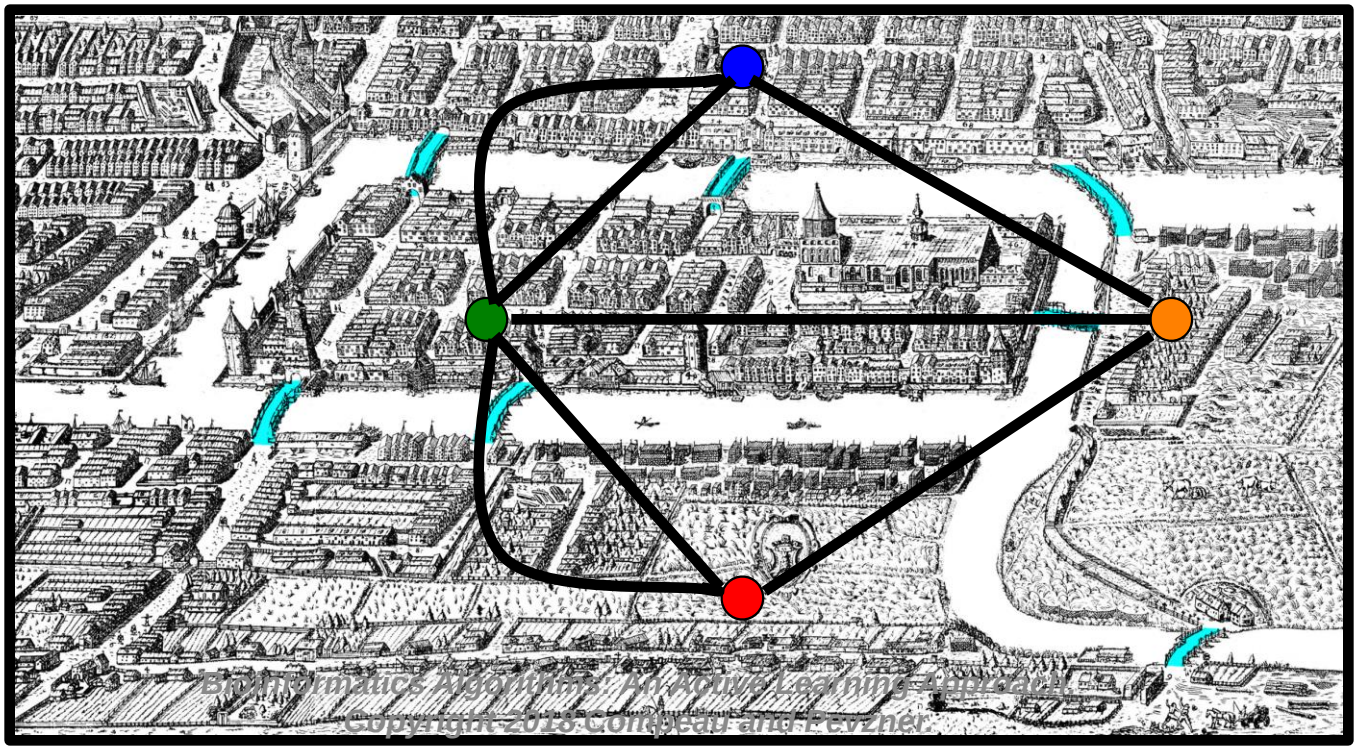
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Eulerian Path Problem

Eulerian Path Problem. Find an Eulerian path in a graph.



- **Input.** A graph.
- **Output.** A path visiting every edge in the graph exactly once.



Copyright 2016 Kevin Wayne

Eulerian Versus Hamiltonian Paths

Eulerian Path Problem. Find an Eulerian path in a graph.

- **Input.** A graph.
- **Output.** A path visiting every edge in the graph exactly once.

Hamiltonian Path Problem. Find a Hamiltonian path in a graph.

- **Input.** A graph.
- **Output.** A path visiting every node in the graph exactly once.

Find a difference!



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- **Similar Problems with Different Fates**
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Eulerian Versus Hamiltonian Paths

Eulerian Path Problem. Find an **Eulerian** path in a graph.



- **Input.** A graph.
- **Output.** A path visiting every **edge** in the graph exactly once.

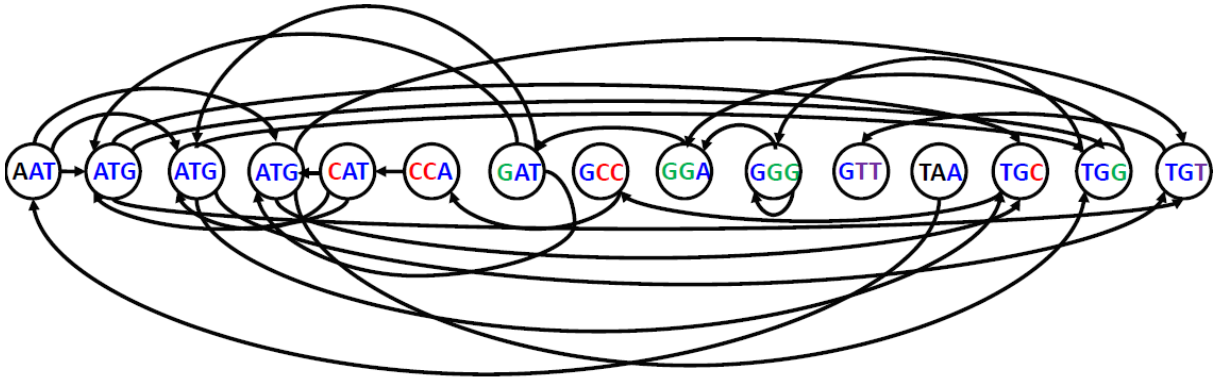
Hamiltonian Path Problem. Find a **Hamiltonian** path in a graph.



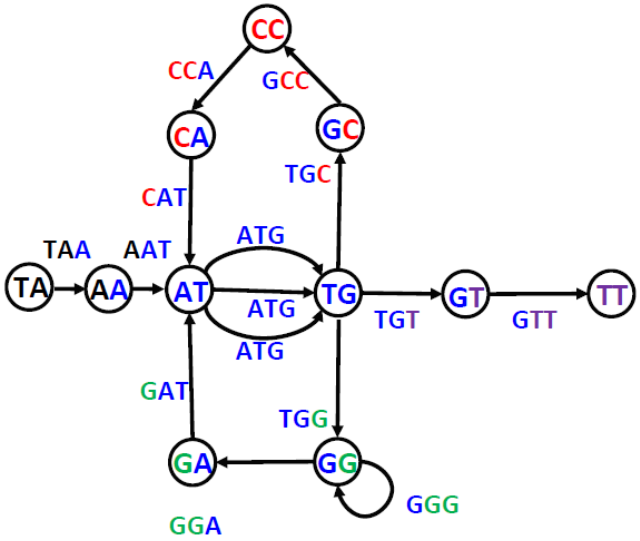
- **Input.** A graph.
- **Output.** A path visiting every **node** in the graph exactly once.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What Problem Would You Prefer to Solve?

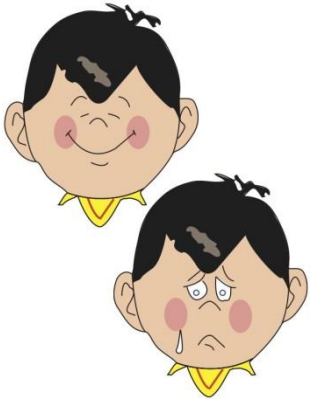


Hamiltonian Path Problem



Eulerian Path Problem

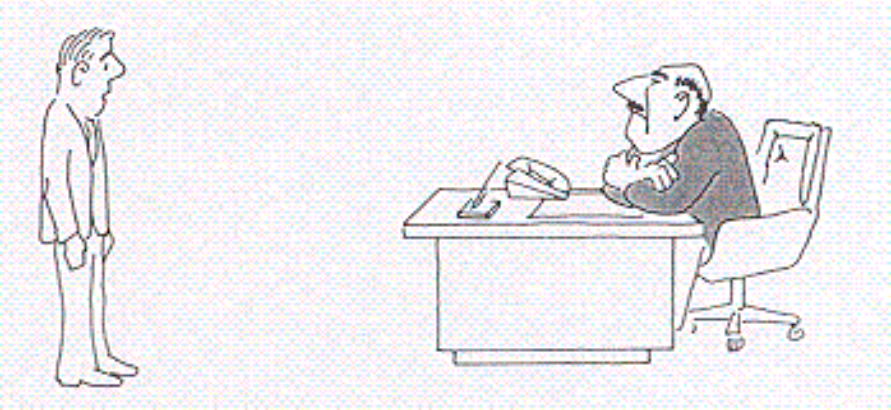
While Euler solved the Eulerian Path Problem (even for a city with a million bridges), nobody has developed a fast algorithm for the Hamiltonian Path Problem yet.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

NP-Complete Problems

- The Hamiltonian Path Problem belongs to a collection containing thousands of computational problems for which no fast algorithms are known.



"I can't find an efficient algorithm, I guess I'm just too dumb."

From Garey and Johnson, Computers and Intractability. 1979

Bioinformatics Algorithms: An Active Learning Approach.
Copyright © 2018 Compeau and Pevzner

Change of Attitude

That would be an excellent argument, but the question of whether or not NP-Complete problems can be solved efficiently is one of seven **Millennium Problems** in mathematics.



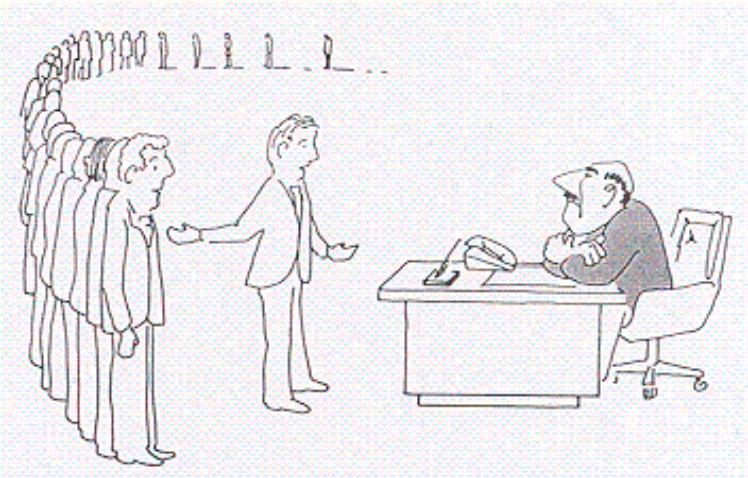
"I can't find an efficient algorithm, because no such algorithm is possible."

From Garey and Johnson, Computers and Intractability. 1979

Copyright © 2018 Compeau and Pevzner

The Modern State of Affairs

NP-Complete problems are all equivalent: find an efficient solution to one, and you have an efficient solution to them all.



"I can't find an efficient algorithm, but neither can all these famous people."

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- **De Bruijn Graphs**
- Euler's Theorem
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

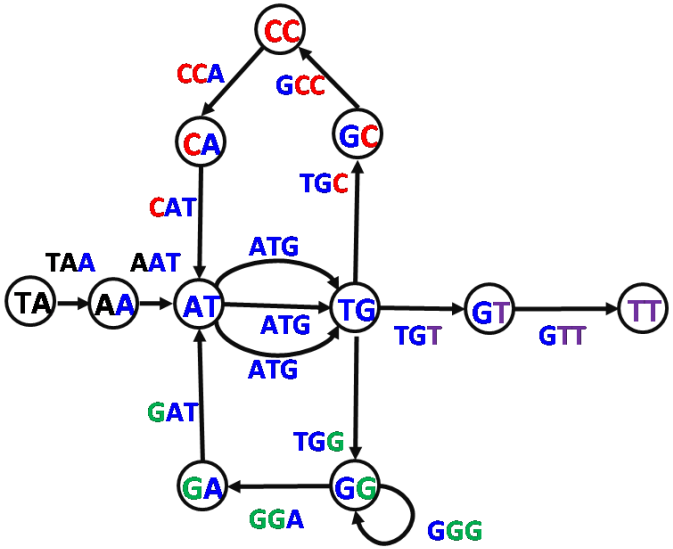
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Eulerian Path Problem

Eulerian Path Problem. Find an **Eulerian** path in a graph.



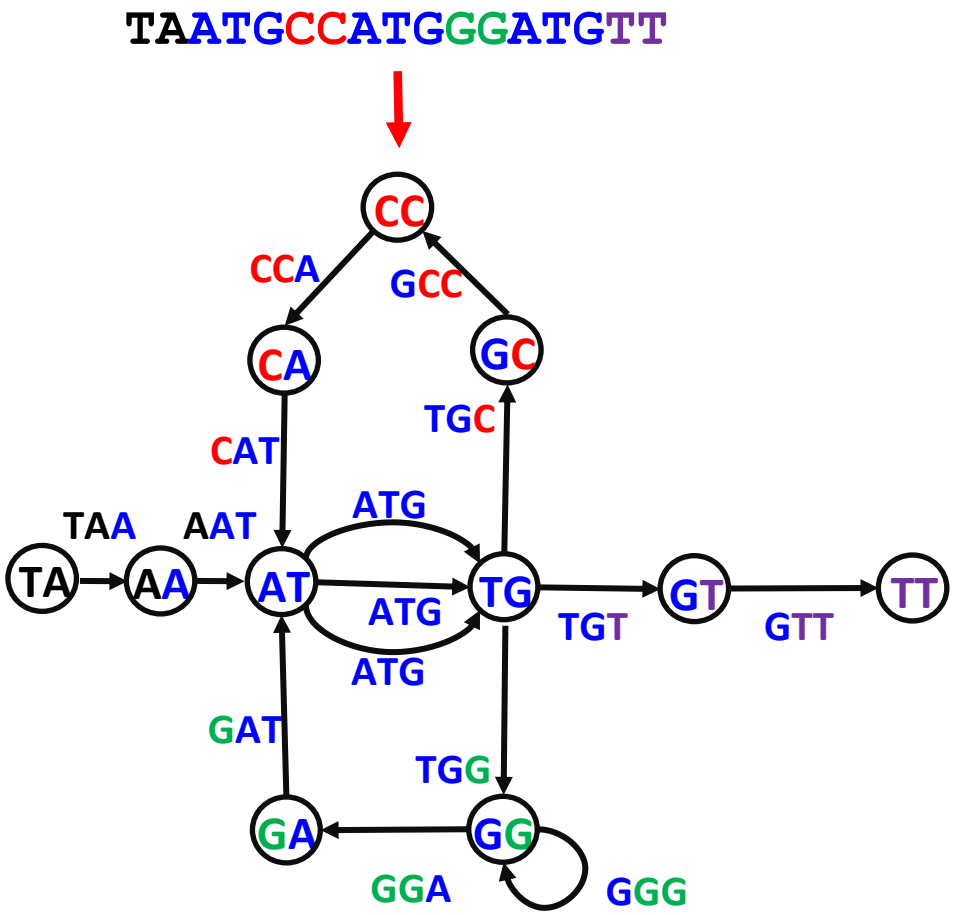
- **Input.** A graph.
- **Output.** A path visiting every **edge** in the graph exactly once.



We constructed the de Bruijn graph from *Genome*, but in reality, *Genome* is unknown!

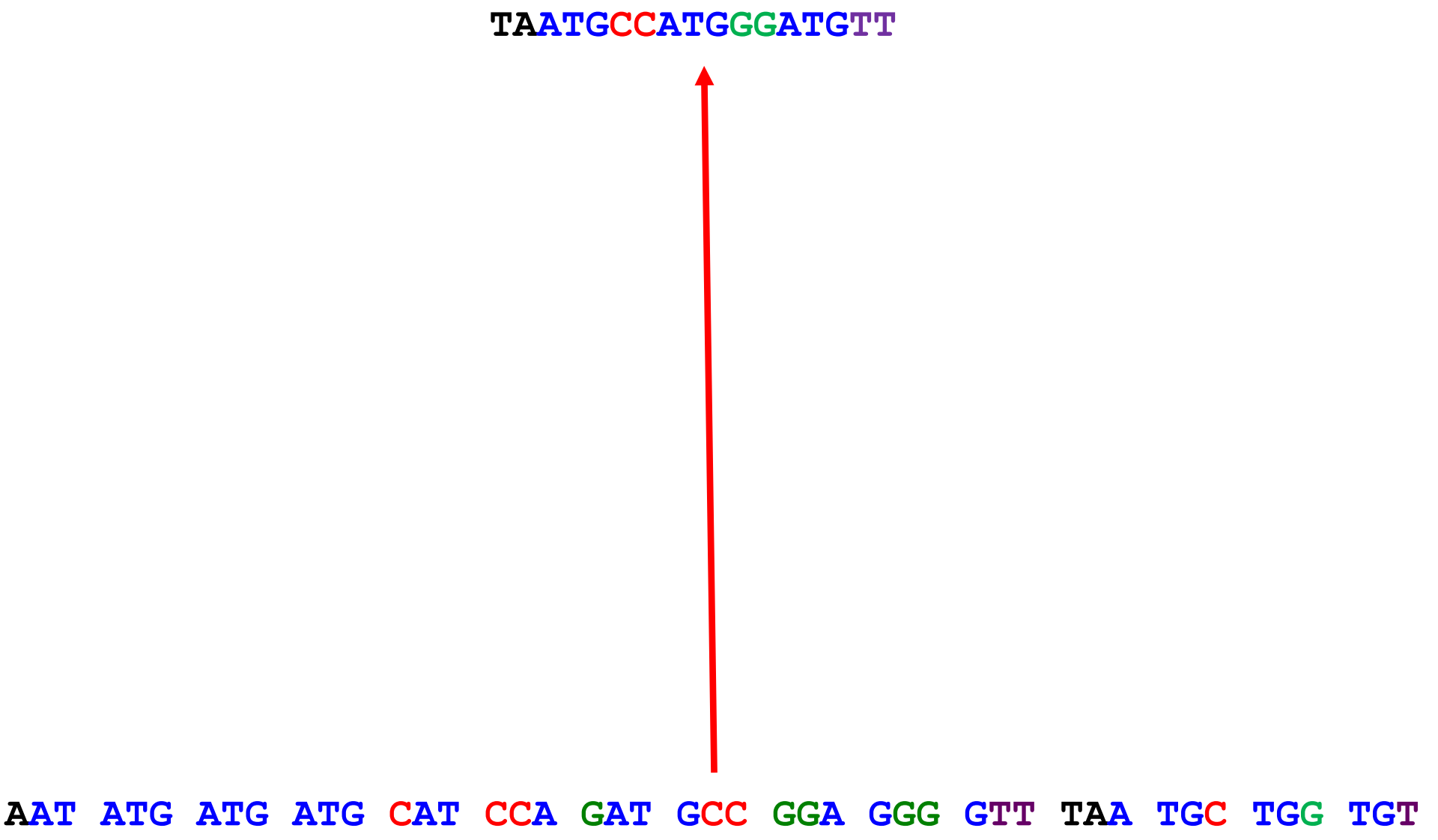
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What We Have Done: From *Genome* to de Bruijn Graph



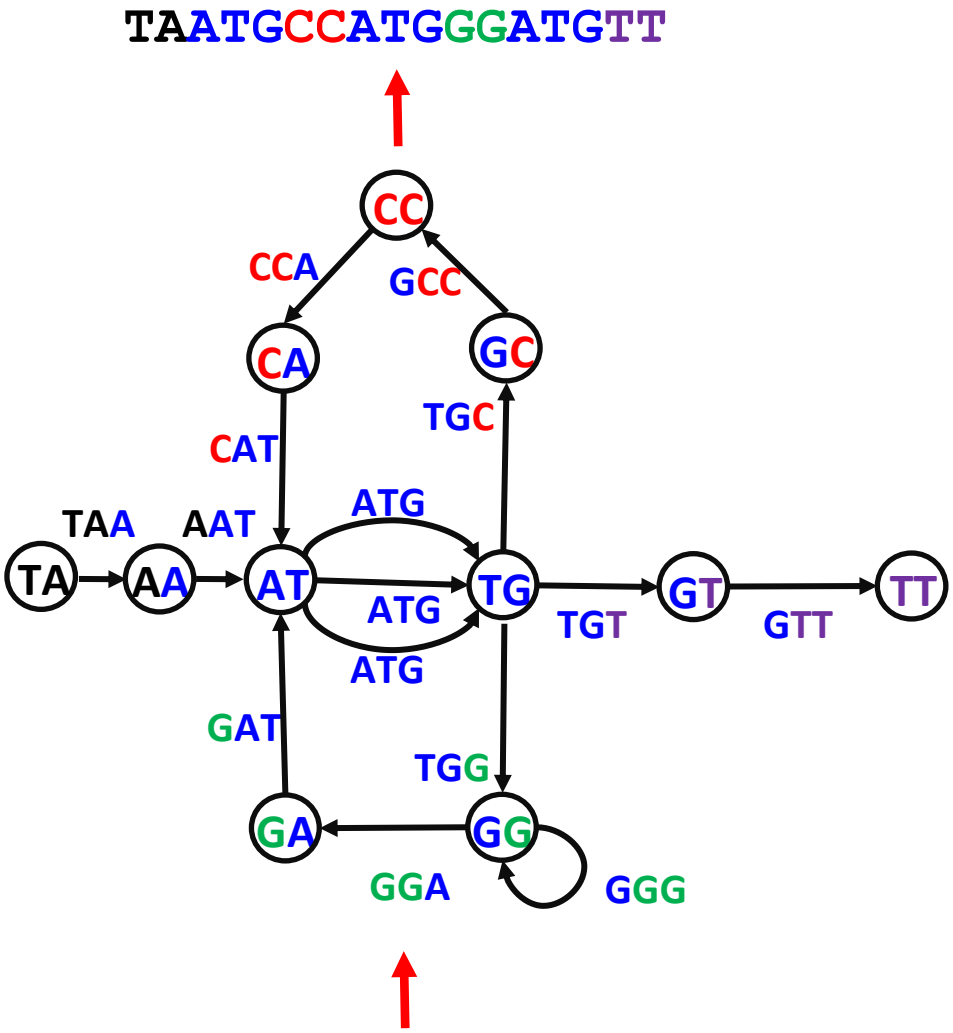
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What We Want: From Reads (*k*-mers) to *Genome*



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What We will Show: From Reads to de Bruijn Graph to *Genome*

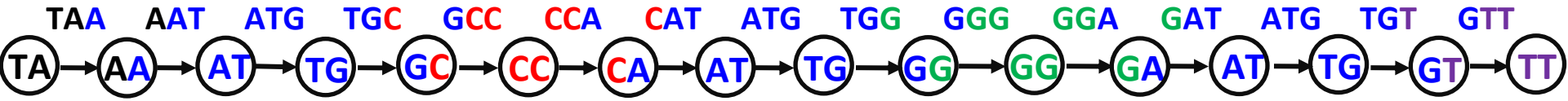


AAT ATG ATG ATG CAT CCA GAT GCC GGA GGG GTT TAA TGC TGG TGT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing de Bruijn Graph when *Genome* Is Known

TAATGCCATGGGATGTT



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing de Bruijn when *Genome* Is Unknown

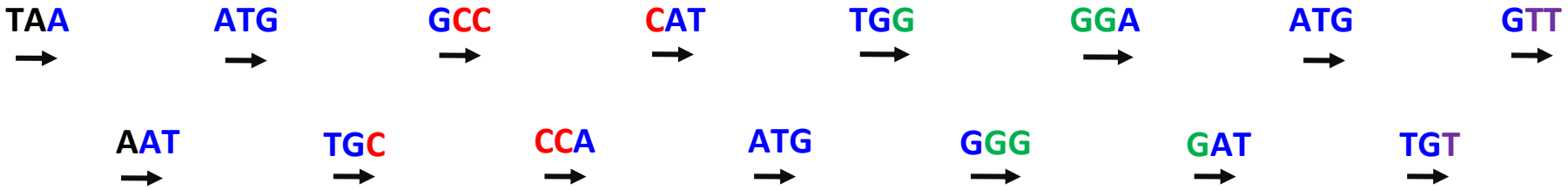
TAA ATG GCC CAT TGG GGA ATG GTT

AAT TGC CCA ATG GGG GAT TGT

Composition$_3$**(TAATGCCATGGGATGTT)**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

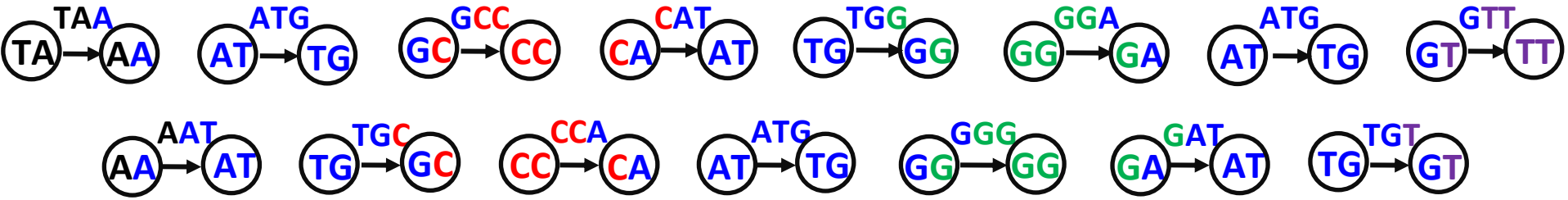
Representing Composition as a Graph Consisting of Isolated Edges



Composition$_3$**(TAATGCCATGGGATGTT)**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

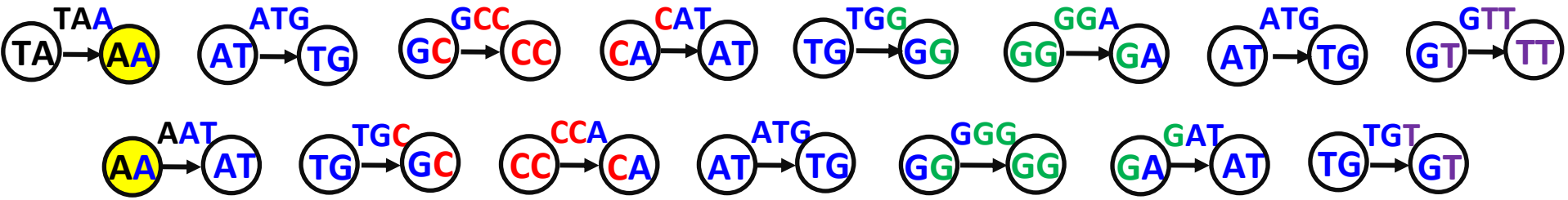
Constructing de Bruijn Graph from *k*-mer Composition



Composition$_3$**(TAATGCCATGGGATGTT)**

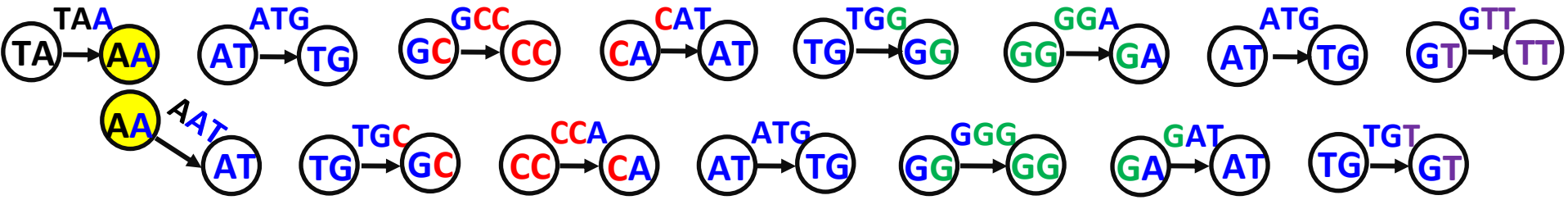
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes

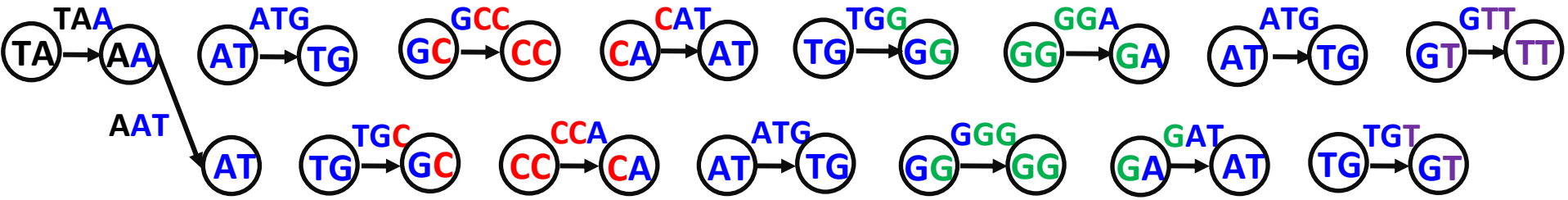


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

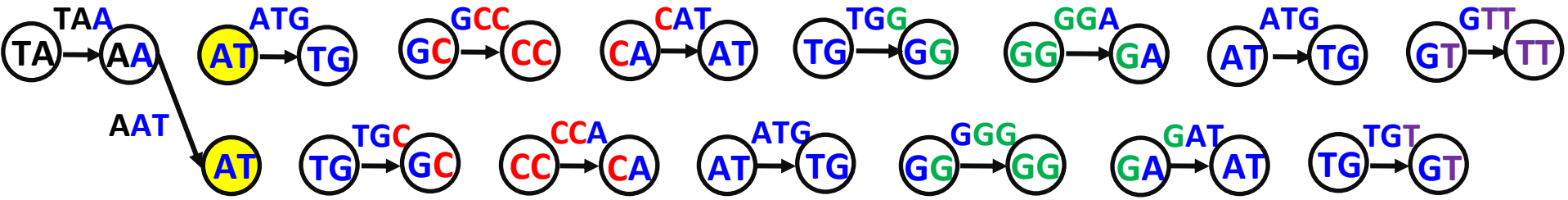
Gluing Identically Labeled Nodes



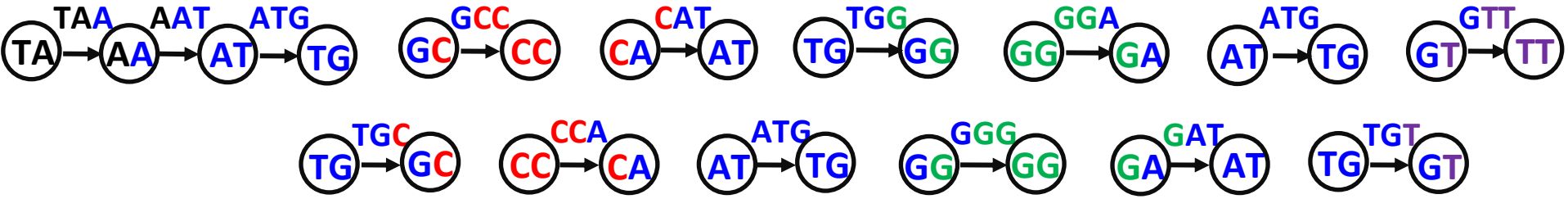
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



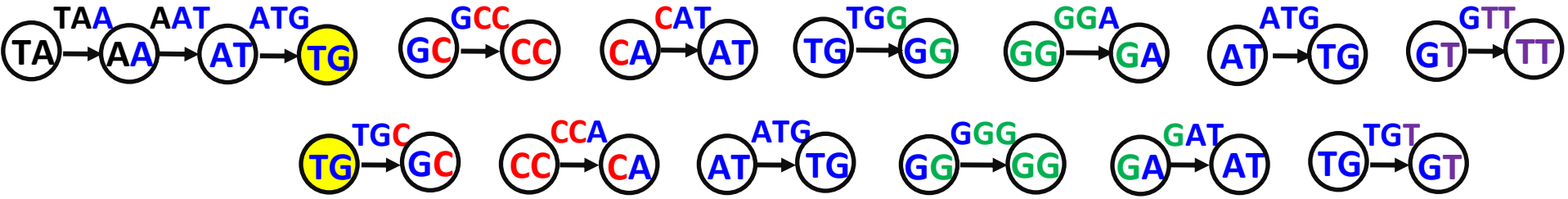
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



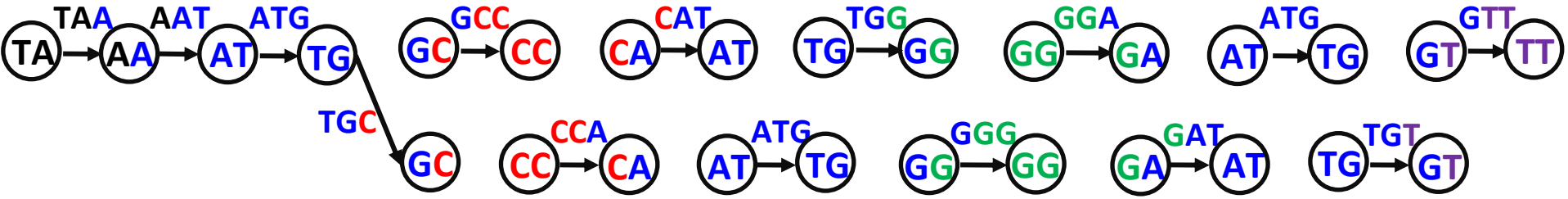
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



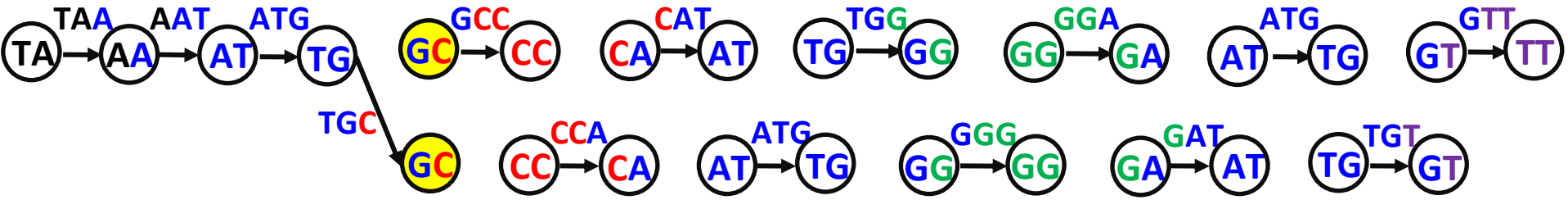
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



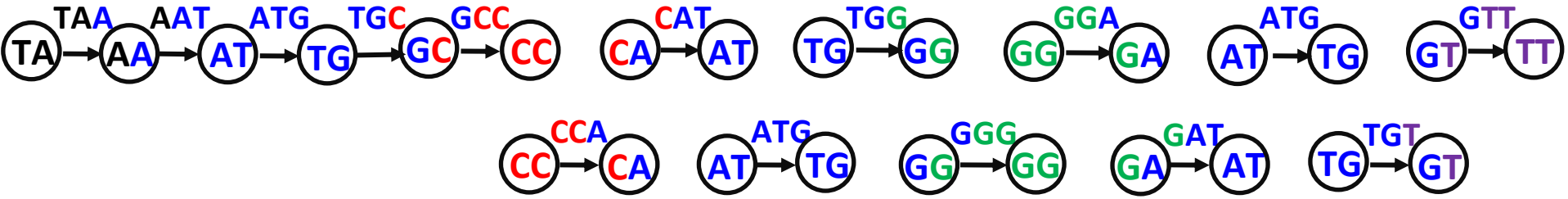
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



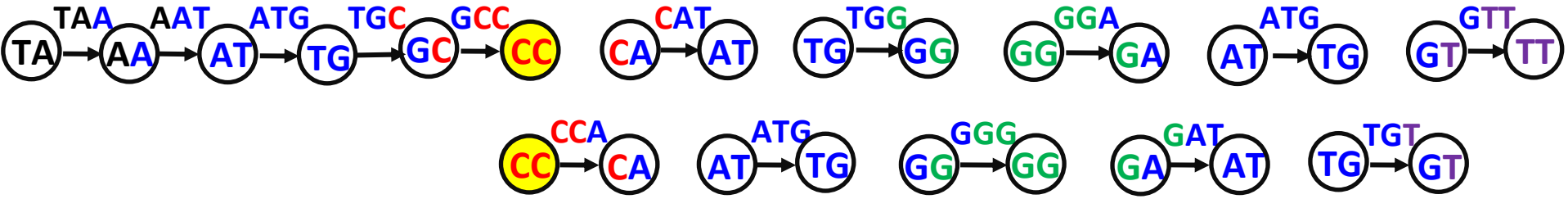
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



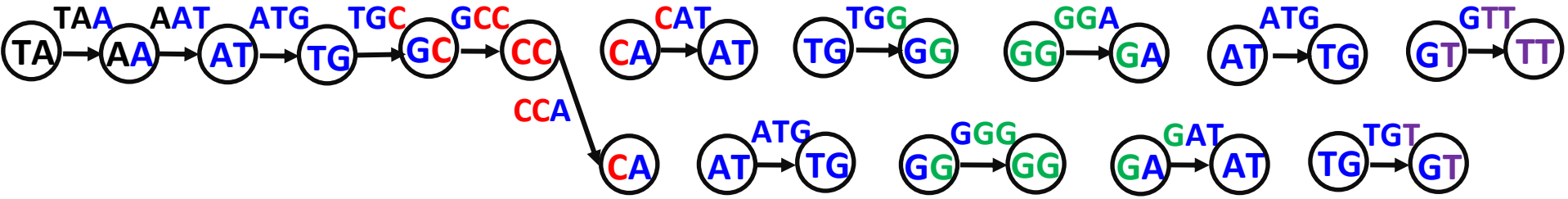
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



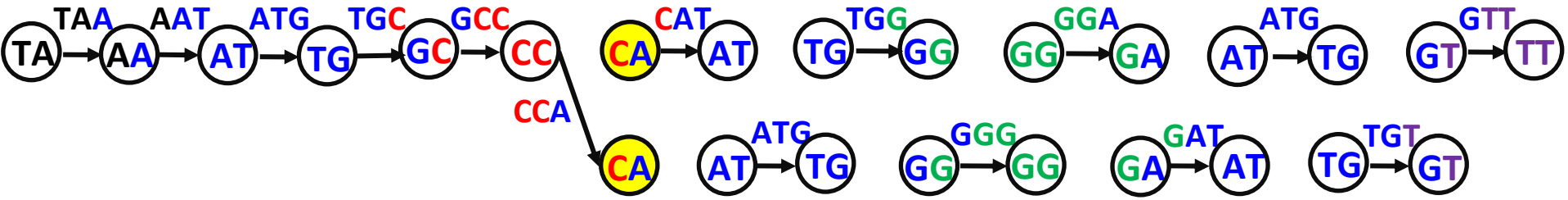
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



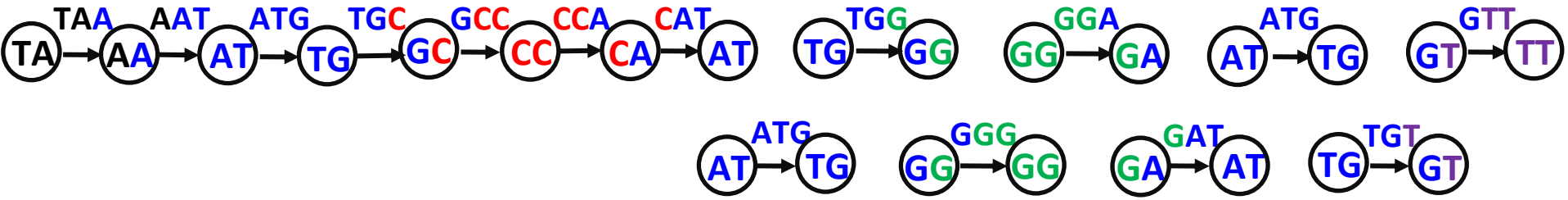
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



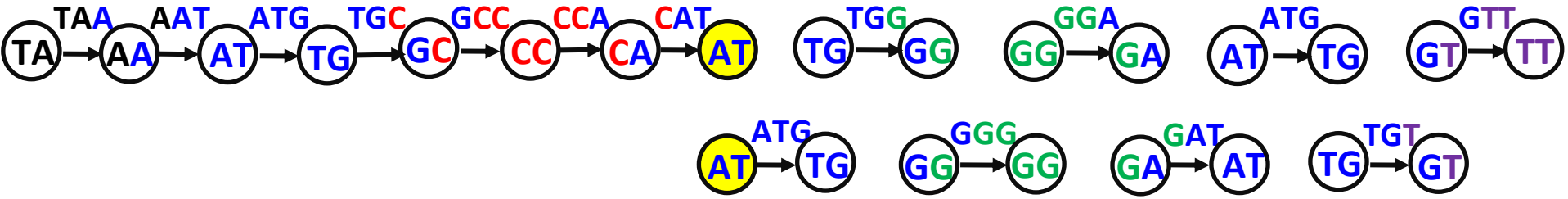
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



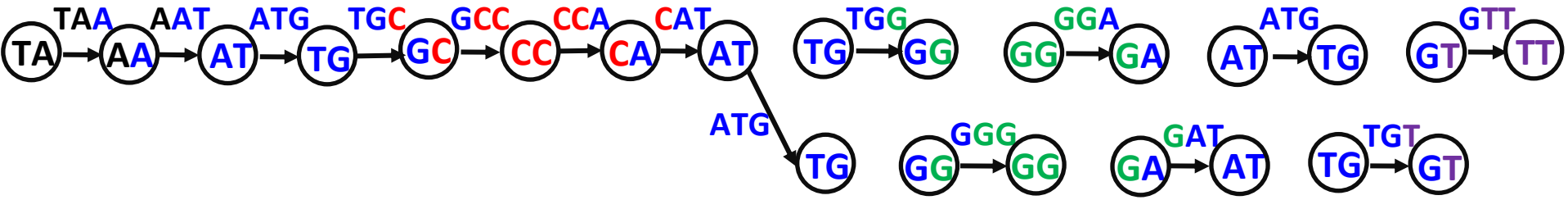
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



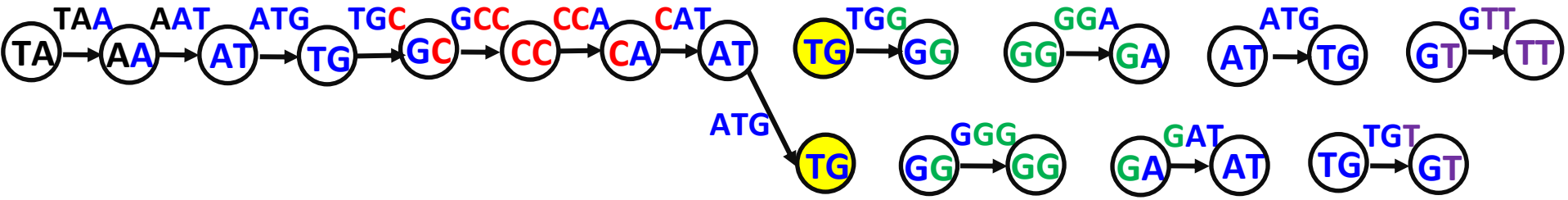
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



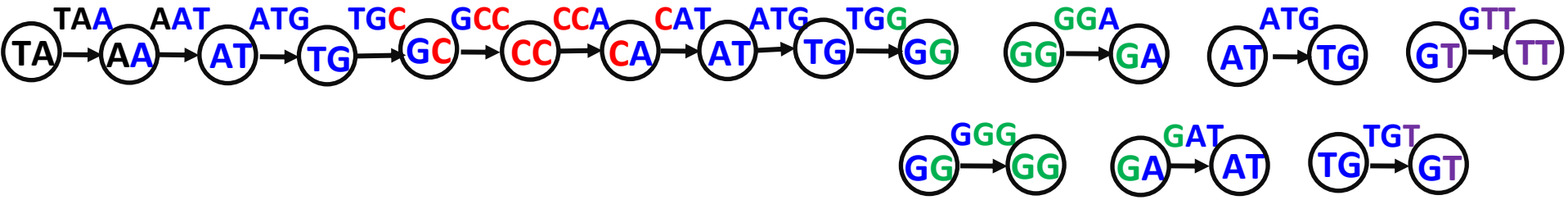
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



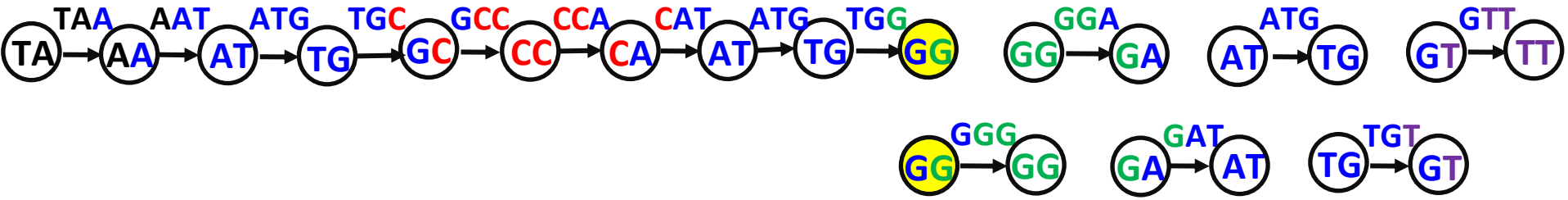
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



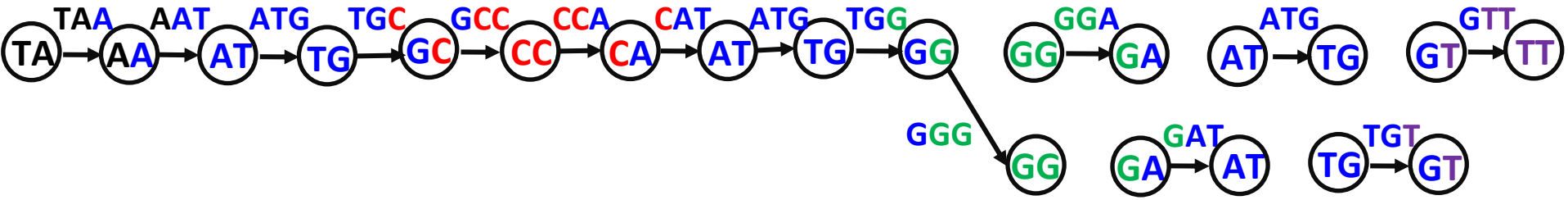
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



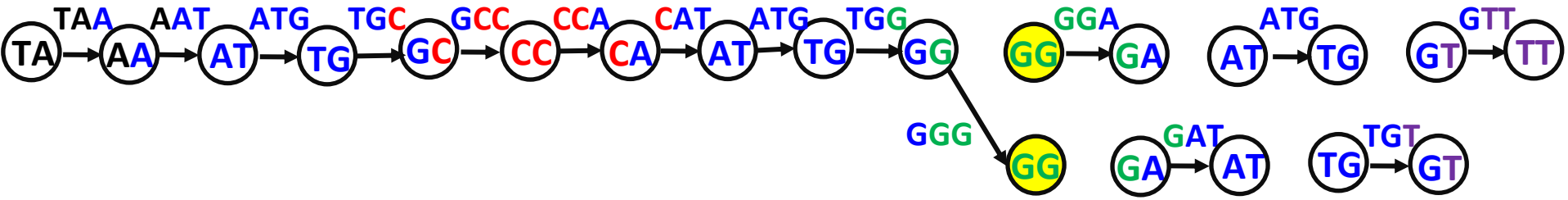
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



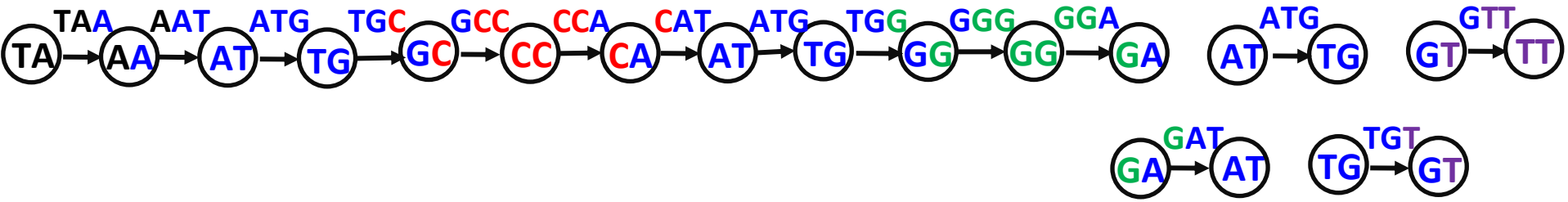
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



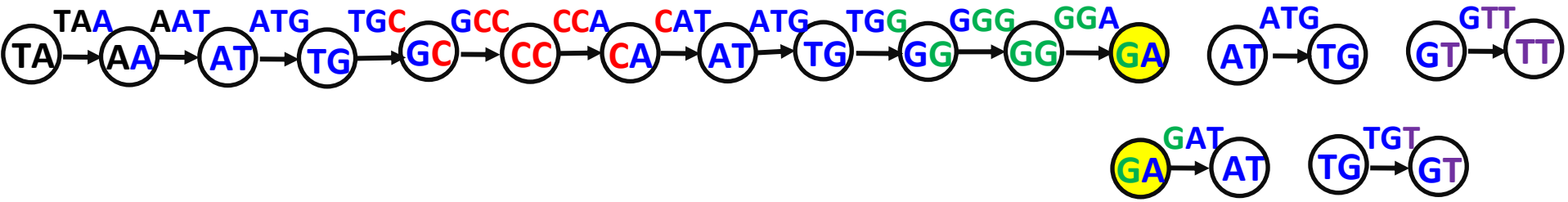
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



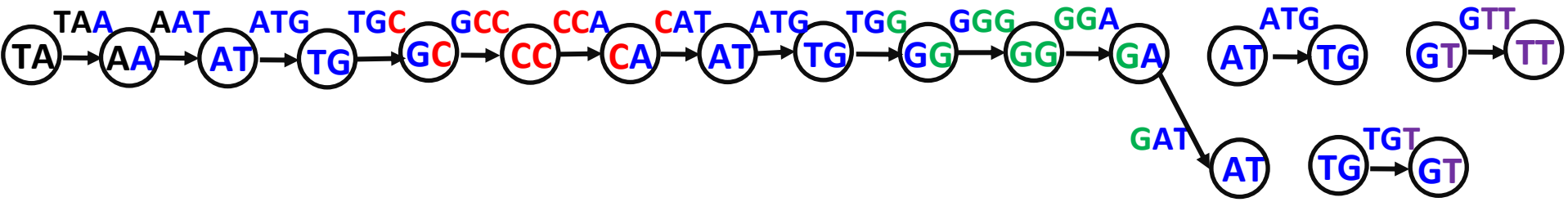
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



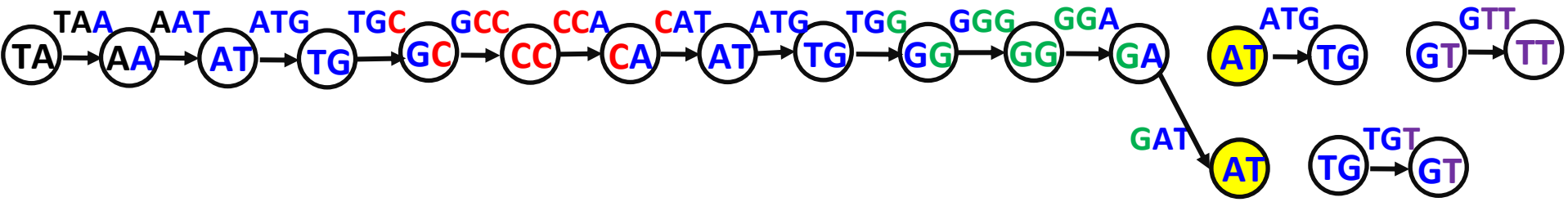
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



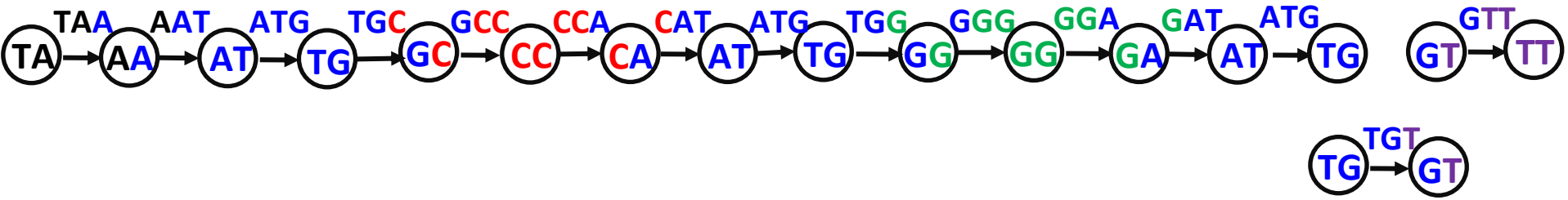
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



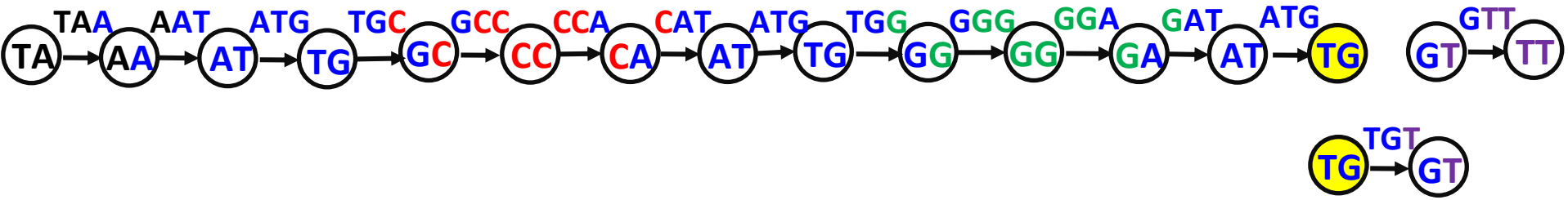
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



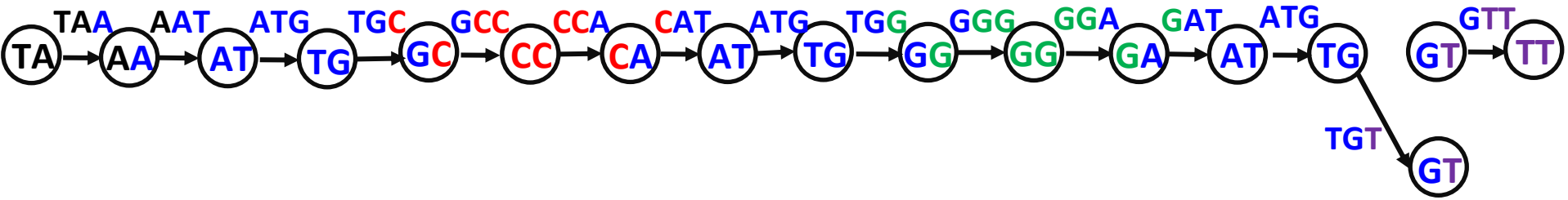
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



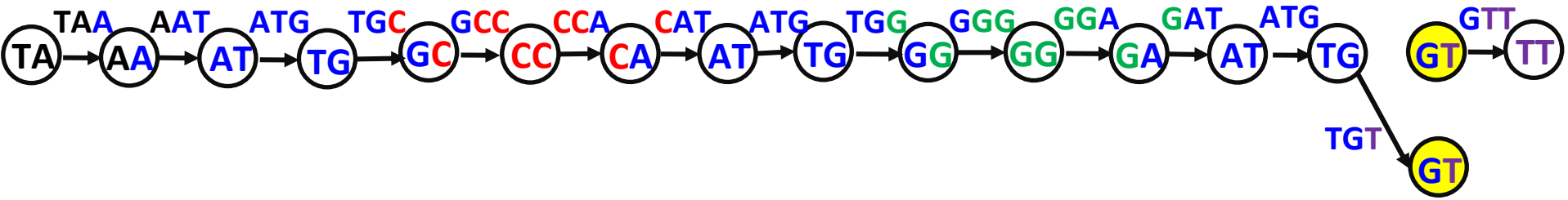
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

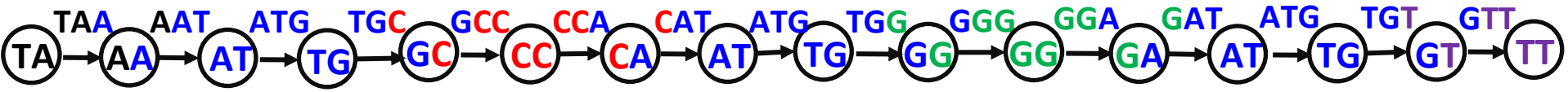


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



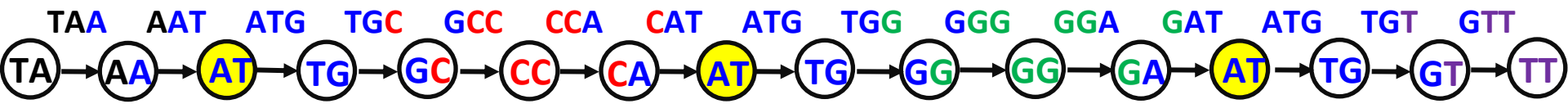
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

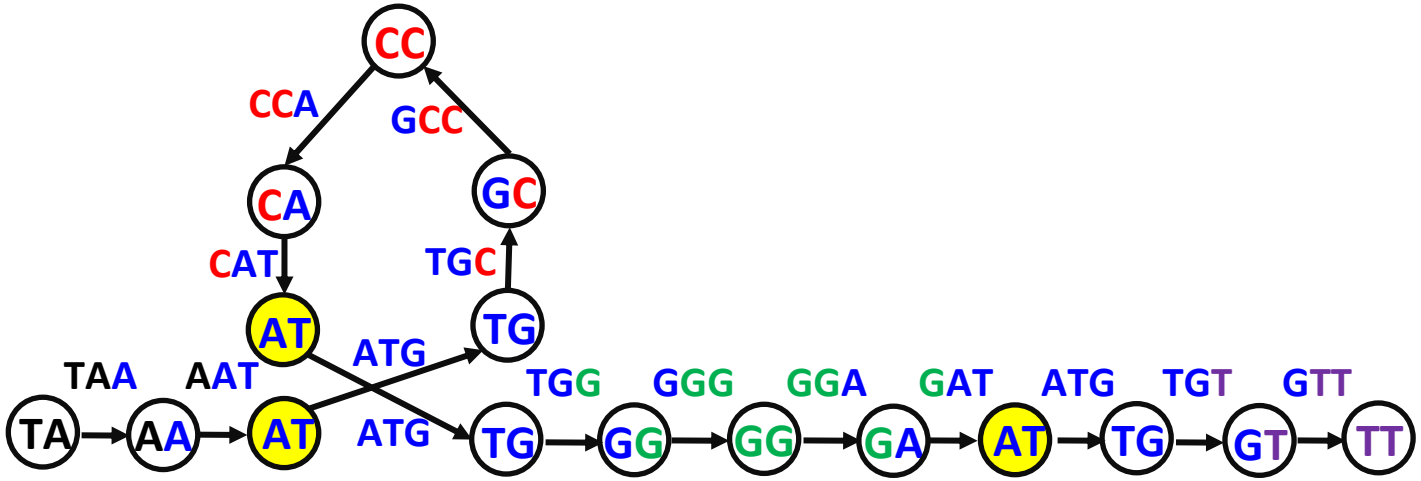
We Are Not Done with Gluing Yet



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

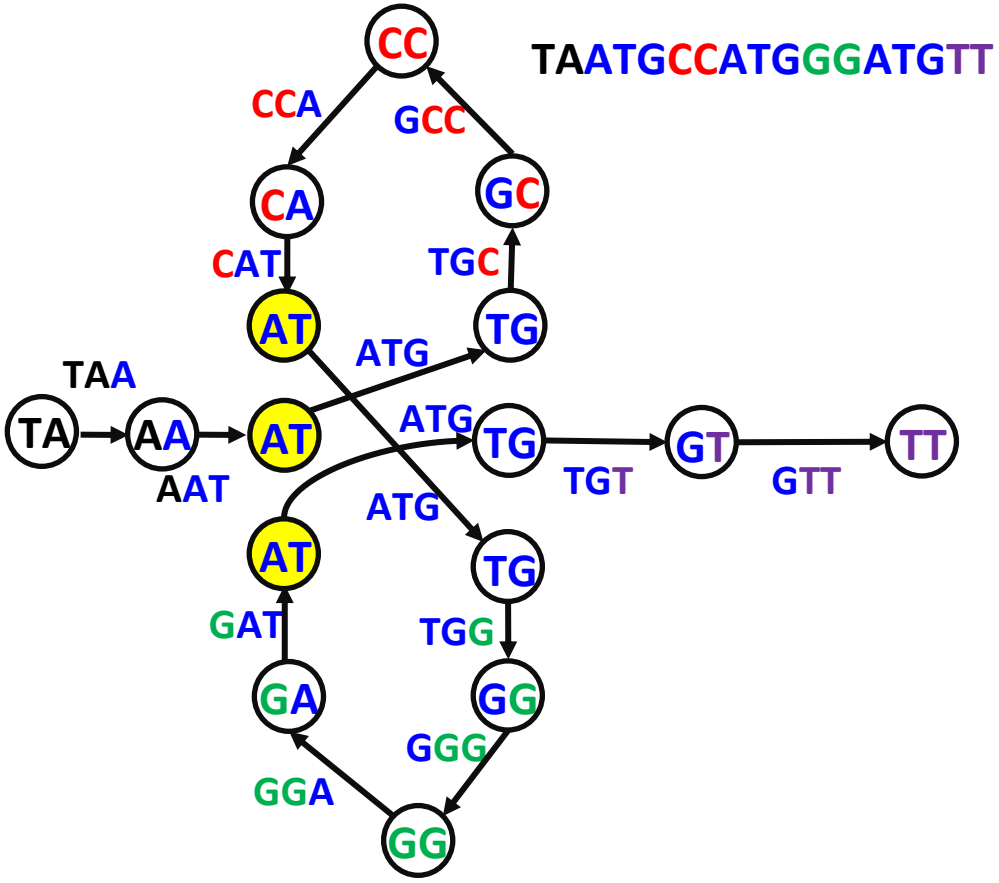
Gluing Identically Labeled Nodes



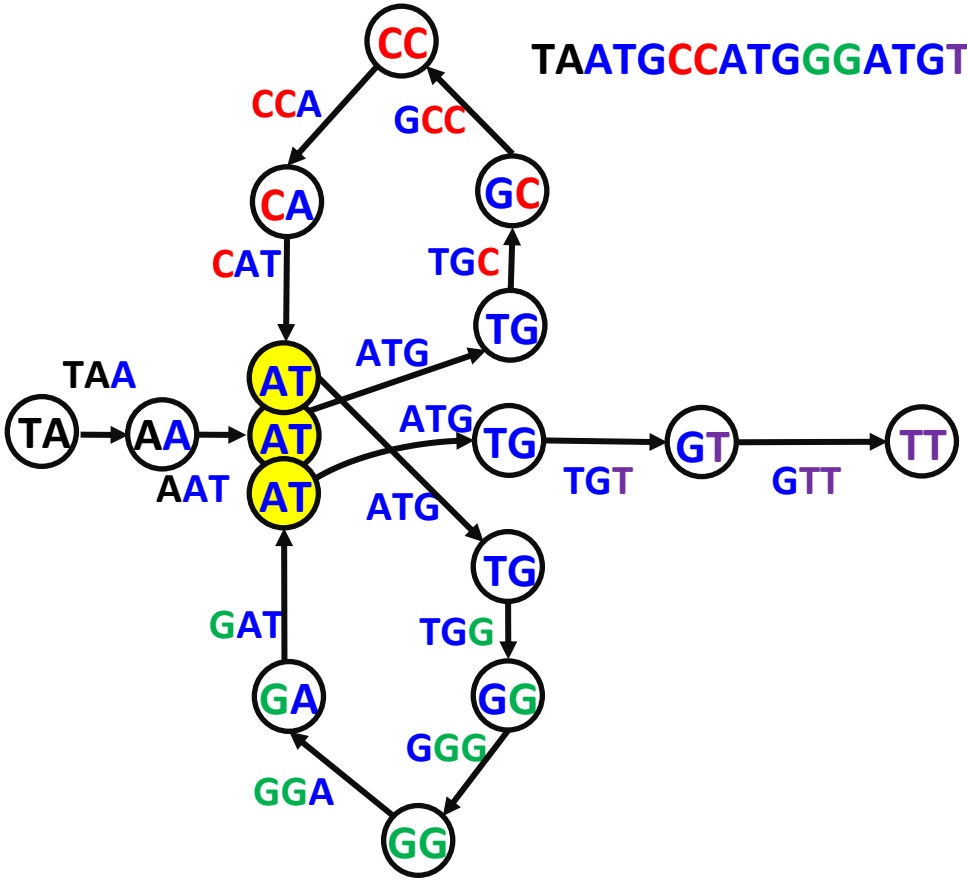


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes

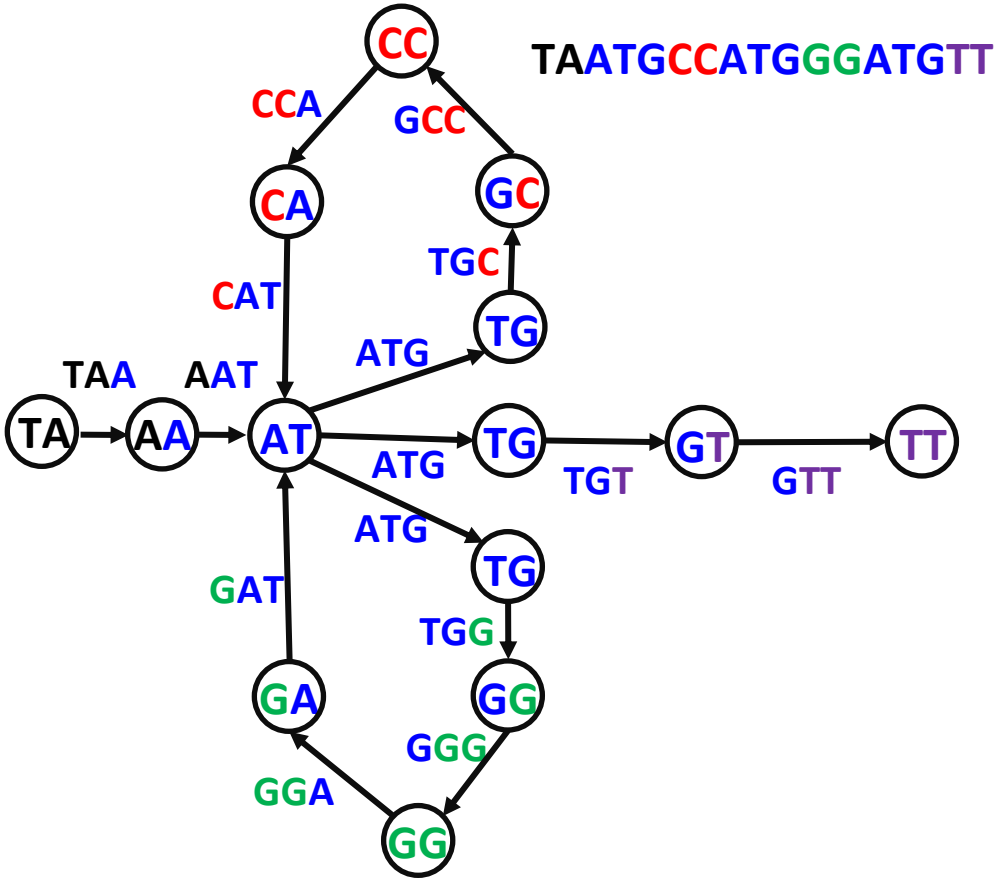


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.



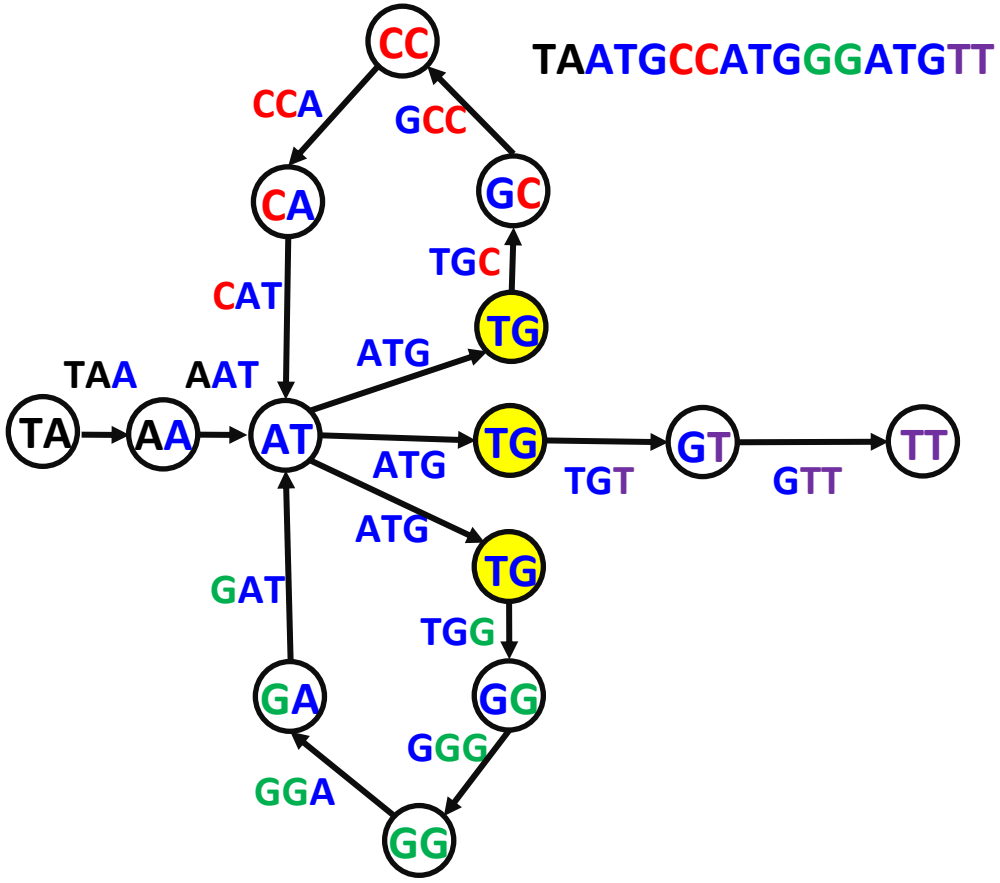
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



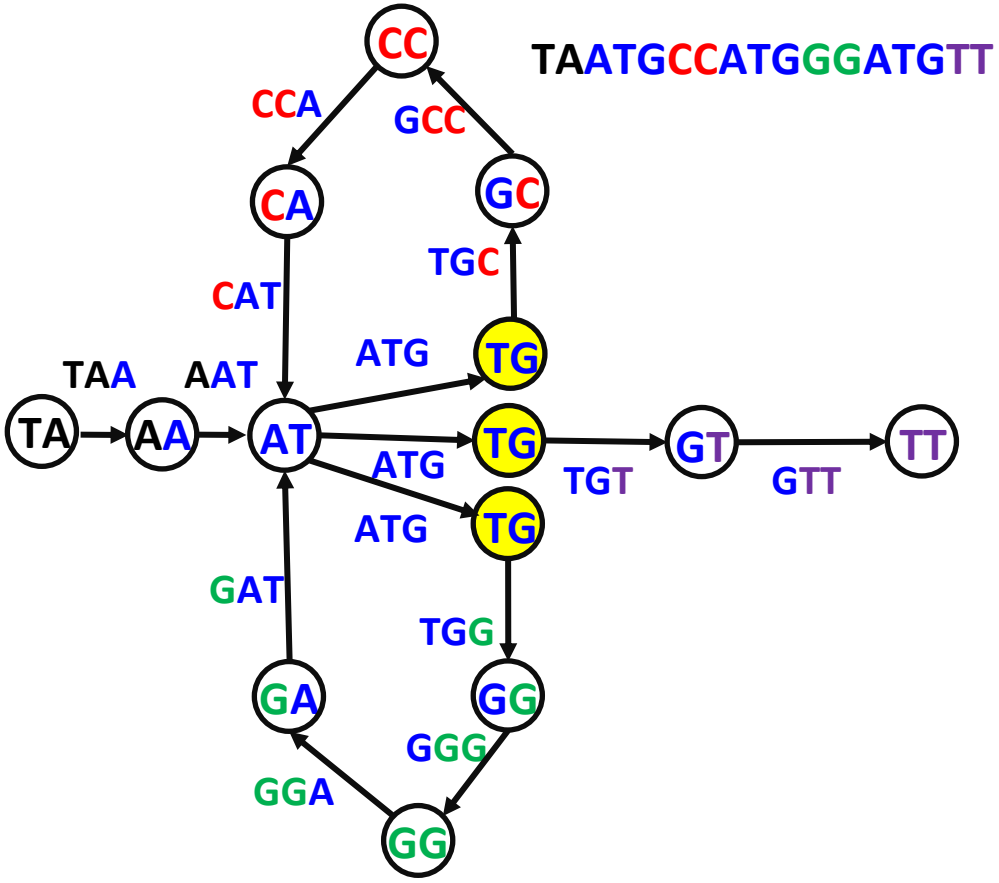
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



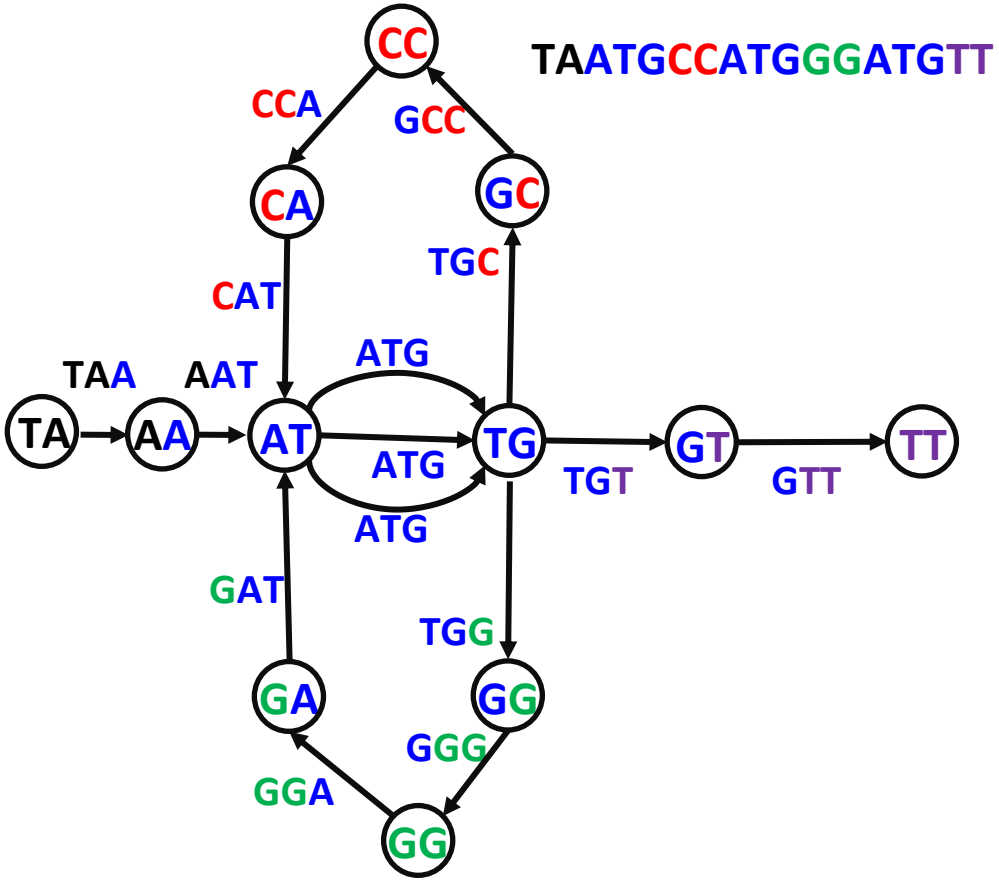
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



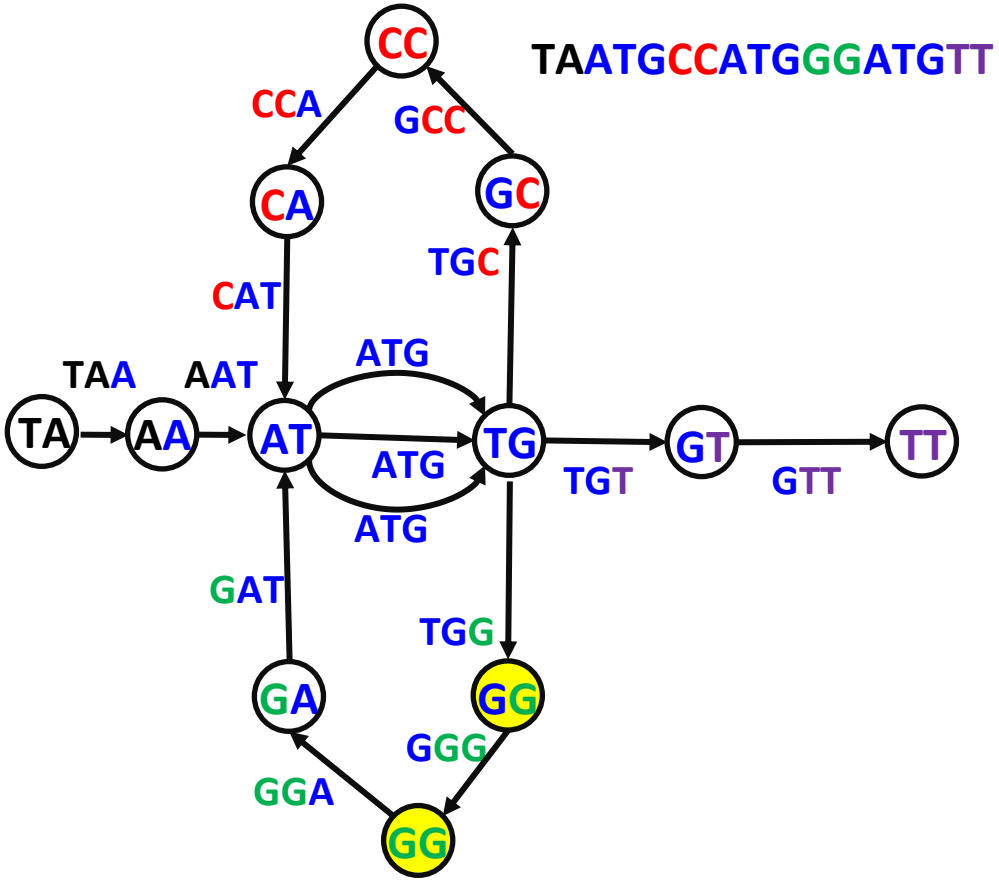
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



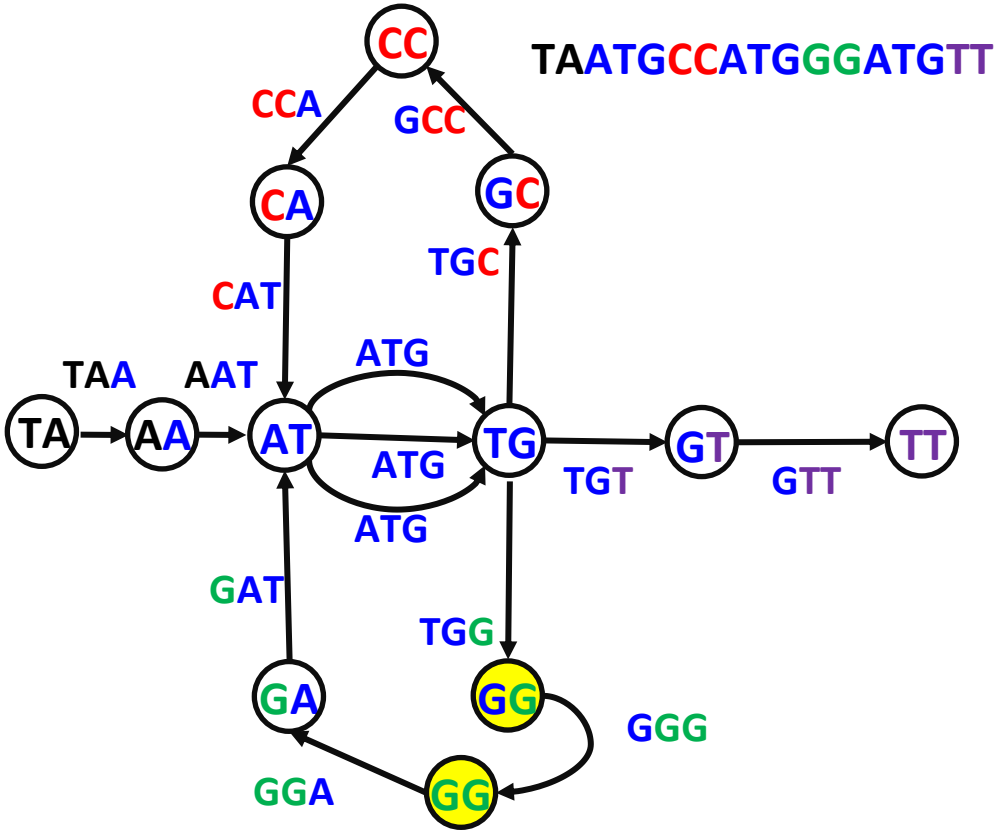
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



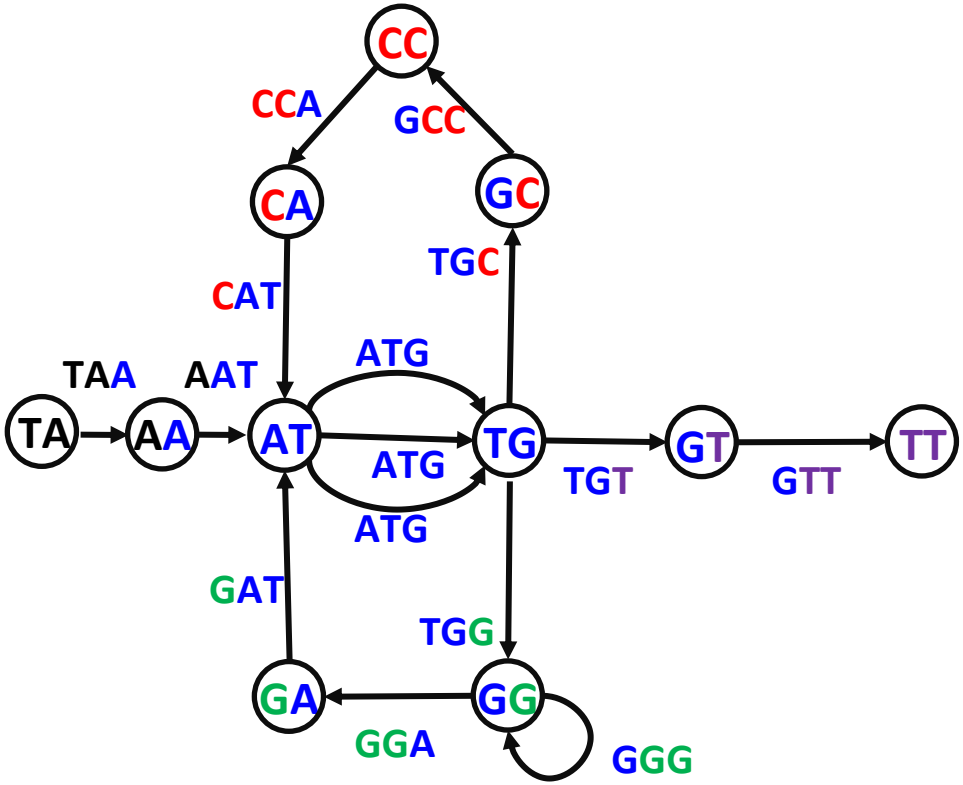
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Gluing Identically Labeled Nodes



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Same de Bruijn Graph:
DeBruin(Genome)=DeBruin(Genome Composition)



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

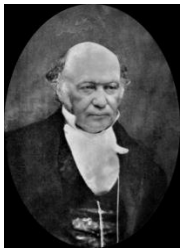
Constructing de Bruijn Graph

De Bruijn graph of a collection of *k*-mers:

- Represent every *k*-mer as an edge between its prefix and suffix
- Glue **ALL** nodes with identical labels.

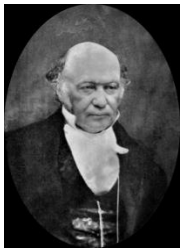
DeBruijn(k-mers)
form a node for each (*k-1*)-mer from ***k-mers***
for each *k*-mer in ***k-mers***
connect its prefix node with its suffix node by an edge

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

From Hamilton  to Euler  to de Bruijn 

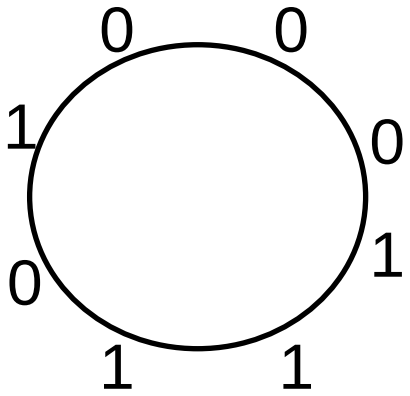
Universal String Problem (De Bruijn, 1946). Find a circular string containing each binary *k*-mer exactly once.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

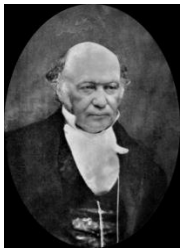
From Hamilton  to Euler  to de Bruijn 

Universal String Problem (De Bruijn, 1946). Find a circular string containing each binary *k*-mer exactly once.

000 001 010 011 100 101 110 111

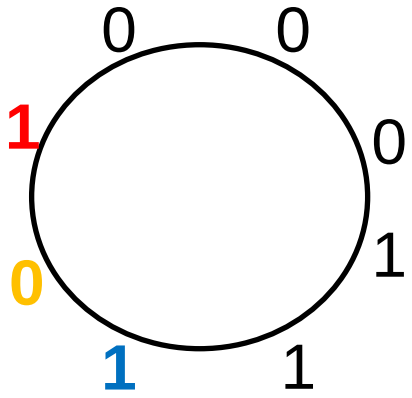


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

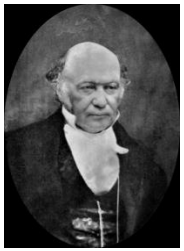
From Hamilton  to Euler  to de Bruijn 

Universal String Problem (Nicolaas de Bruijn, 1946). Find a circular string containing each binary *k*-mer exactly once.

000 001 010 011 100 **101** 110 111

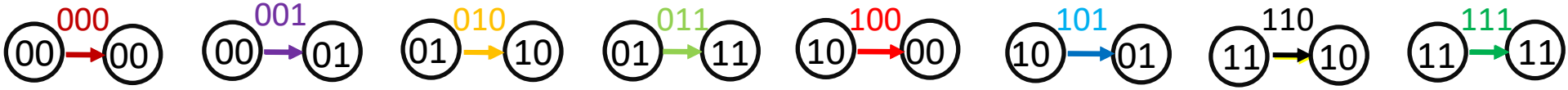


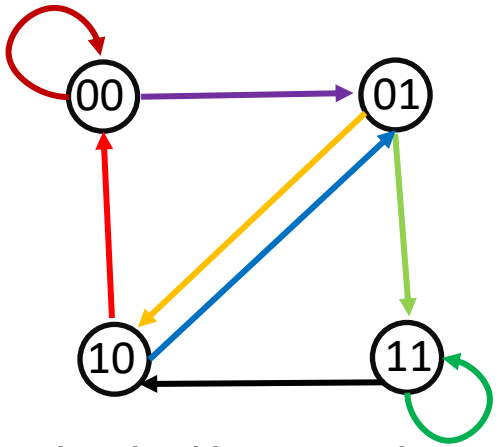
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

From Hamilton  to Euler  to de Bruijn 

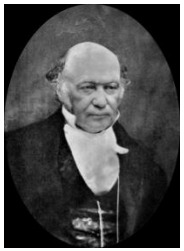
Universal String Problem (Nicolaas de Bruijn, 1946). Find a circular string containing each binary *k*-mer exactly once.

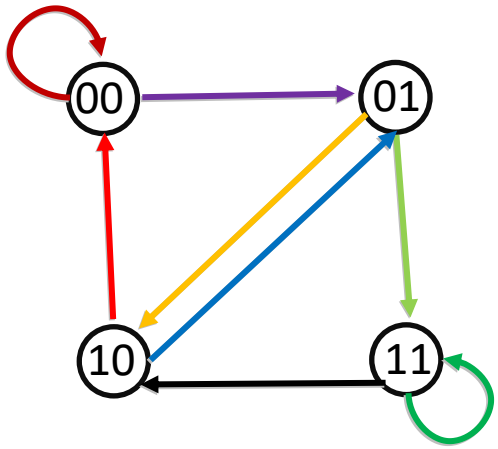
000 001 010 011 100 101 110 111

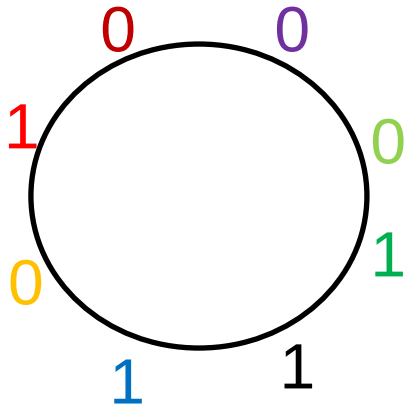




Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

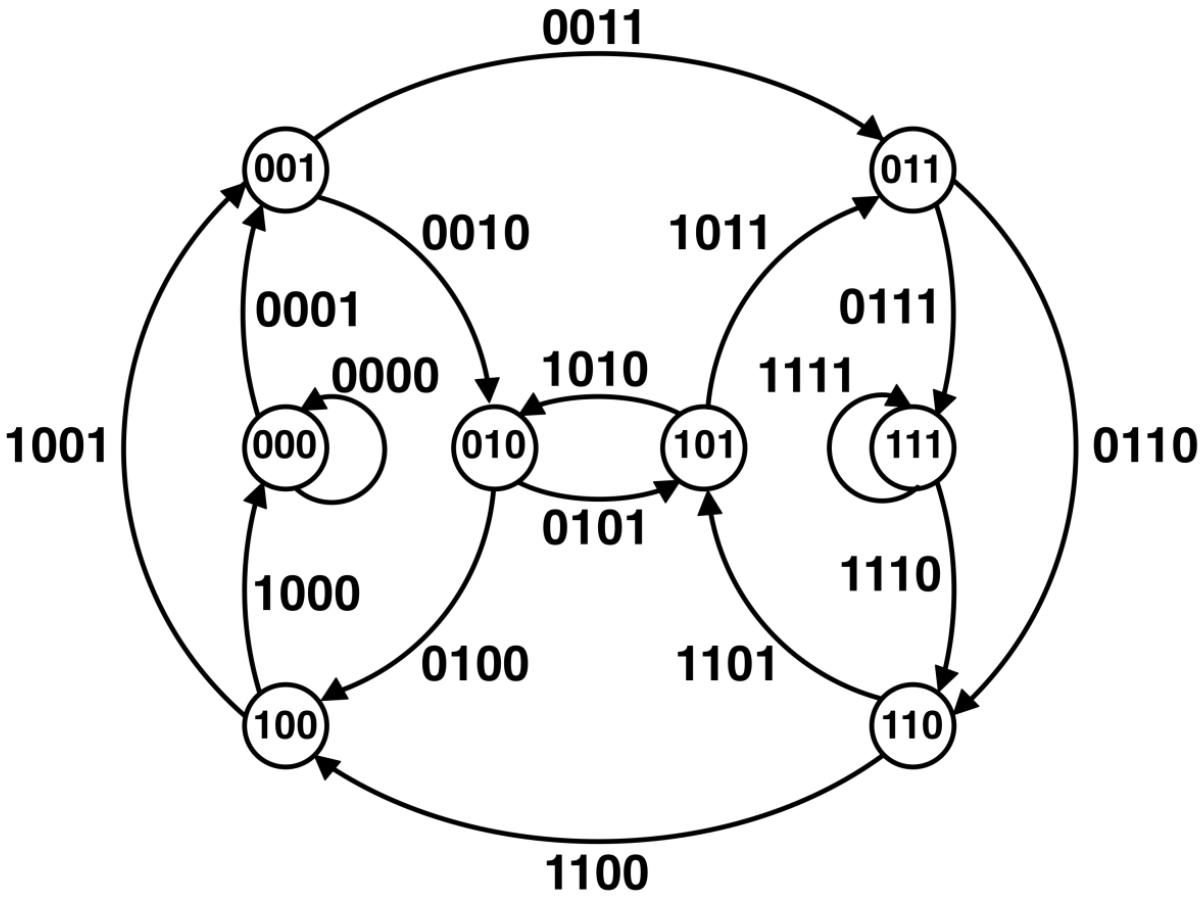
From Hamilton  to Euler  to de Bruijn 





Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

De Bruijn Graph for 4-Universal String



Does it have an Eulerian cycle? If yes, how can we find it?

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

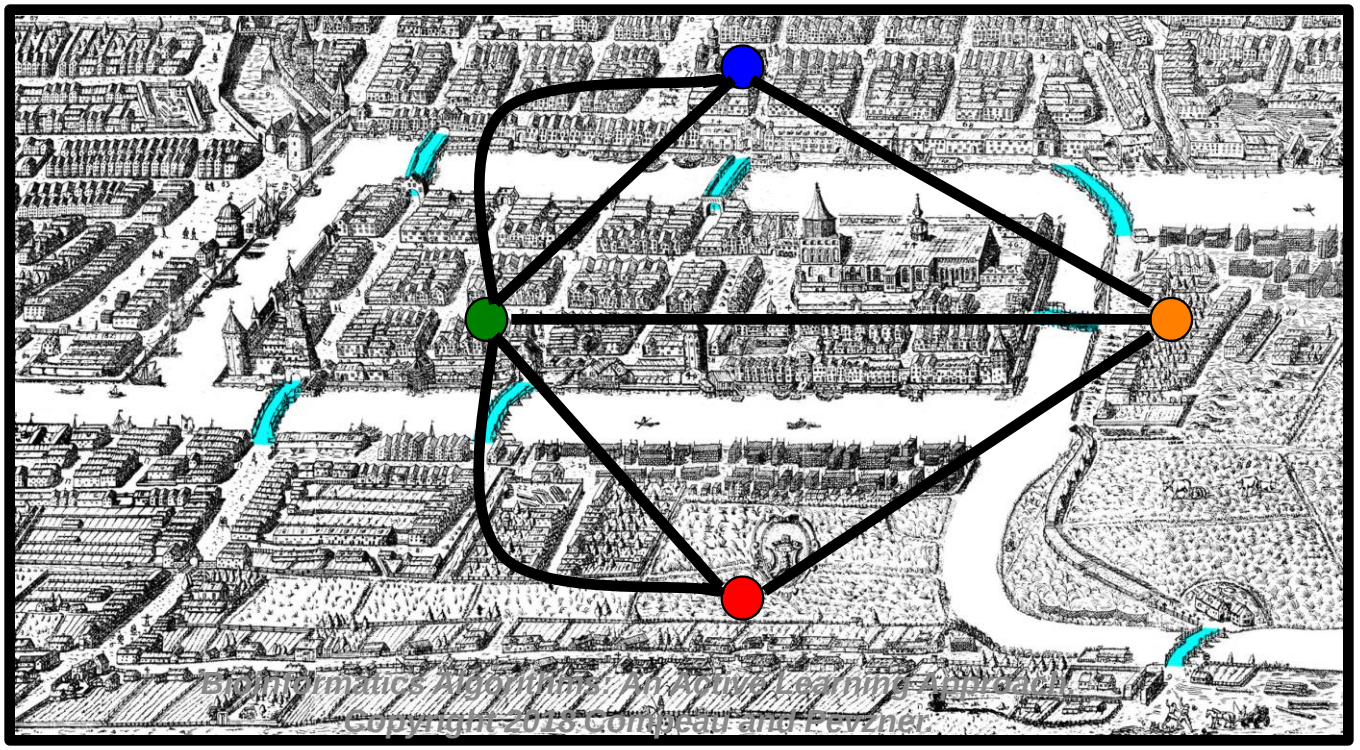
- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- **Euler's Theorem**
- Assembling Read-Pairs
- De Bruijn Graphs Face Harsh Realities of Assembly

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Eulerian CYCLE Problem

Eulerian CYCLE Problem. Find an Eulerian cycle in a graph.

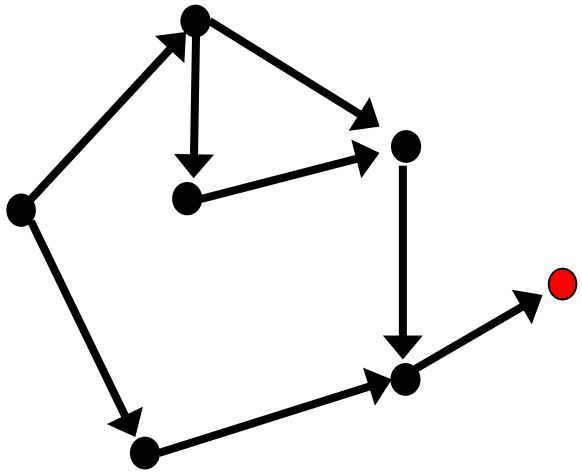
- **Input.** A graph.
- **Output.** A cycle visiting every edge in the graph exactly once.

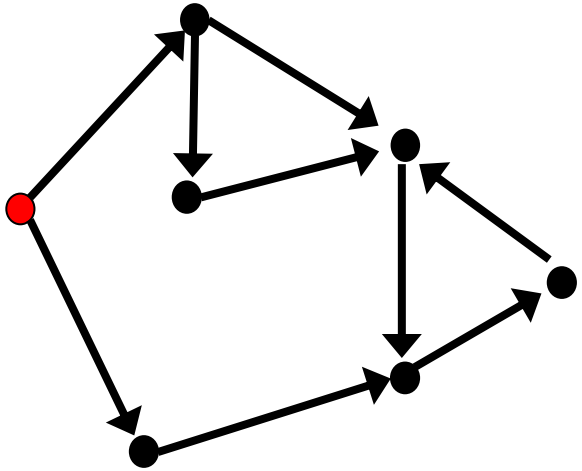


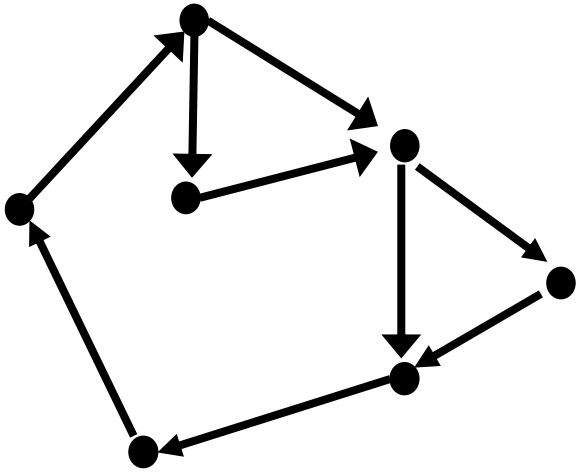
Bioinformatics Algorithms, 2nd Edition Lecture Slides
Copyright 2013 Compeau and Pevzner

A Graph is **Eulerian** if It Contains an Eulerian Cycle.

Is this graph Eulerian?



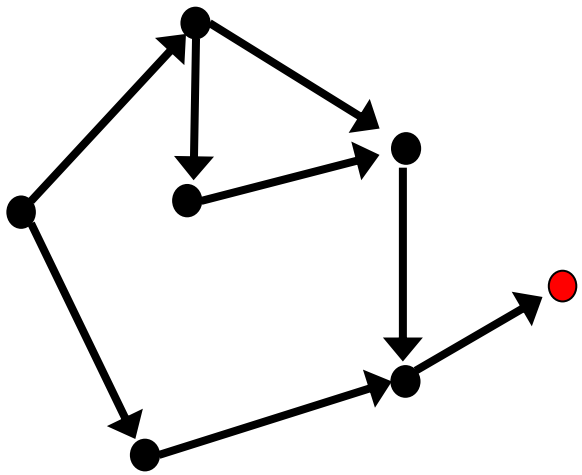


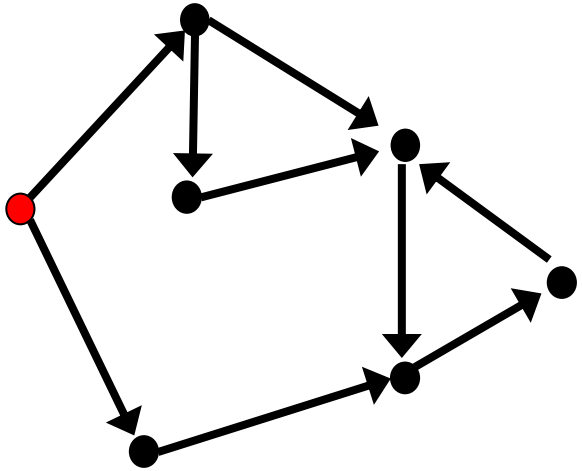


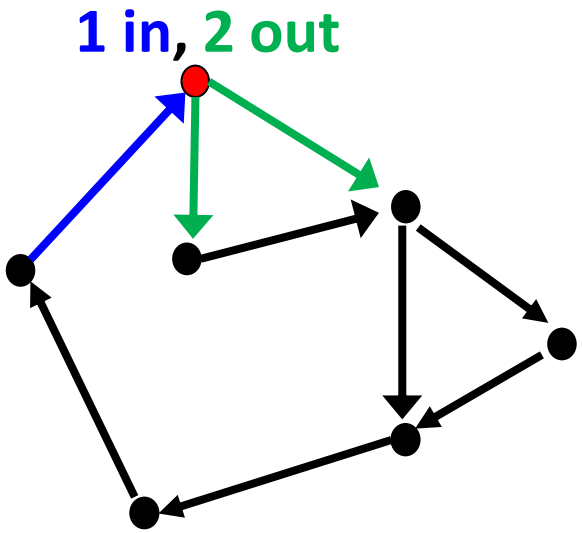
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Graph is **Eulerian** if It Contains an Eulerian Cycle.

Is this graph Eulerian?



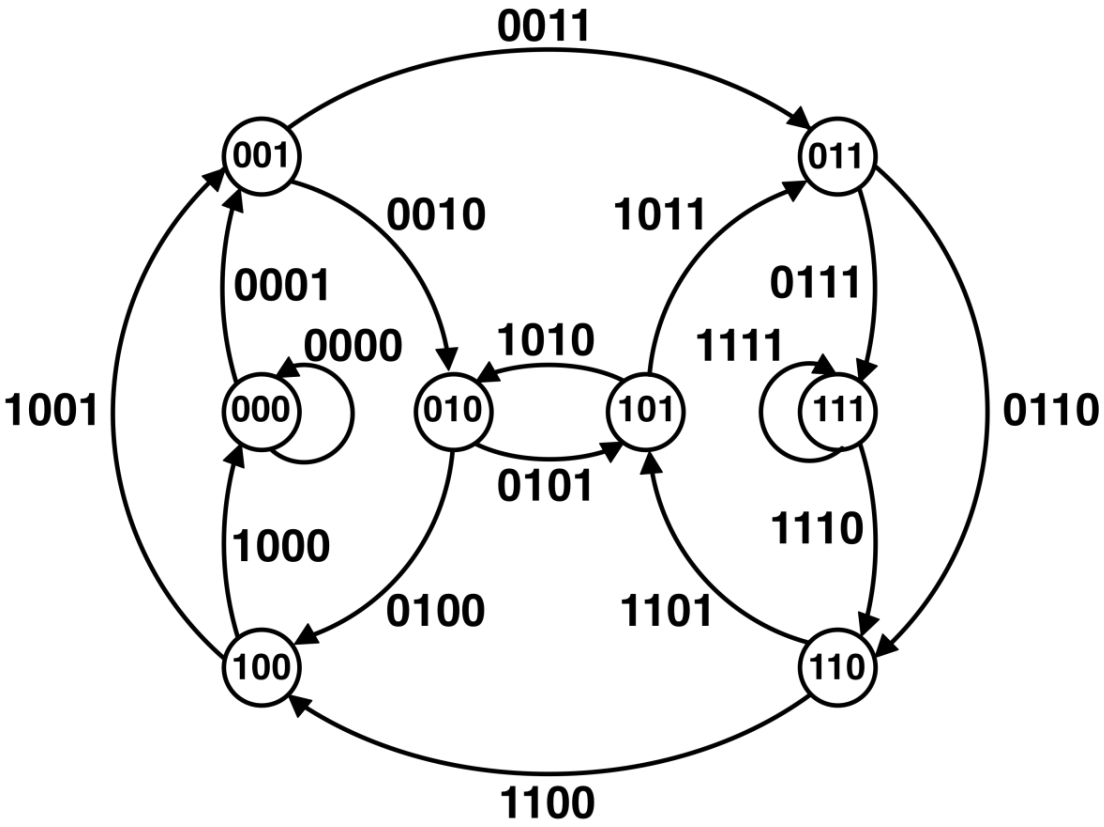




A graph is **balanced** if ***indegree*** = ***outdegree*** for each node

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

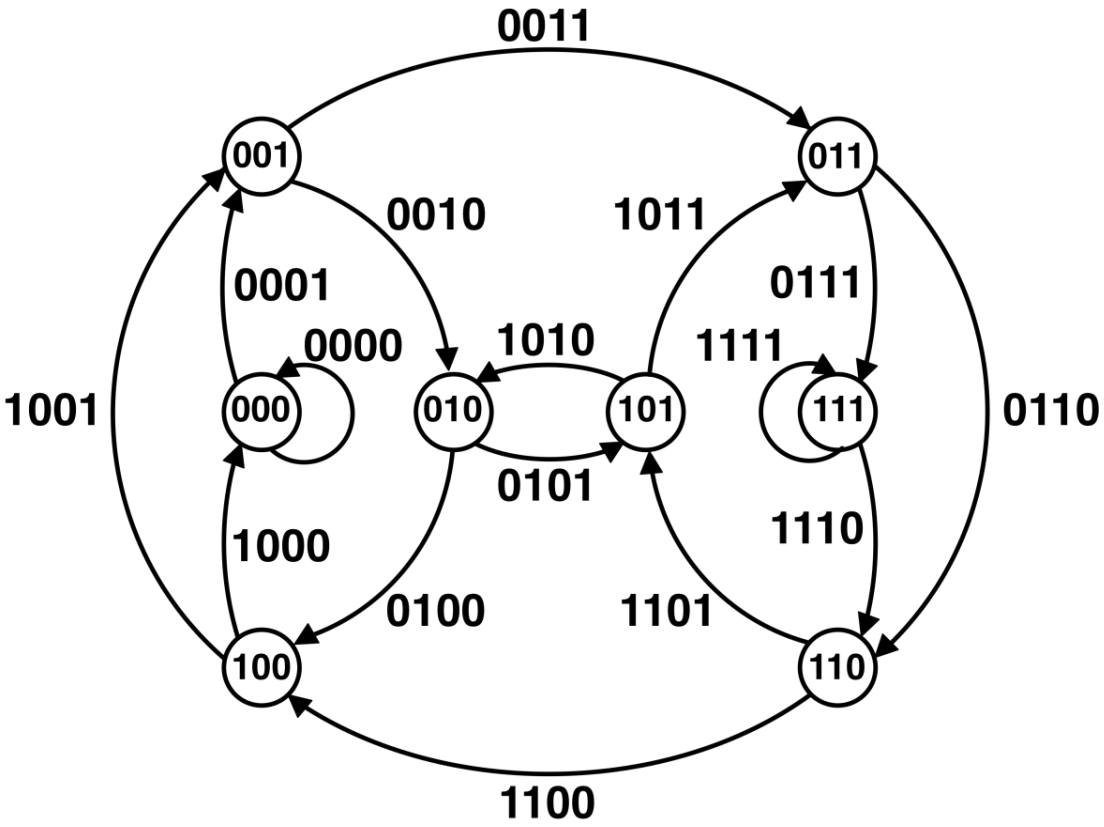
Is the Graph for 4-Universal String Balanced?



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Euler's Theorem

- Every Eulerian graph is balanced

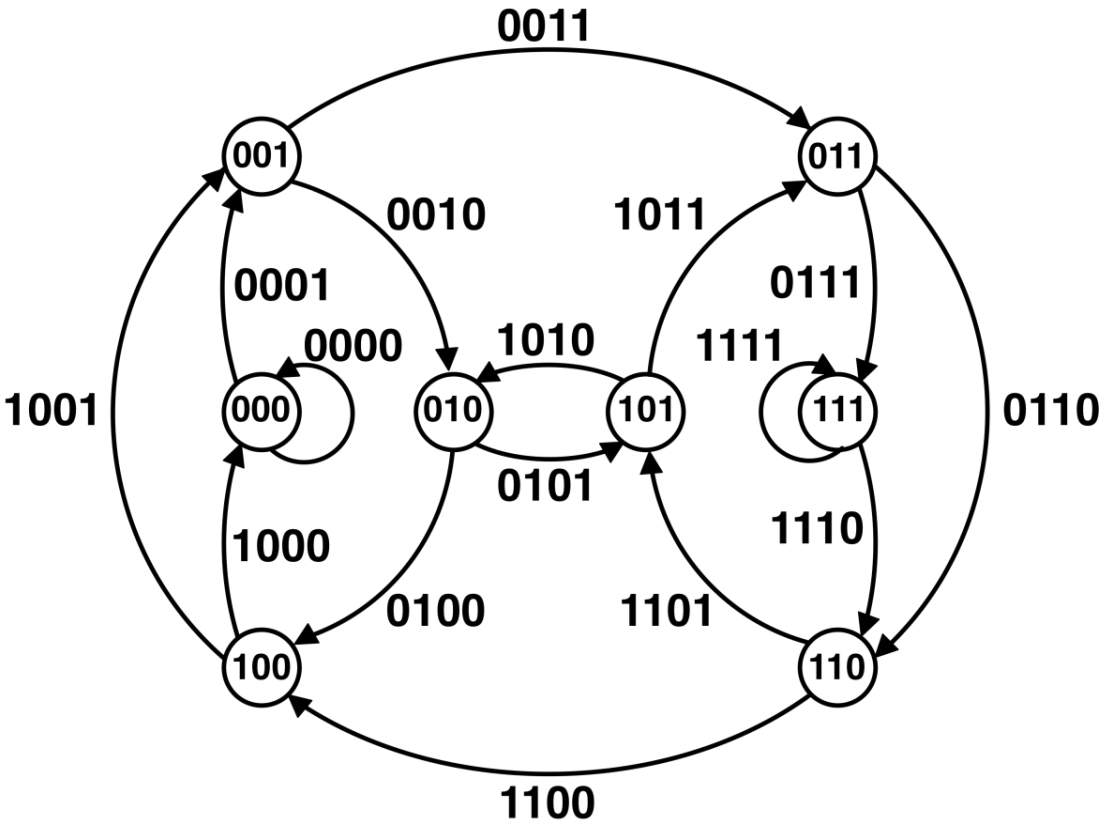


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Euler's Theorem



- Every Eulerian graph is balanced
- **Every balanced* graph is Eulerian**



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2013 Compeau and Pevzner.

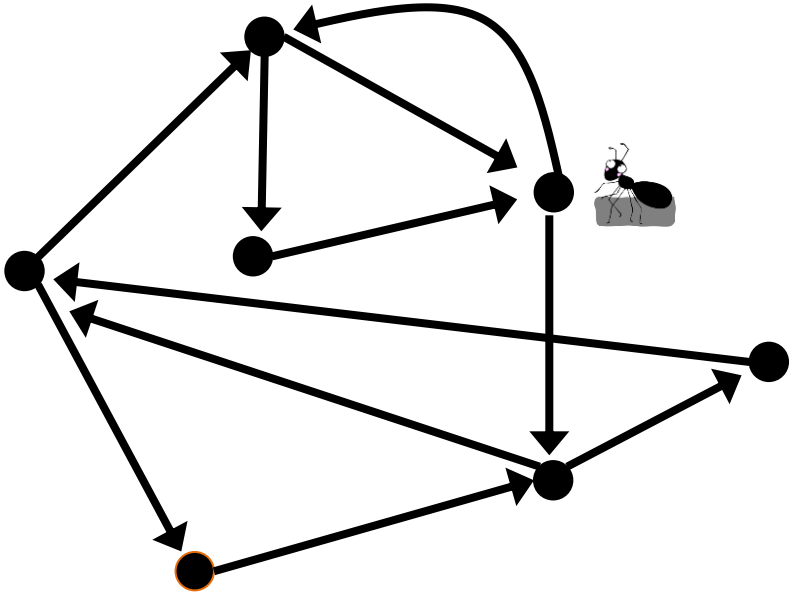
(*) and strongly connected, of course!

Recruiting an Ant to Prove Euler's Theorem

Let an ant randomly walk through the graph.

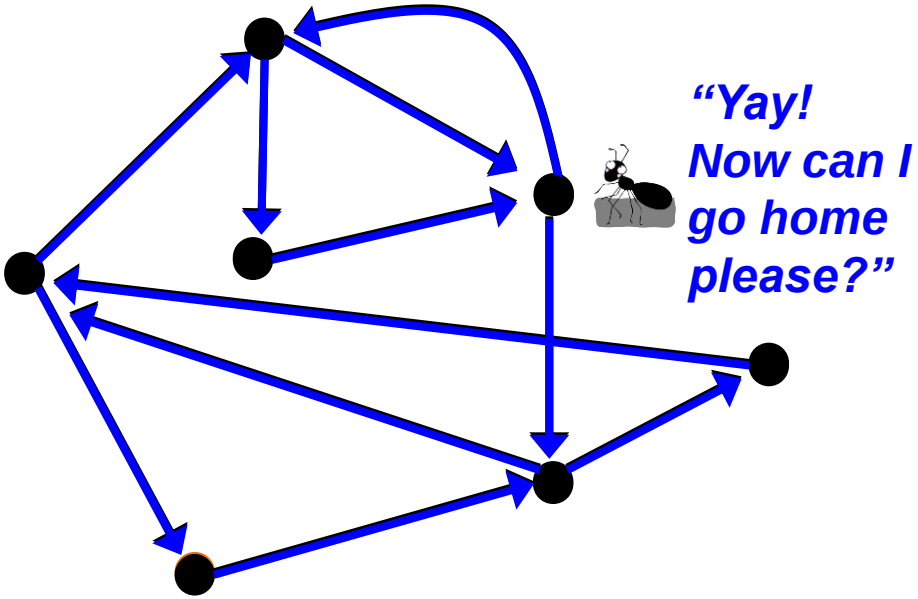
The ant cannot use the same edge twice!





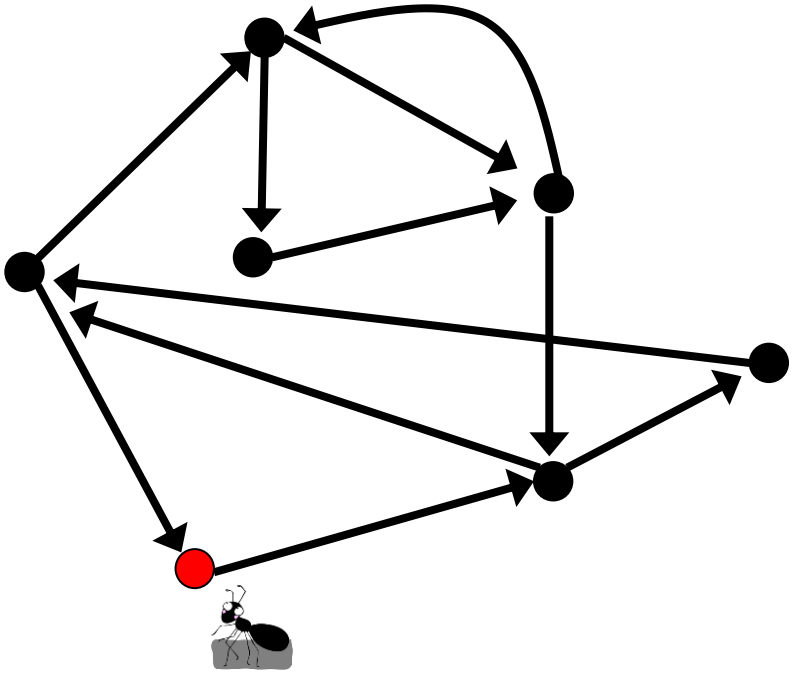
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

If Ant Was a Genius…



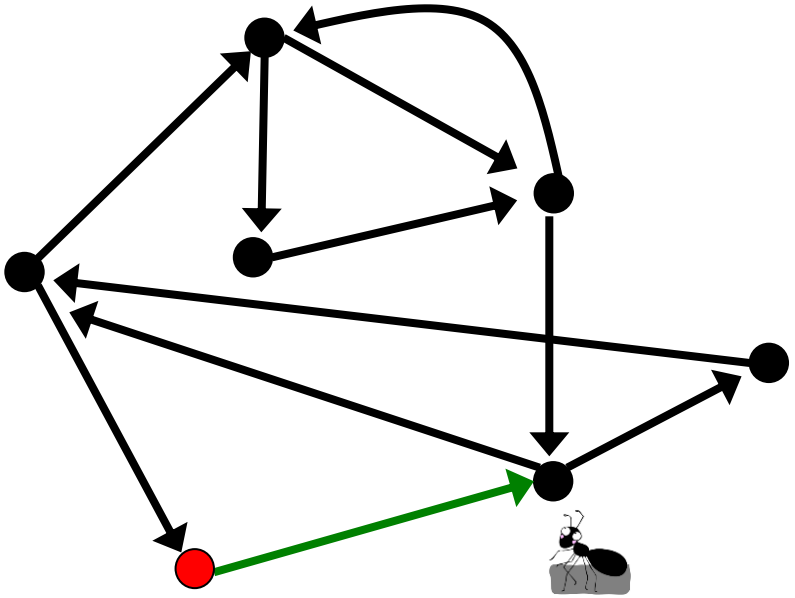
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

A Less Intelligent Ant Would Randomly Choose a Node and Start Walking…



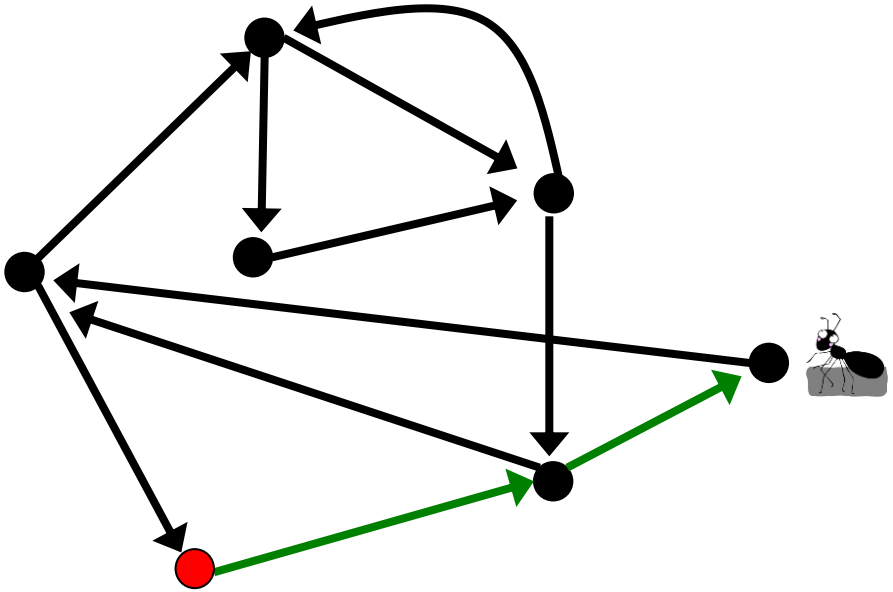
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Walking…



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

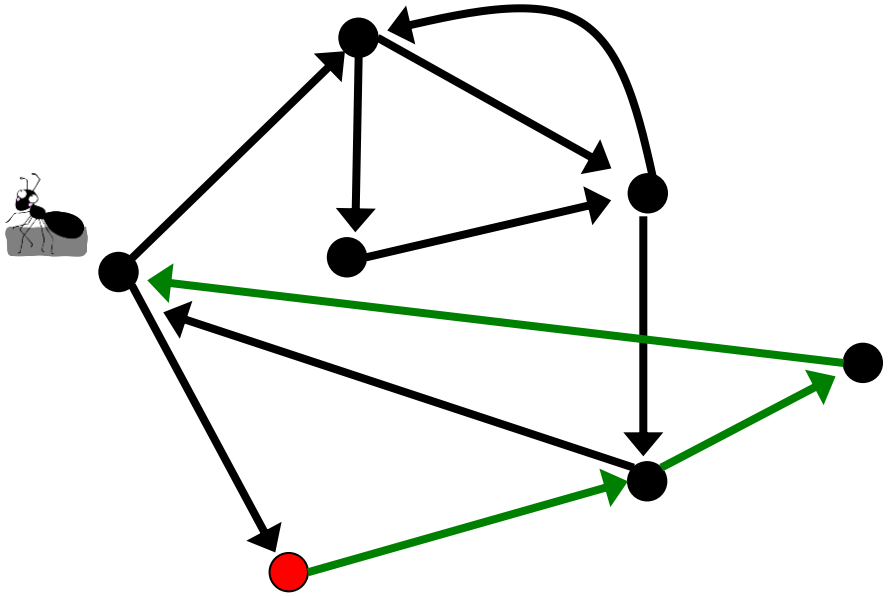
Walking… and Walking…



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

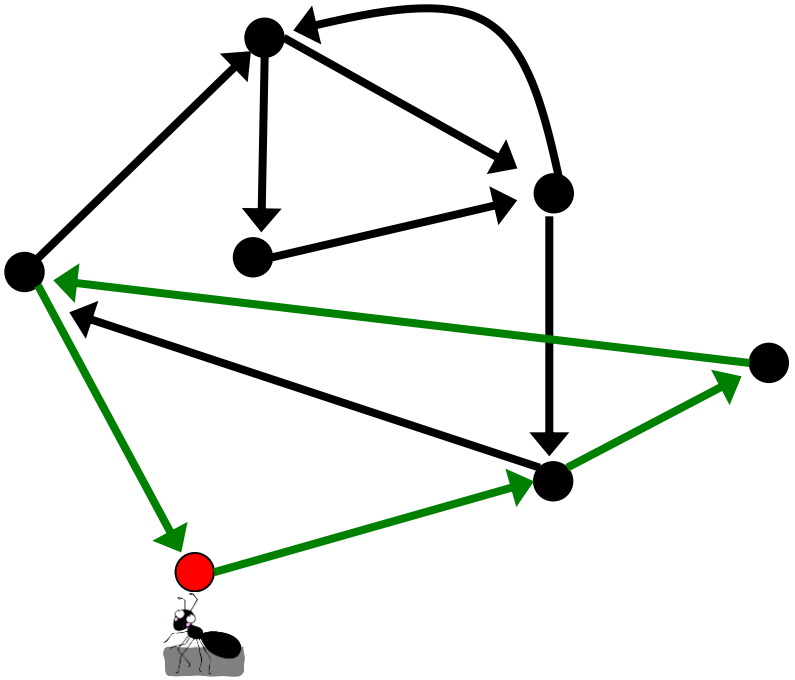
Walking… and Walking… and Walking…

Can it get stuck? **In what node?**



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

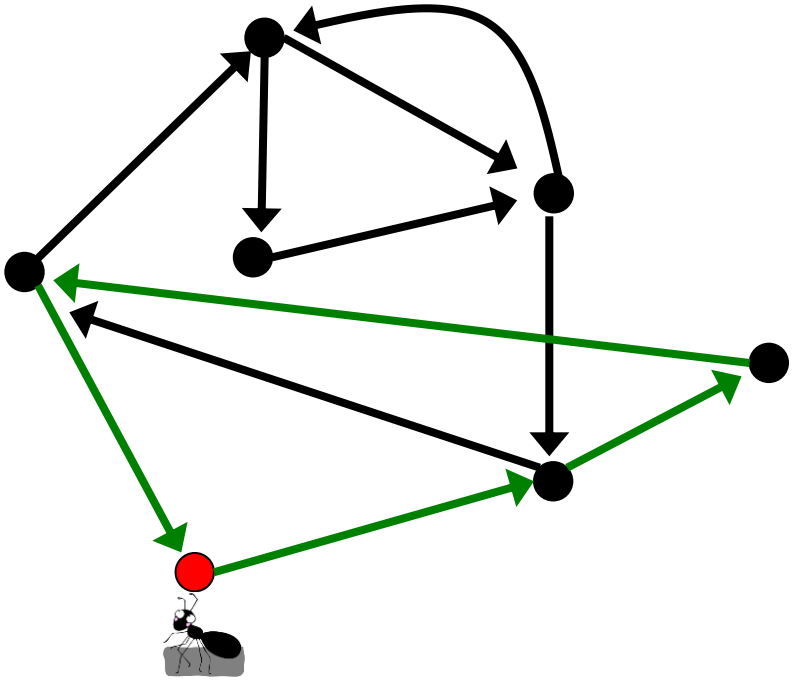
The Ant Can Only Get Stuck at the Starting Node



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

The Ant Has Completed a Cycle BUT has not Proven Euler's theorem yet…

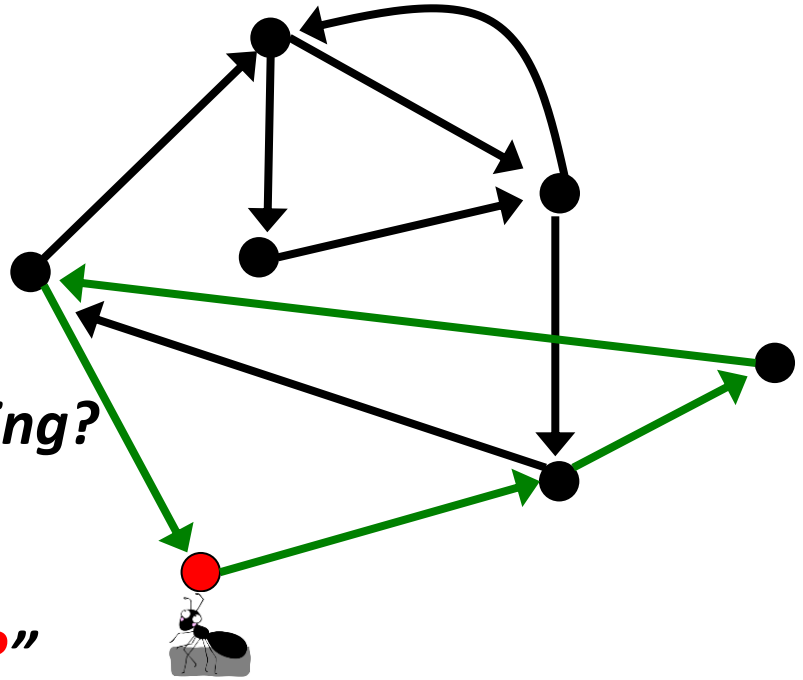
The constructed cycle is not Eulerian. **Can we enlarge it?**



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Let's Start at a Different Node in the Green Cycle

Let's start at a node with still unexplored edges.

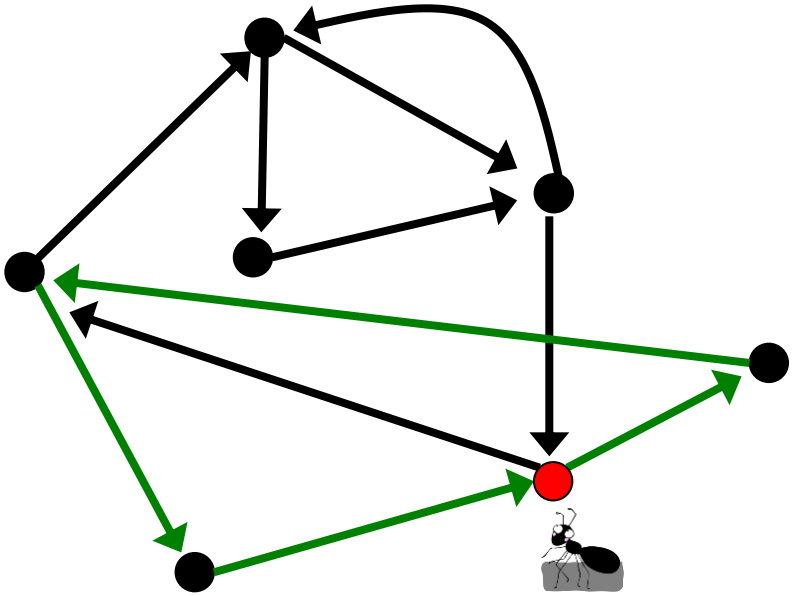


"Why should I start at a different node? Backtracking? I'm not evolved to walk backwards! And what difference does it make???"

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

New Instructions for the Ant:

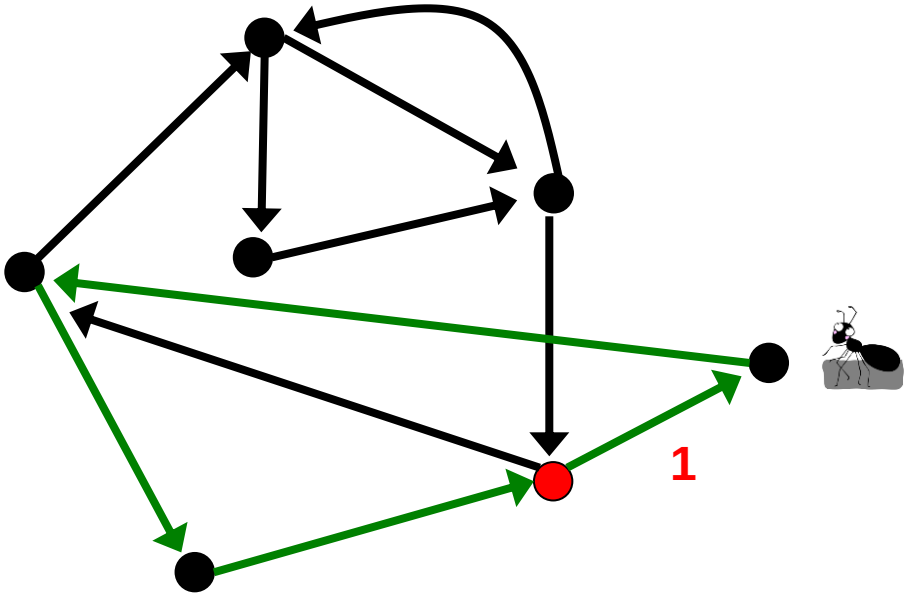
Starting at a node that has an unused edge, traverse the already constructed (green cycle) and return back to the starting node.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

An Ant Traversing Previously Constructed Cycle

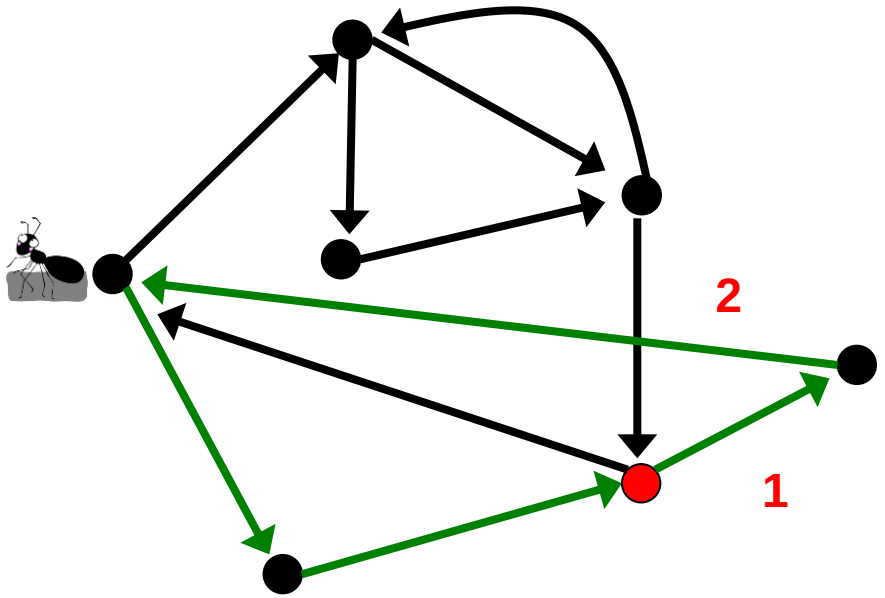
Starting at a node that has an unused edge, traverse the already constructed (green cycle) and return back to the starting node.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

An Ant Traversing Previously Constructed Cycle

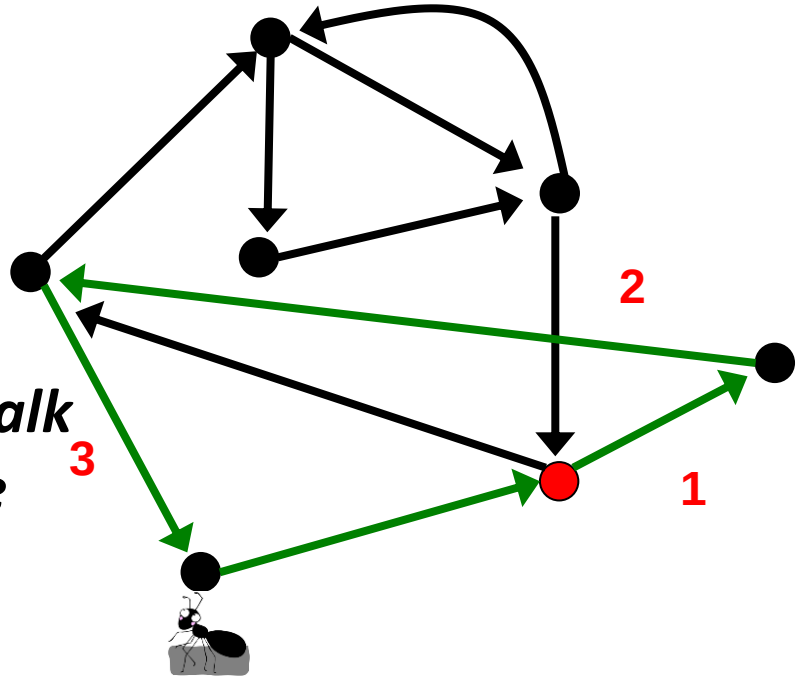
Starting at a node that has an unused edge, traverse the already constructed (green cycle) and return back to the starting node.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

An Ant Traversing Previously Constructed Cycle

Starting at a node that has an unused edge, traverse the already constructed (green cycle) and return back to the starting node.



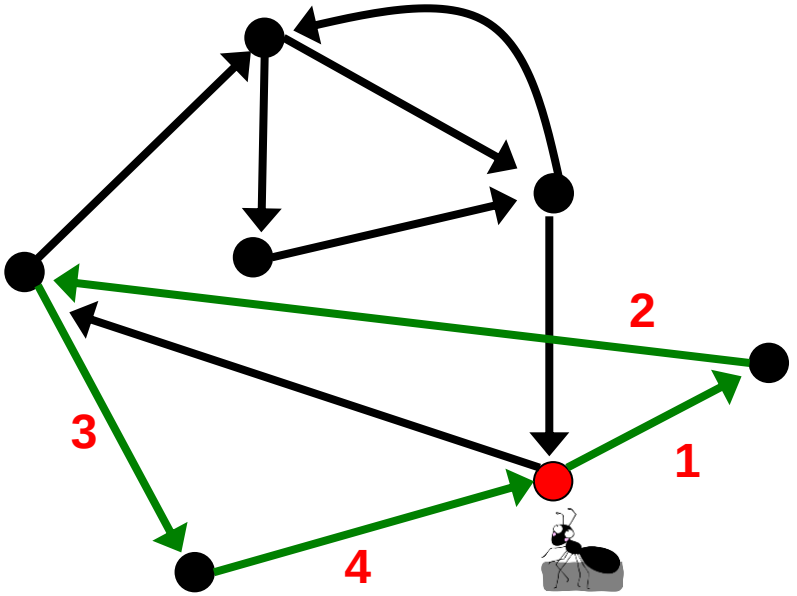
"Why do I have to walk along the same cycle again??? Can I see something new?"

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

I Returned Back BUT… I Can Continue Walking!

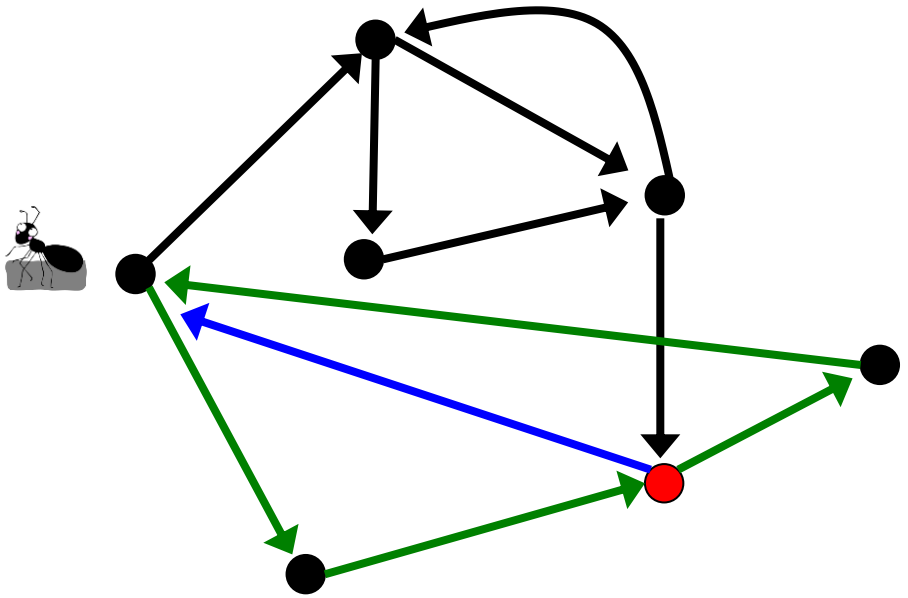
Starting at a node that has an unused edge, traverse the already constructed (green cycle) and return back to the starting node.

After completing the cycle, start random exploration of still untraversed edges in the graph.



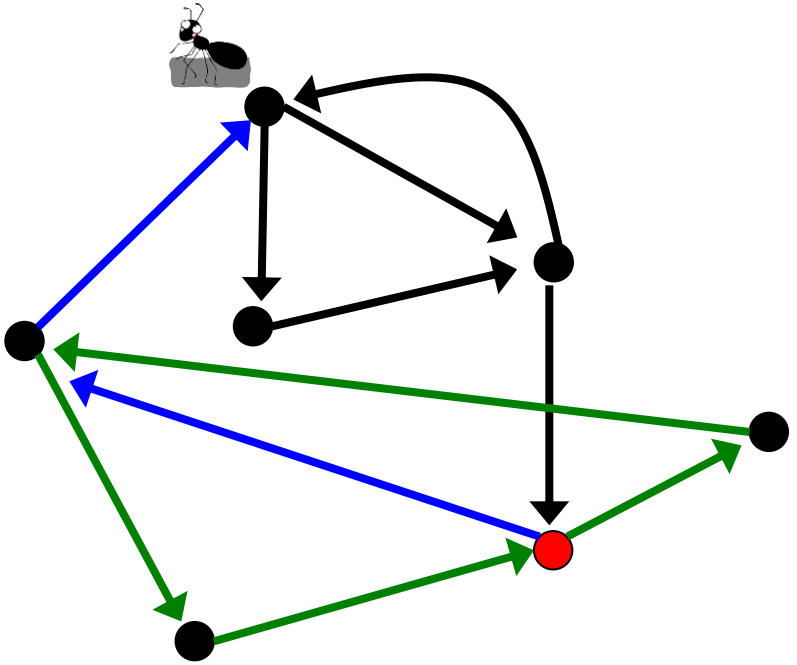
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Enlarging the Previously Constructed Cycle



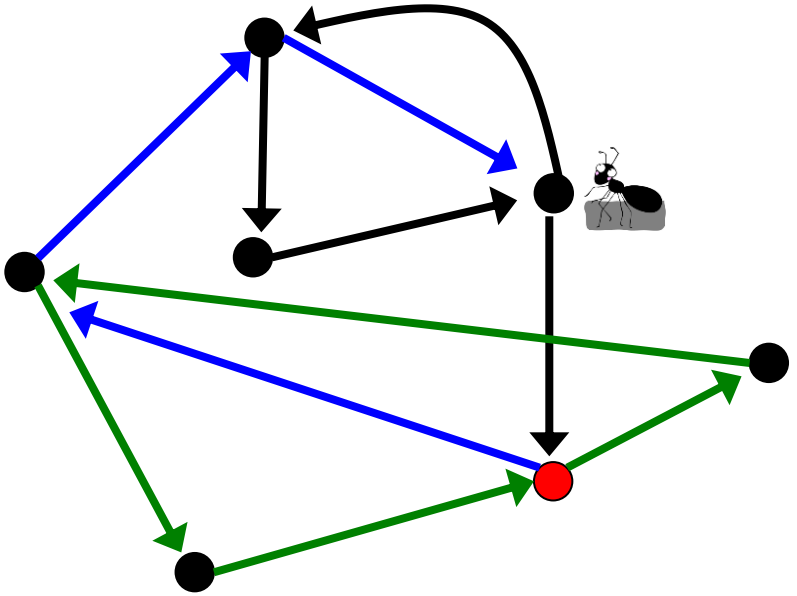
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Enlarging the Previously Constructed Cycle



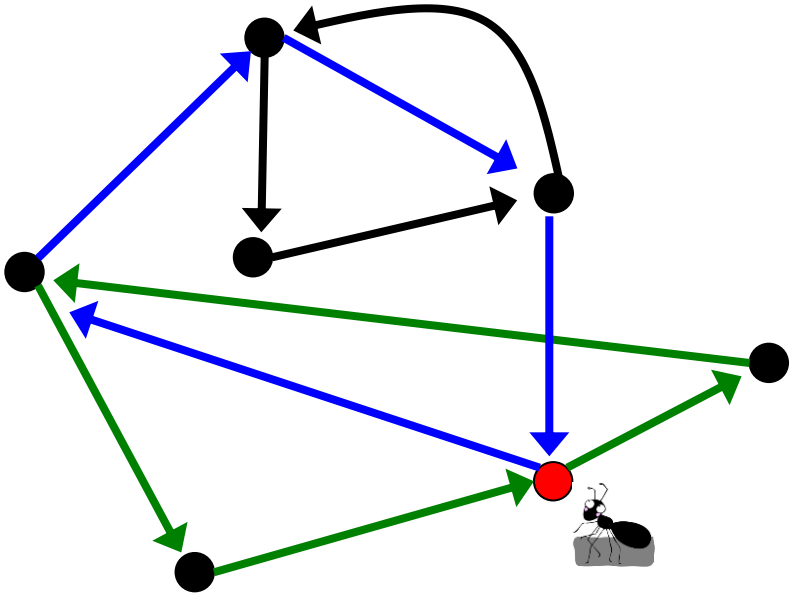
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Enlarging the Previously Constructed Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Enlarging the Previously Constructed Cycle

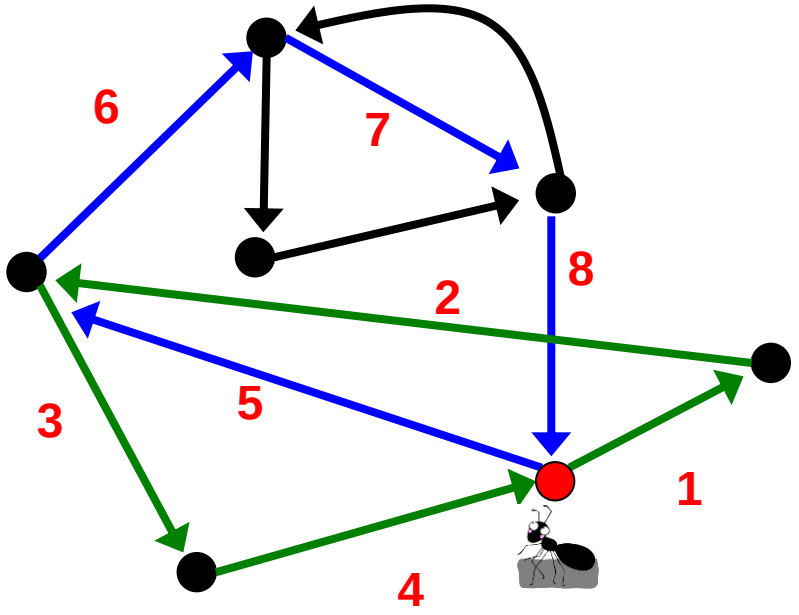


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Stuck Again!

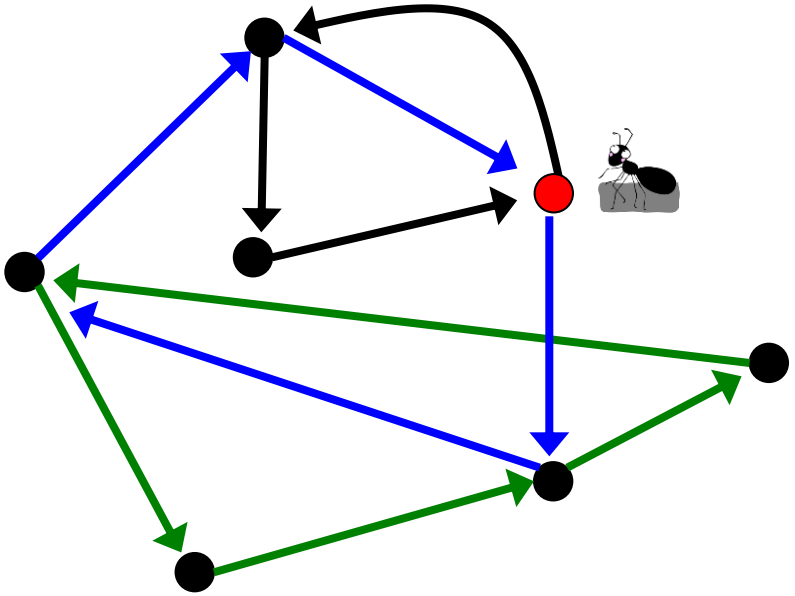
No Eulerian cycle yet… can we enlarge the green-blue cycle?

The ant should walk along the constructed cycle starting at yet another node. **Which one?**



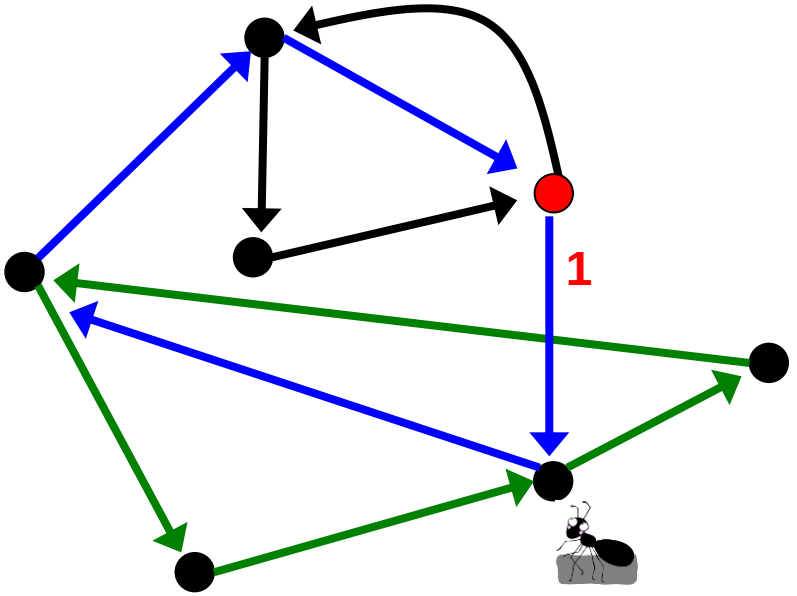
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Starting at a New Node, Again…



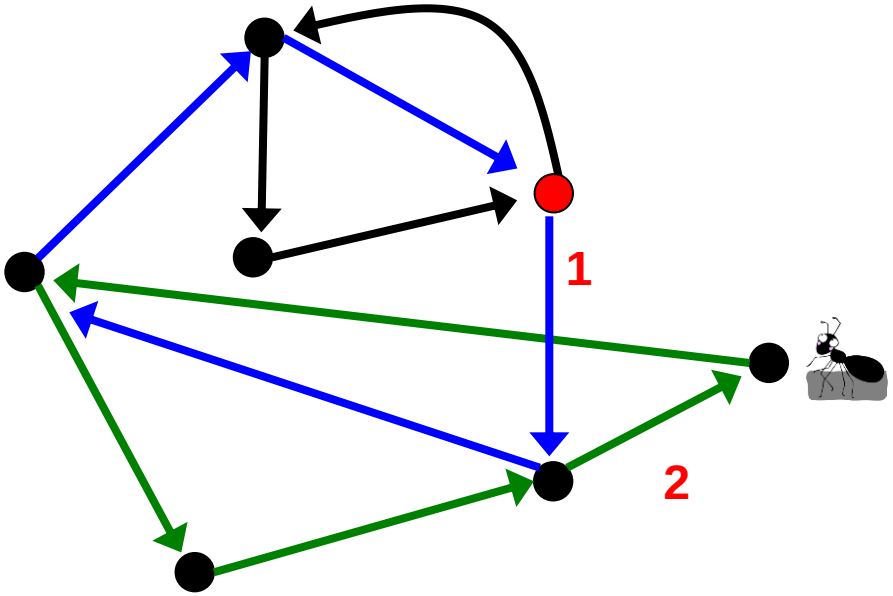
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Traversing the Previously Constructed Green-Blue Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

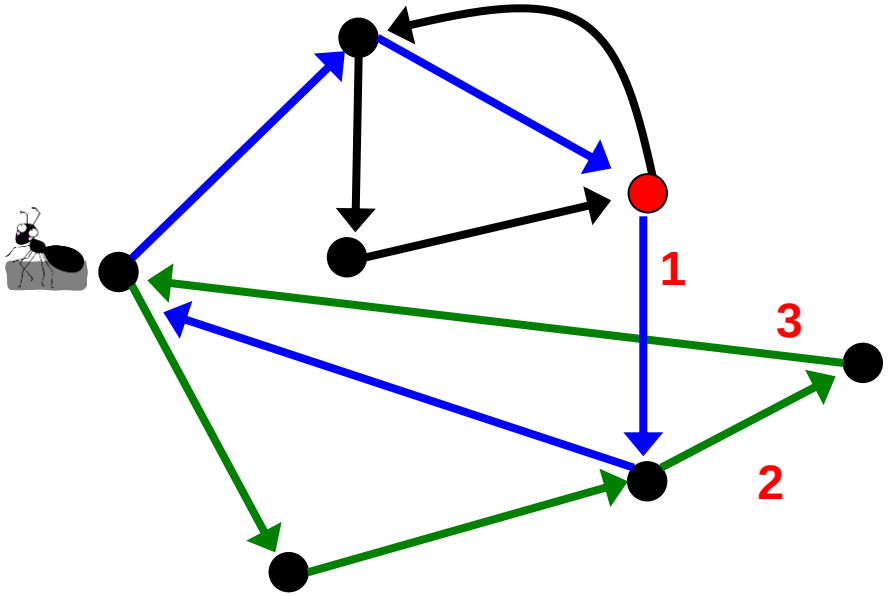
Traversing the Previously Constructed Green-Blue Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

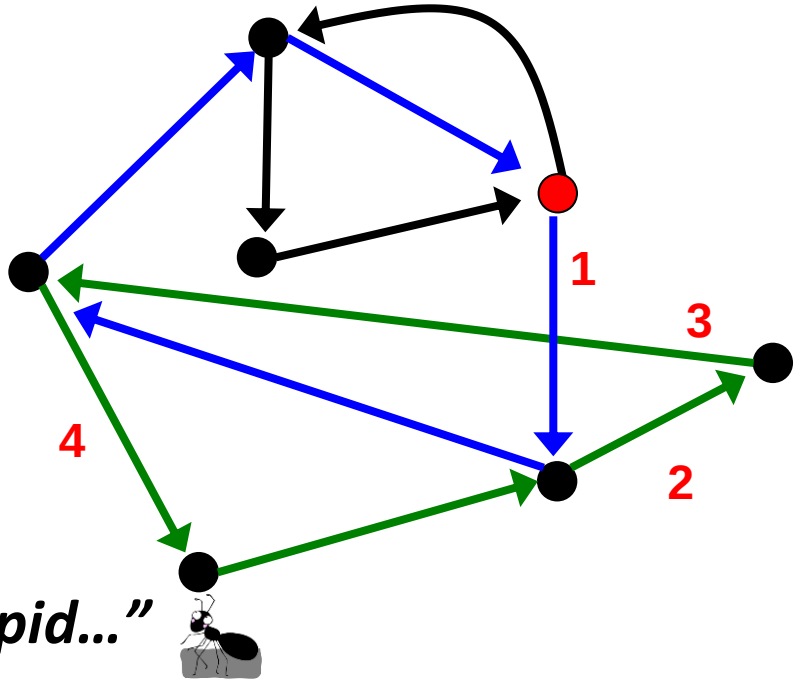
Traversing the Previously Constructed Green-Blue Cycle

"I hate to traverse the same cycle! What difference does it make where I start my walk???



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

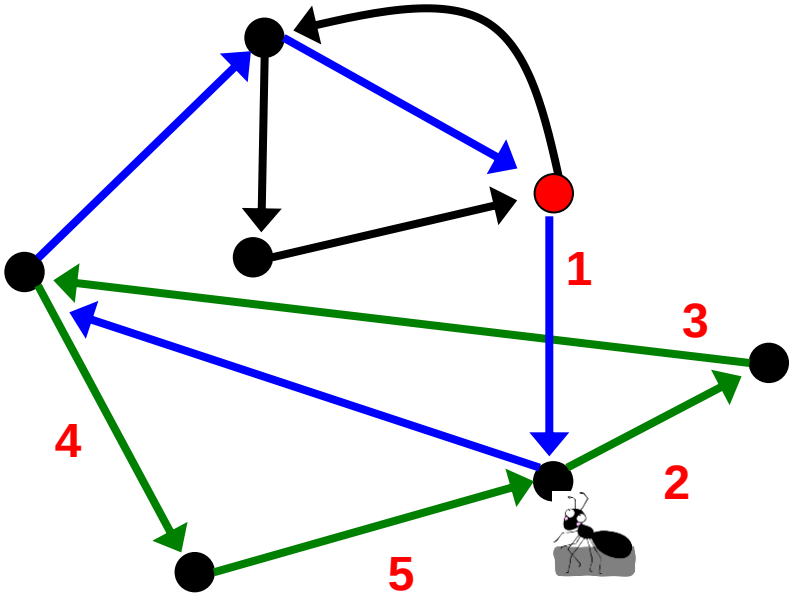
Traversing the Previously Constructed Green-Blue Cycle



"These instructions are stupid…"

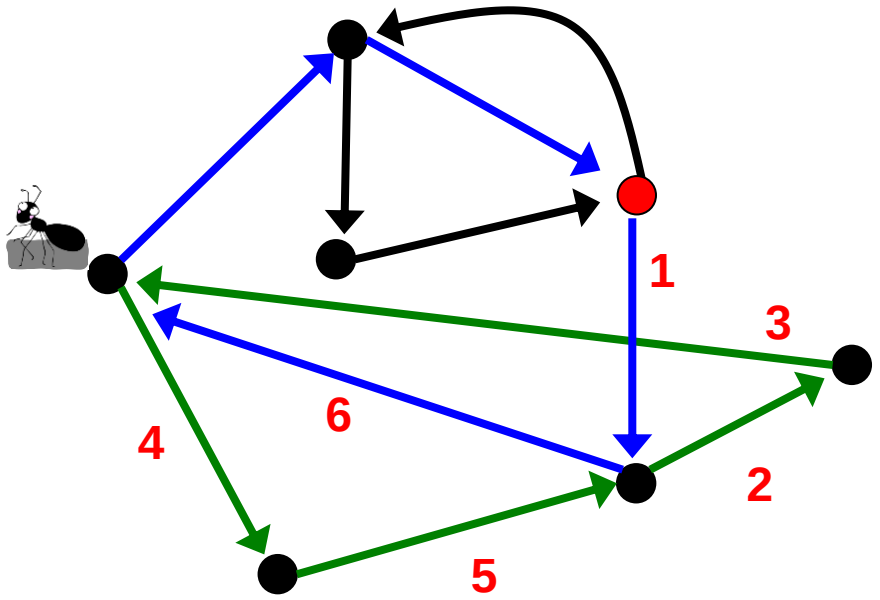
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Traversing the Previously Constructed Green-Blue Cycle



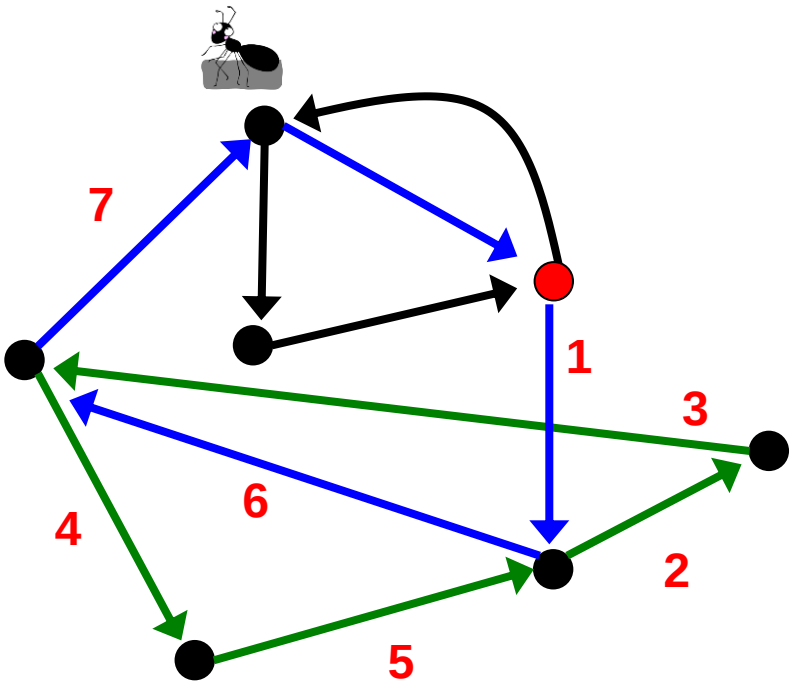
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Traversing the Previously Constructed Green-Blue Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

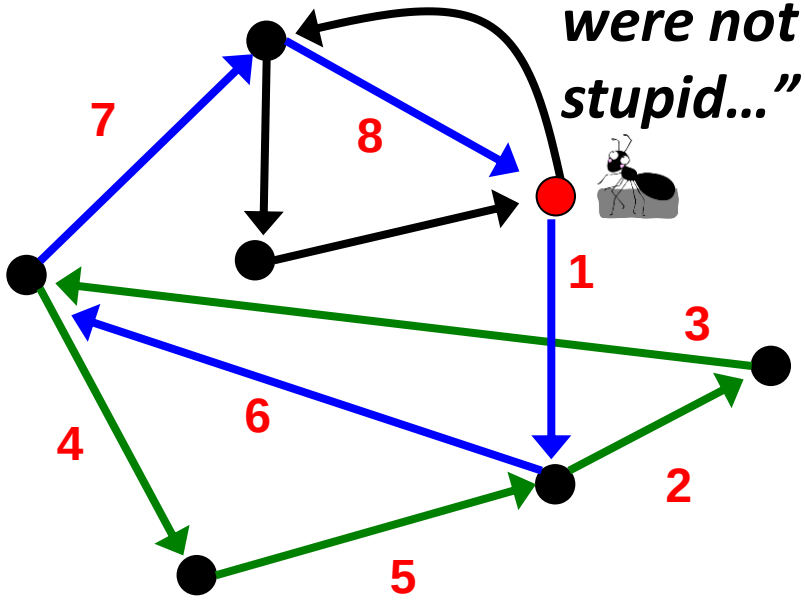
Traversing the Previously Constructed Green-Blue Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

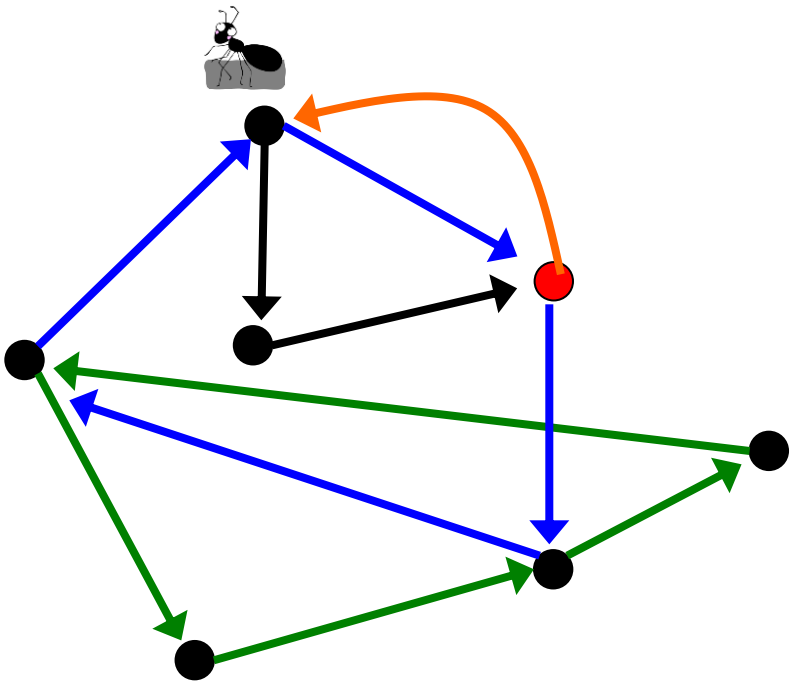
I Returned Back BUT… I Can Continue Walking!

"Hmm, maybe these instructions were not that stupid…"



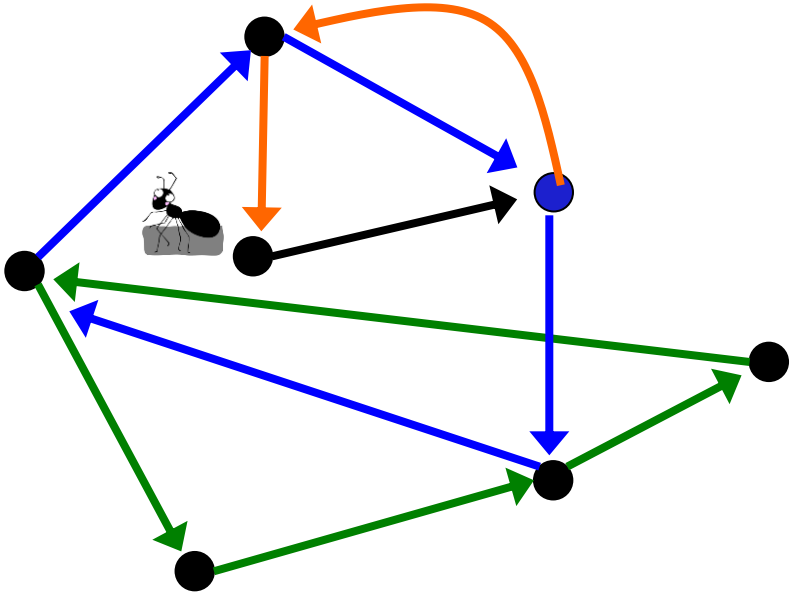
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Enlarging the Green-Blue Cycle



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

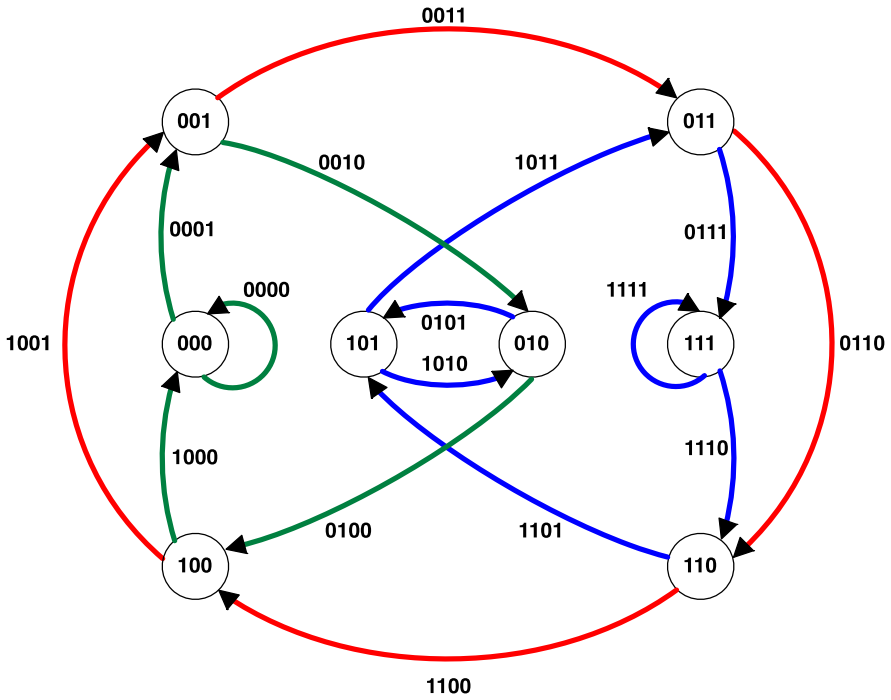
Enlarging the Green-Blue Cycle

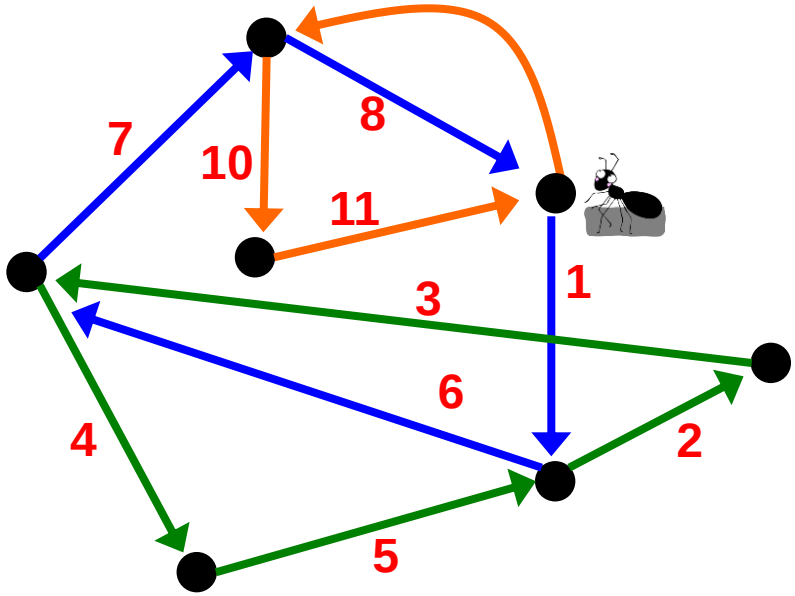


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

I Proved Euler's Theorem! Can I Go Home Please?

EulerianCycle(*BalancedGraph*)
  form a *Cycle* by randomly walking in *BalancedGraph* (avoiding already visited edges)
    while *Cycle* is not Eulerian
      select a node *newStart* in *Cycle* with still unexplored outgoing edges
      form a *Cycle'* by traversing *Cycle* from *newStart* and randomly walking afterwards
      Cycle ← *Cycle'*
  return *Cycle*





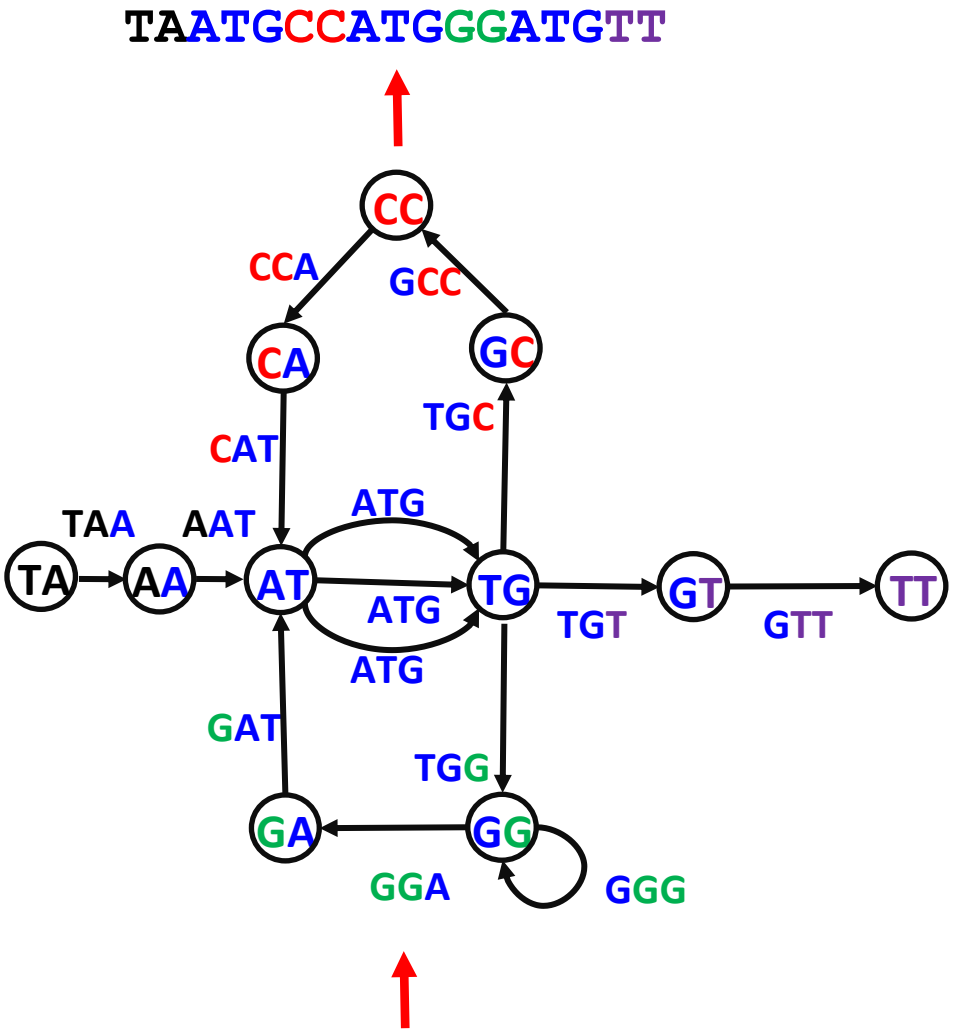
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- **Assembling Read-Pairs**
- De Bruijn Graphs Face Harsh Realities of Assembly

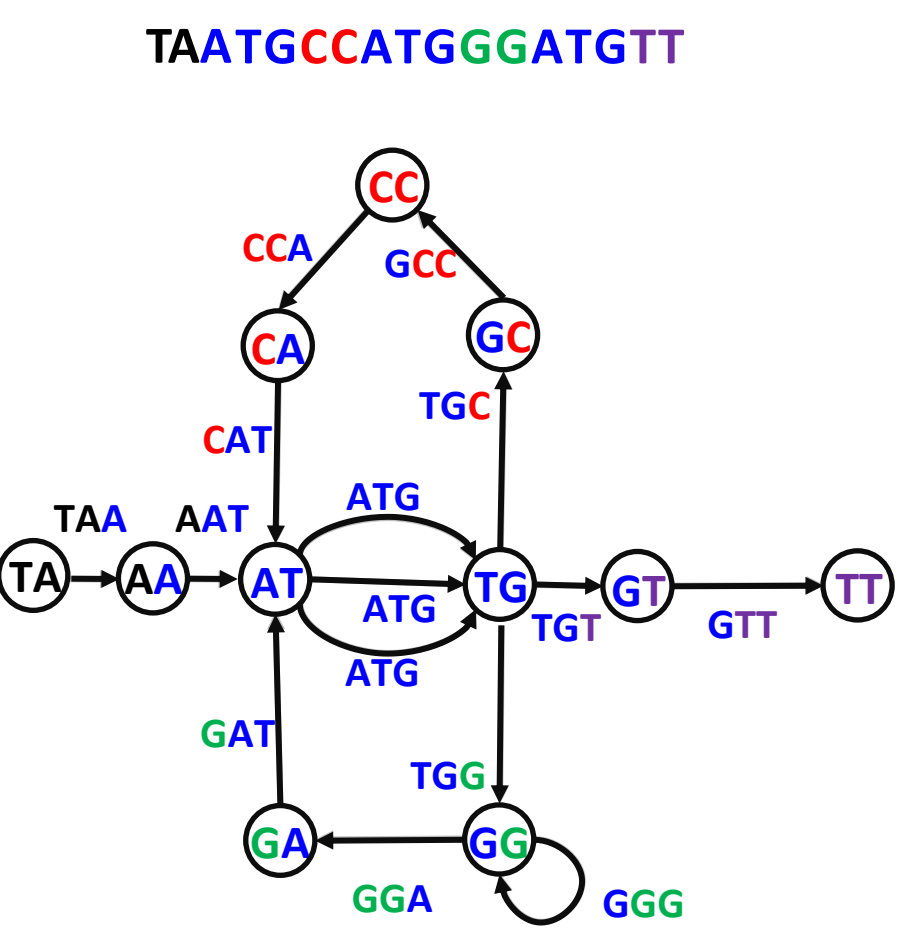
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

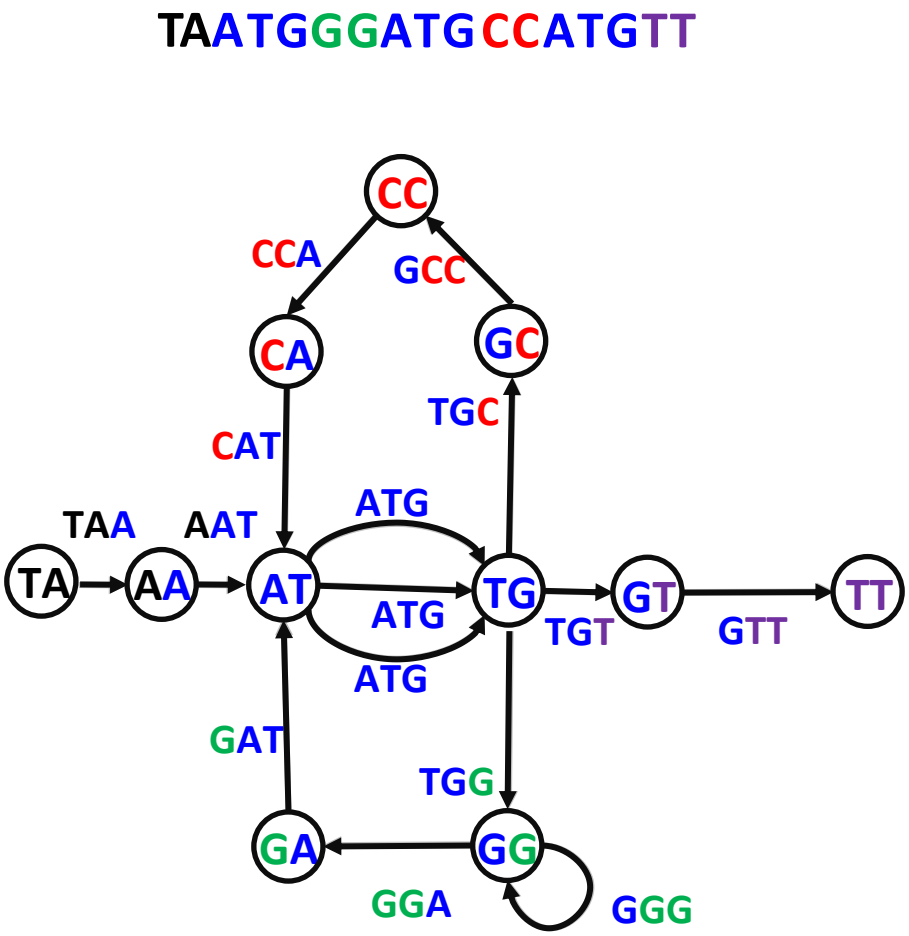
From Reads to de Bruijn Graph to *Genome*



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Multiple Eulerian Paths

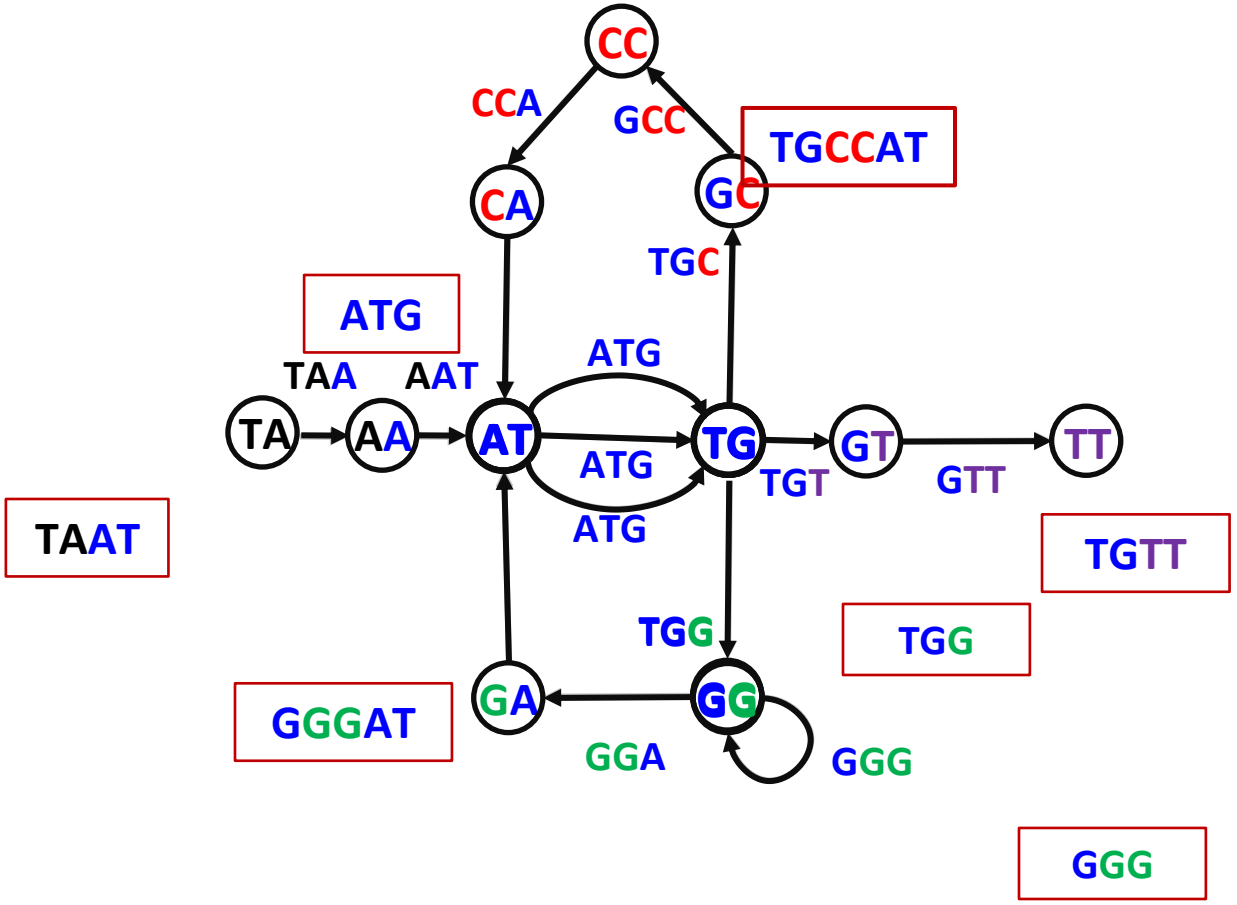




Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Breaking Genome into Contigs

TAATGCCATGGGATGTT



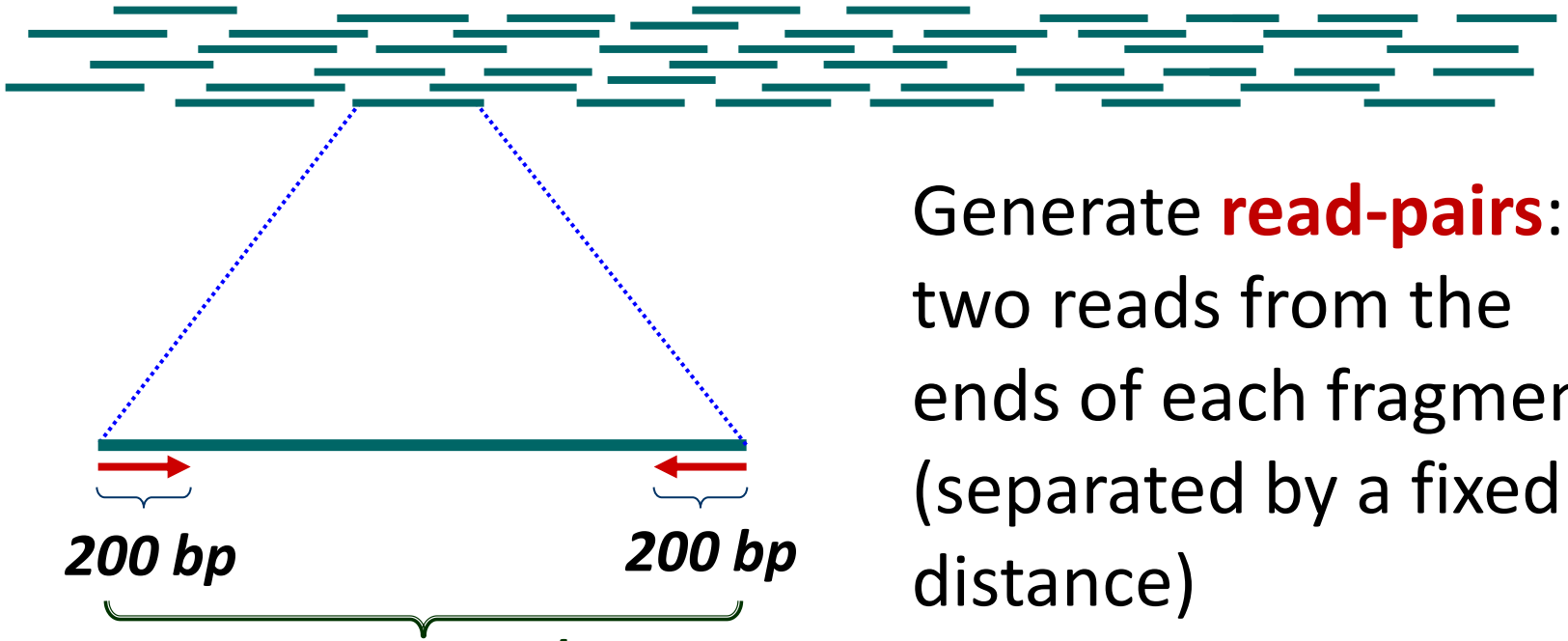
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

DNA Sequencing with Read-pairs

Multiple identical copies of genome



Randomly cut genomes into large equally sized fragments of size ***InsertLength***



Generate **read-pairs**: two reads from the ends of each fragment (separated by a fixed distance)

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

From *k*-mers to Paired *k*-mers



A **paired *k*-mer** is a pair of *k*-mers at a fixed distance *d* apart in *Genome*.
E.g. **TCA** and **TCC** are at distance *d*=11 apart.

Disclaimers:

1. In reality, *Read1* and *Read2* are typically sampled from different strands:
   (→ ..... ← rather than → ..... →)
2. In reality, the distance *d* between reads is measured with errors.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What is *PairedComposition*(**TAATGCCATGGGATGTT**)?

TAA GCC

paired 3-mer

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

What is *PairedComposition*(**TAATGCCATGGGATGTT**)?

TAA GCC
AAT CCA
ATG CAT
TGC ATG
GCC TGG
CCA GGG
CAT GGA
ATG GAT
TGG ATG
GGG TGT
GGA GTT

Representing a **paired 3-mer TAA GCC** as a 2-line expression: **TAA / GCC**

TAA	AAT	ATG	TGC	GCC	CCA	CAT	ATG	TGG	GGG	GGA
GCC	CCA	CAT	ATG	TGG	GGG	GGA	GAT	ATG	TGT	GTT

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

PairedComposition(TAATGCCATGGGATGTT)

TAA GCC
AAT CCA
ATG CAT
TGC ATG
GCC TGG
CCA GGG
CAT GGA
ATG GAT
TGG ATG
GGG TGT
GGA GTT

TAA	AAT	ATG	TGC	GCC	CCA	CAT	ATG	TGG	GGG	GGA
GCC	CCA	CAT	ATG	TGG	GGG	GGA	GAT	ATG	TGT	GTT

AAT	ATG	ATG	CAT	CCA	GCC	GGA	GGG	TAA	TGC	TGG
CCA	CAT	GAT	GGA	GGG	TGG	GTT	TGT	GCC	ATG	ATG

Representing *PairedComposition* in lexicographic order

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

String Reconstruction from Read-Pairs Problem

String Reconstruction from Read-Pairs Problem. Reconstruct a string from its paired *k*-mers.

- **Input.** A collection of paired *k*-mers.
- **Output.** A string *Text* such that *PairedComposition*(*Text*) is equal to the collection of paired *k*-mers.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

How Would de Bruijn Assemble Paired *k*-mers?



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

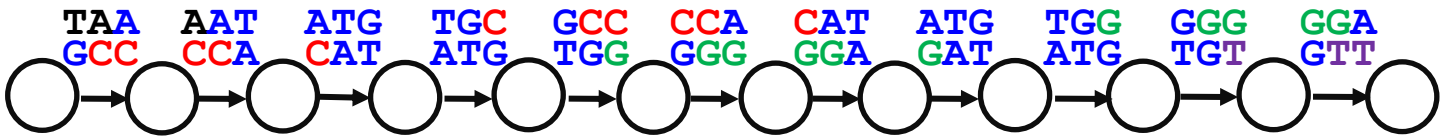
Paired de Bruijn Graphs

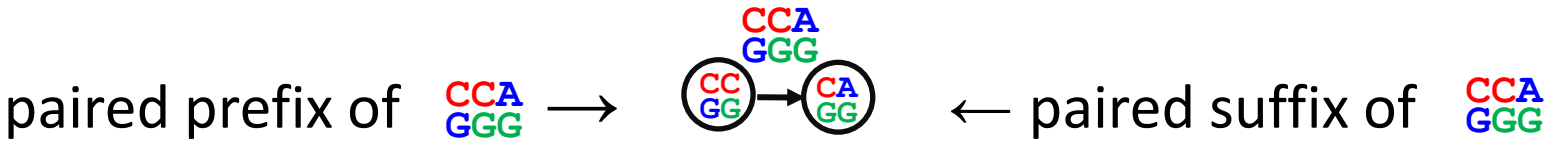


Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Representing Genome TAATGCCATGGGATGTT as a Path

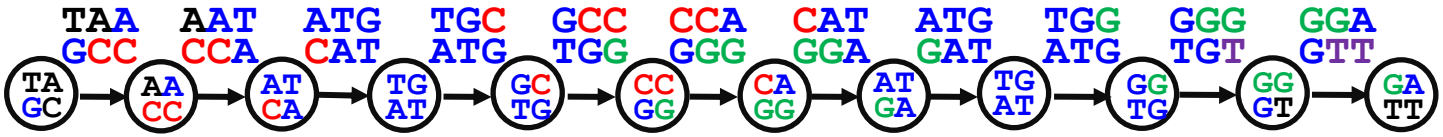
TAA GCC
AAT CCA
ATG CAT
TGC ATG
GCC TGG
CCA GGG
CAT GGA
ATG GAT
TGG ATG
GGG TGT
GGA GTT

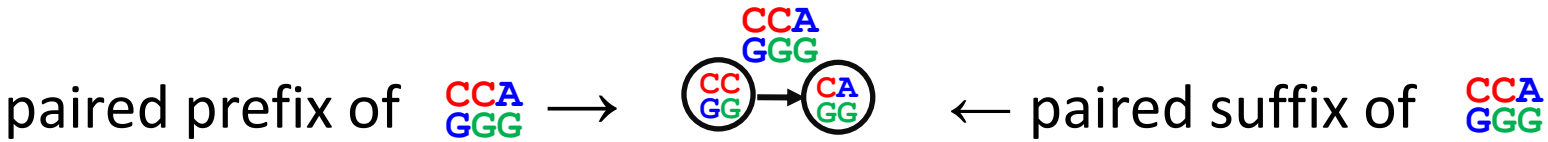




Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

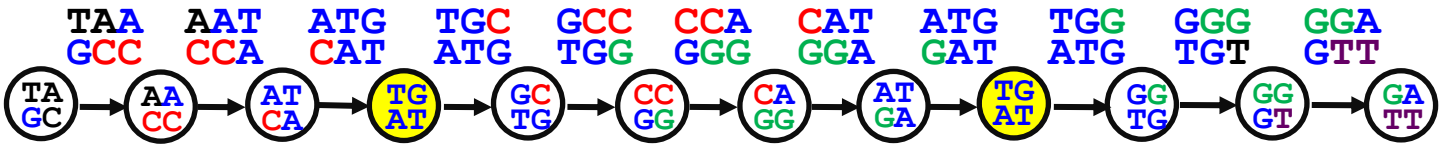
Labeling Nodes by Paired Prefixes and Suffixes





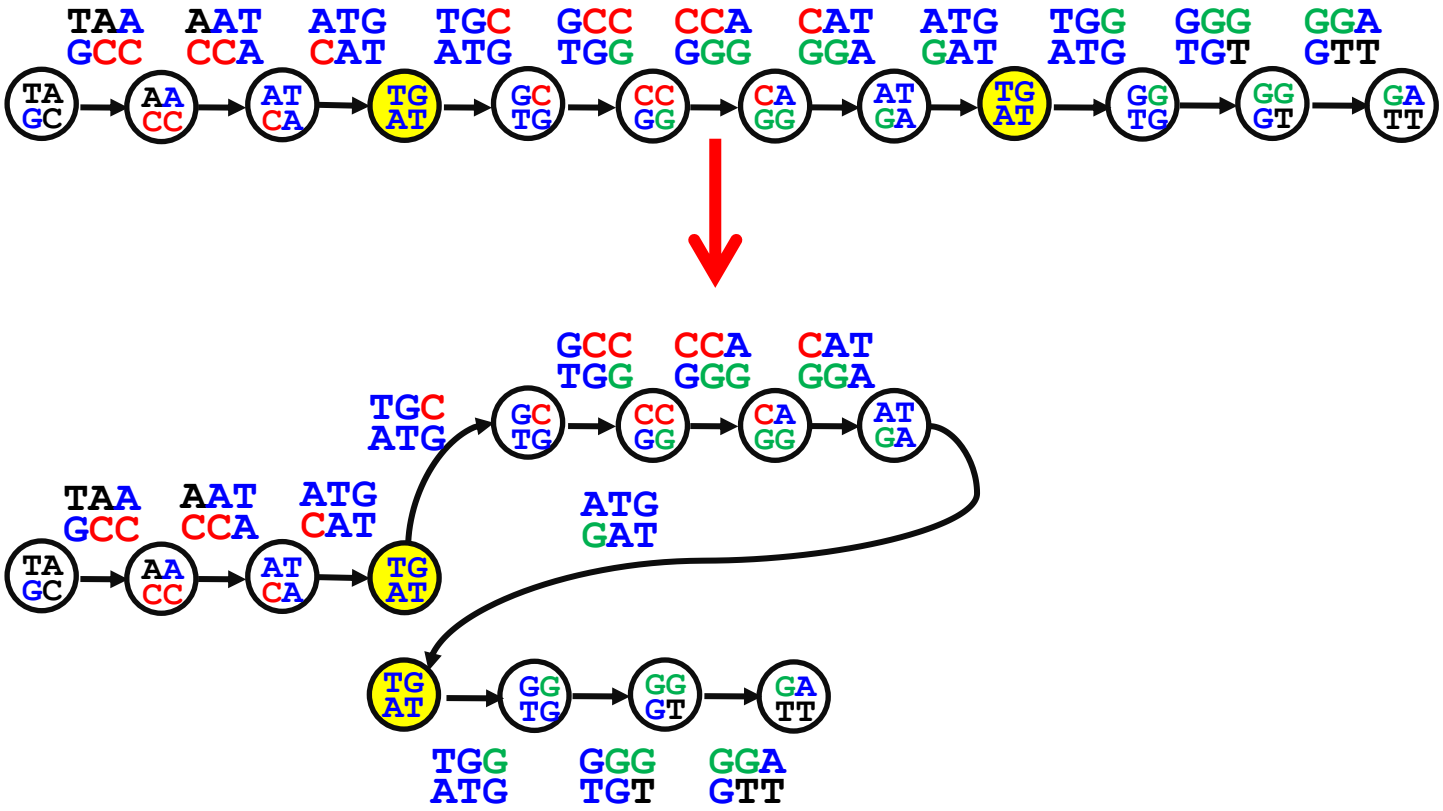
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Glue nodes with identical labels



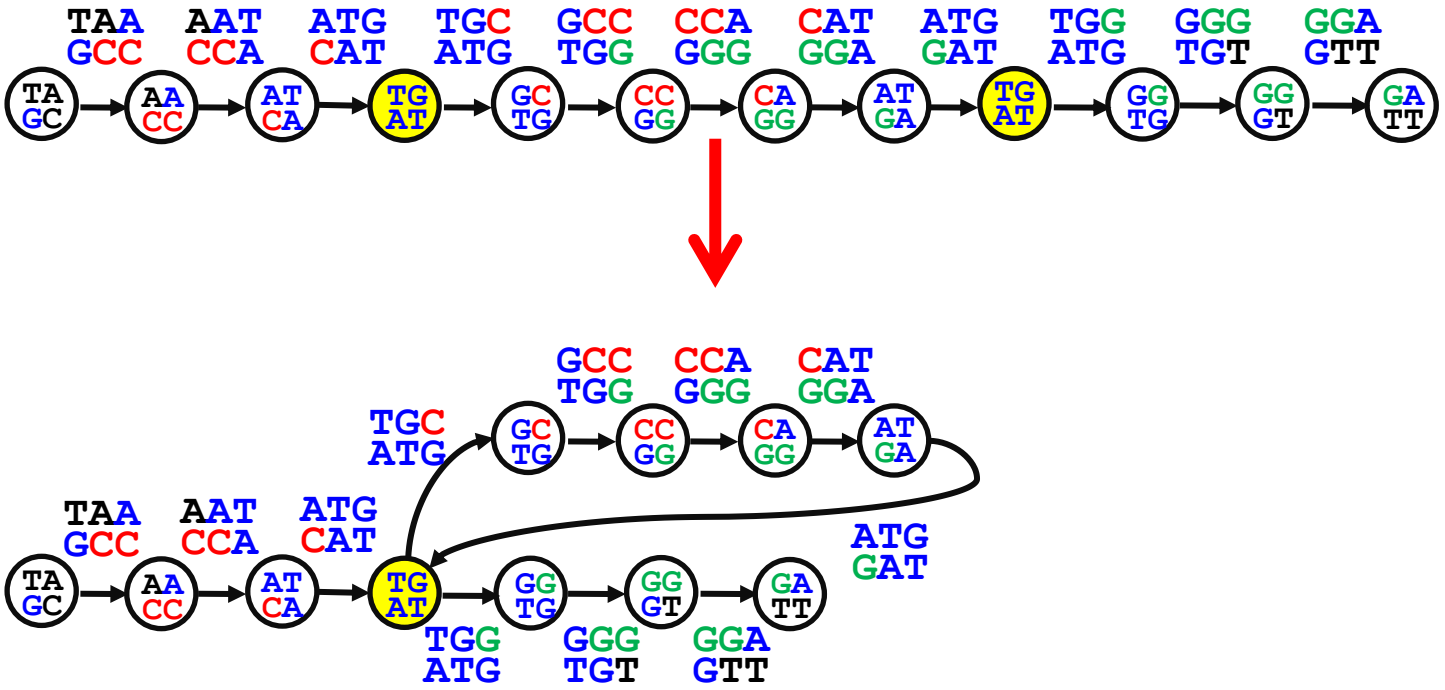
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Glue nodes with identical labels



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

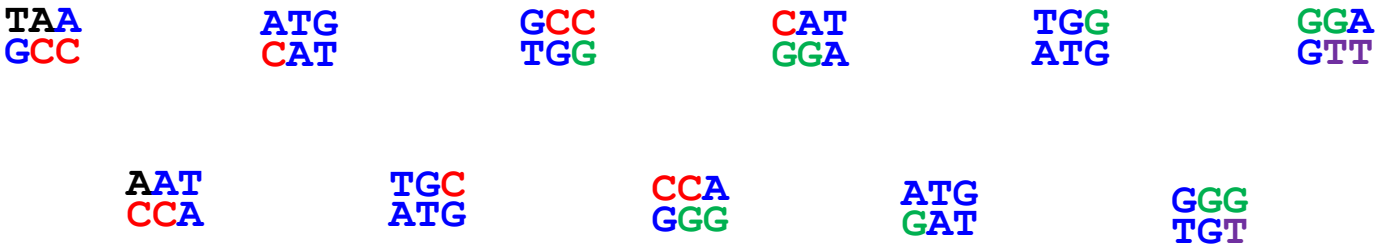
Glue nodes with identical labels



Paired de Bruijn Graph from the Genome

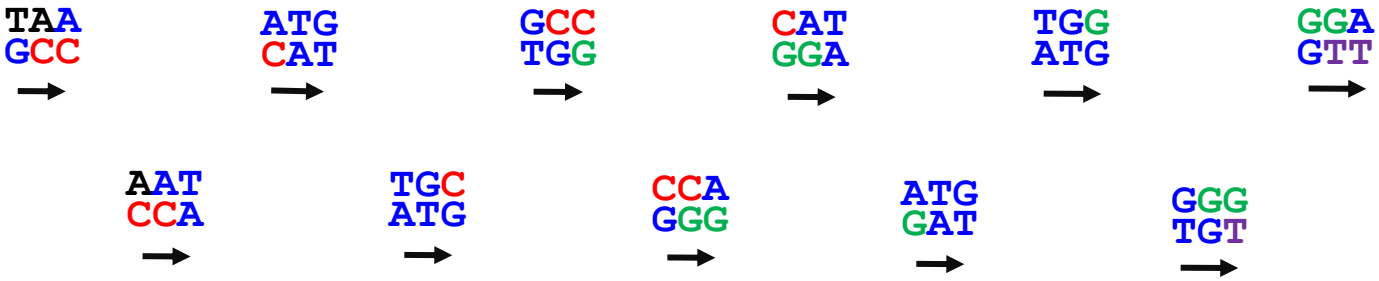
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

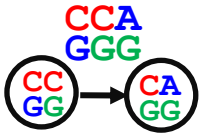
Constructing Paired de Bruijn Graph from paired k-mers



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

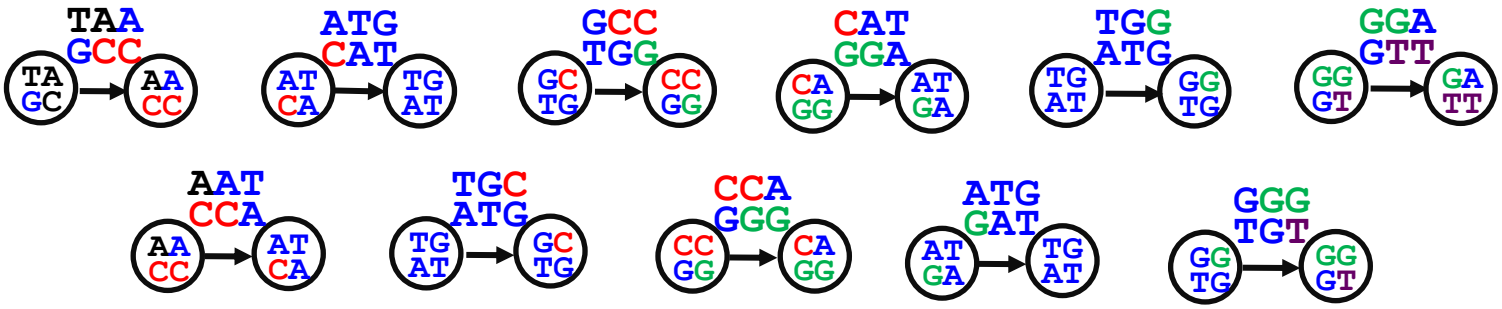
Constructing Paired de Bruijn Graph from paired k-mers

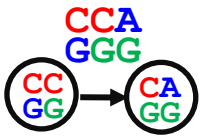


paired prefix of  →  ← paired suffix of 

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

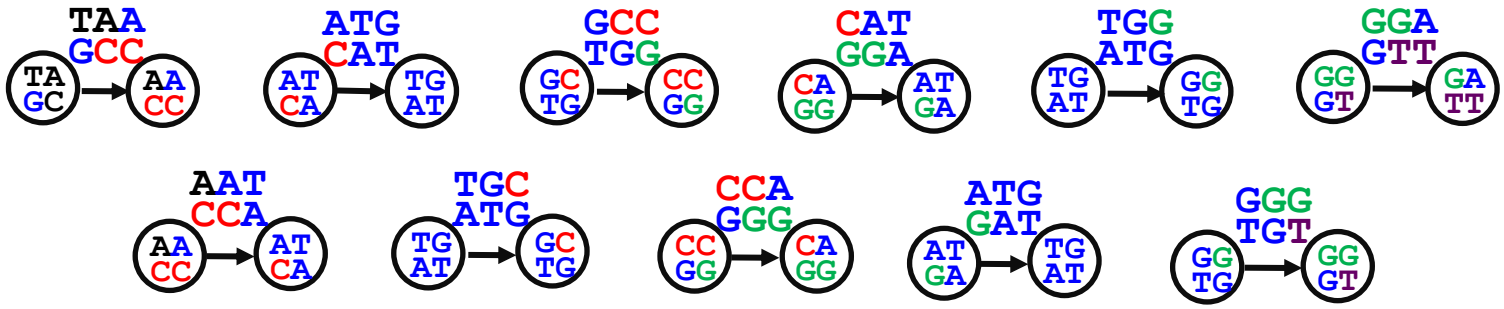
Constructing Paired de Bruijn Graph



paired prefix of CCA / GGG →  ← paired suffix of CCA / GGG

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

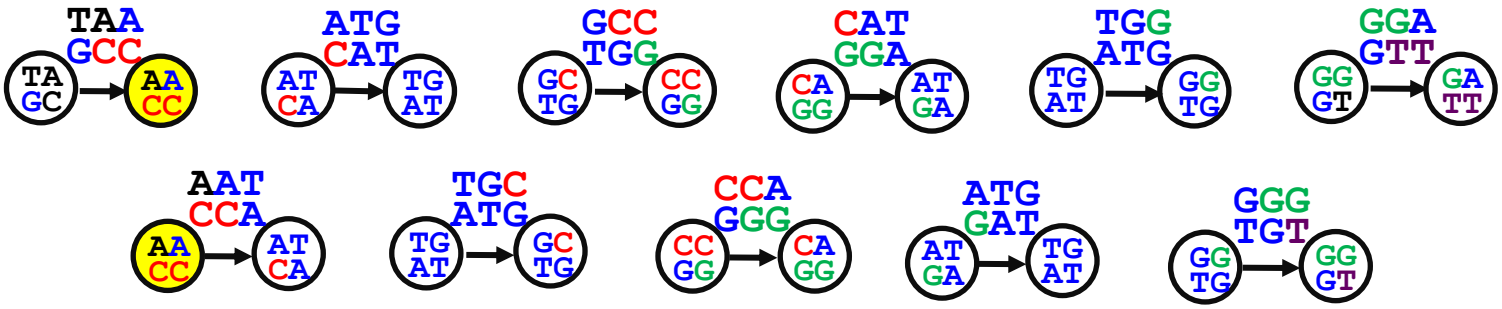
Constructing Paired de Bruijn Graph



- **Paired de Bruijn graph for a collection of paired *k*-mers:**
  - Represent every paired *k*-mer as an edge between its paired prefix and paired suffix.
  - Glue **ALL** nodes with identical labels.

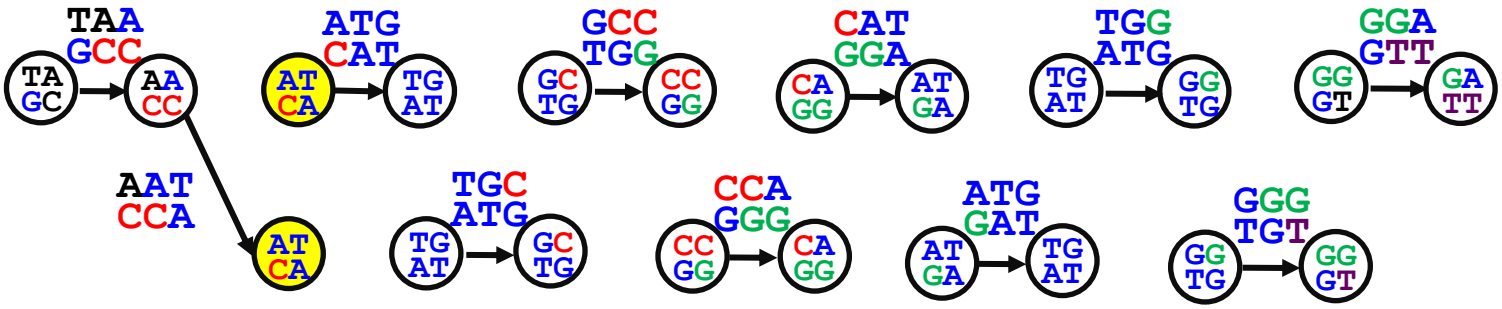
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



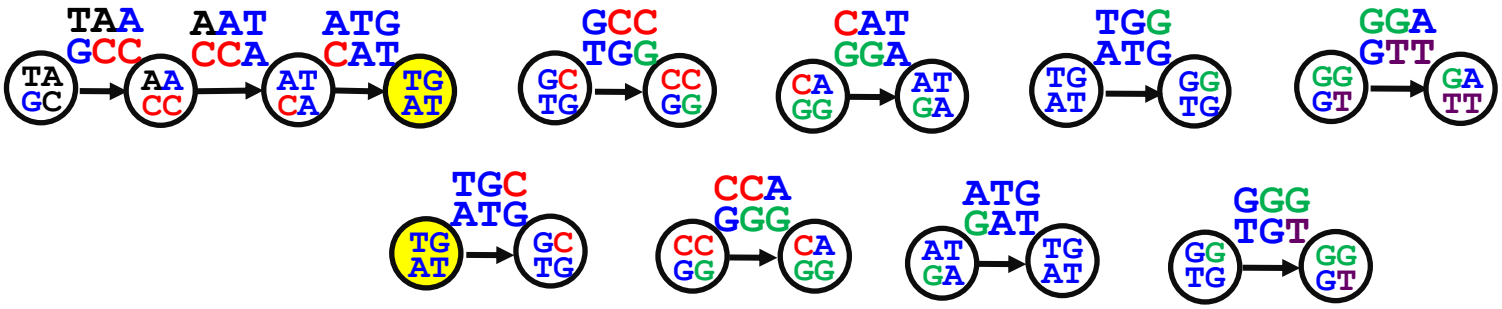
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



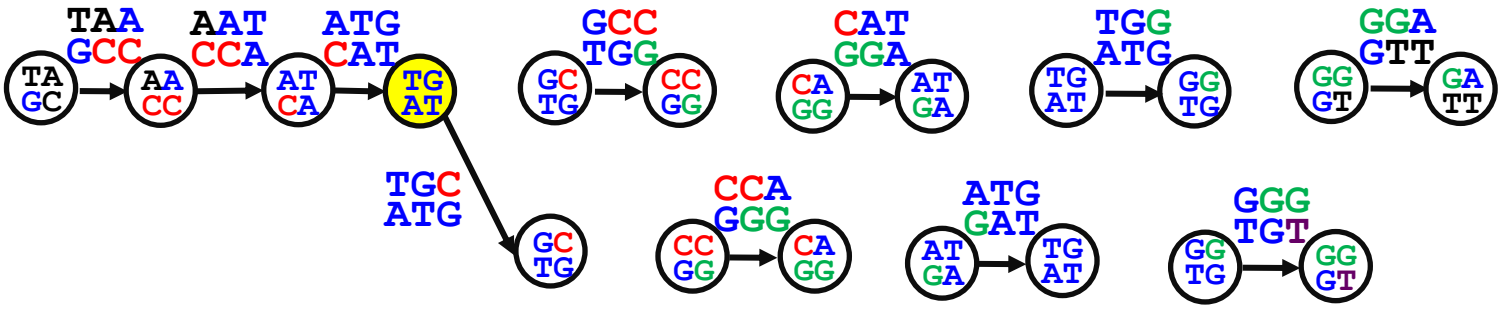
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



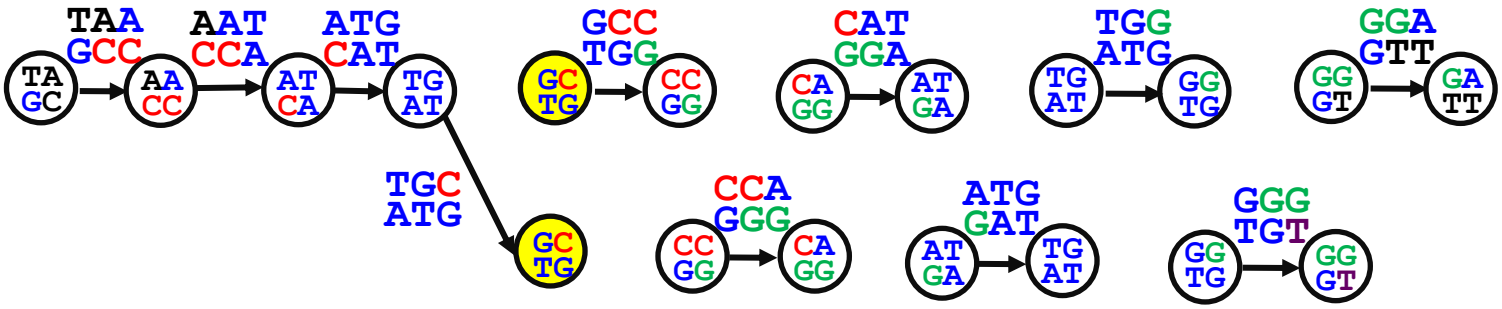
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



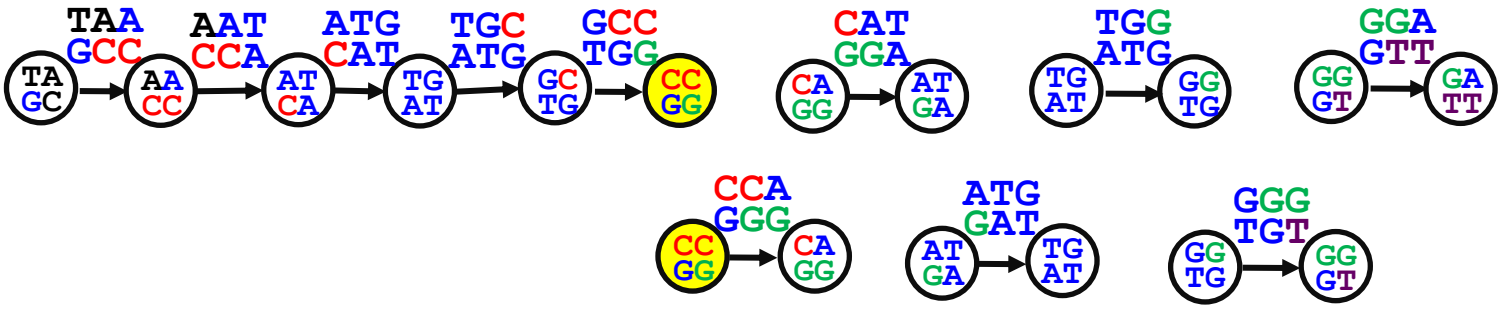
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



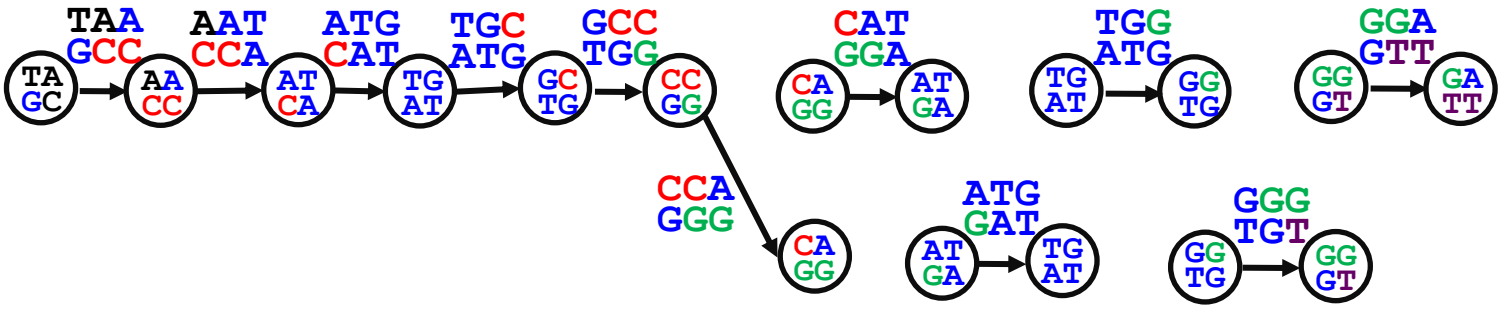
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



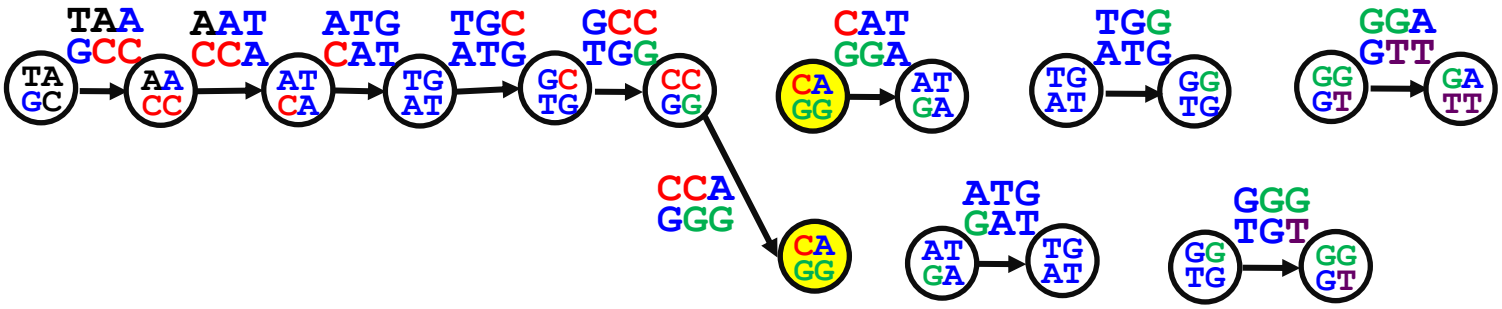
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



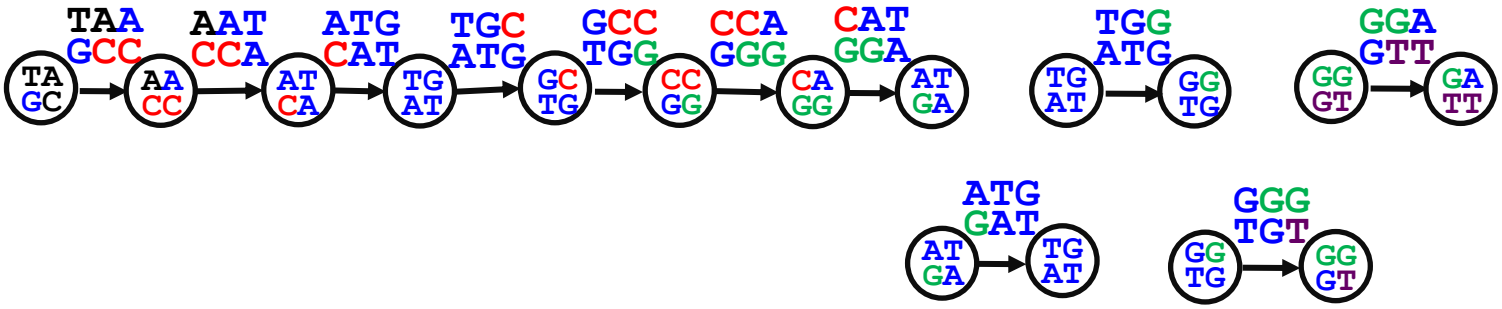
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



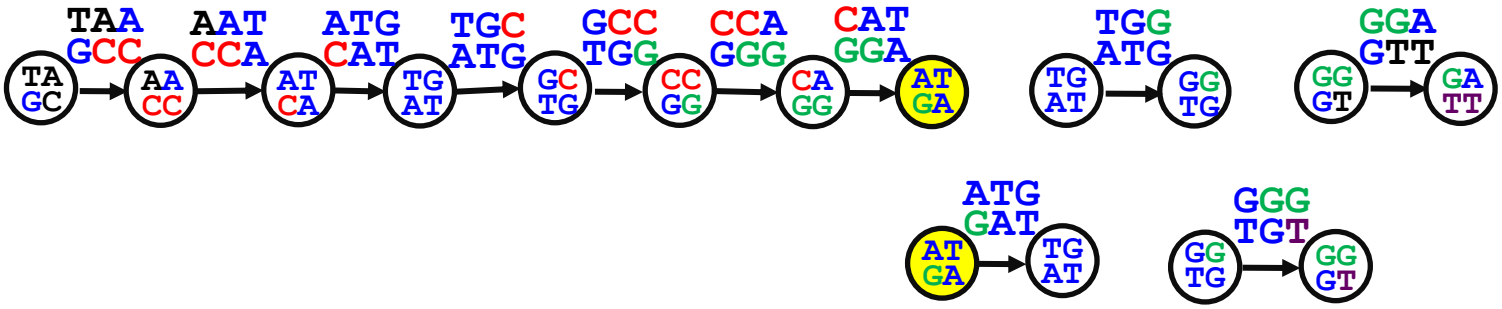
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



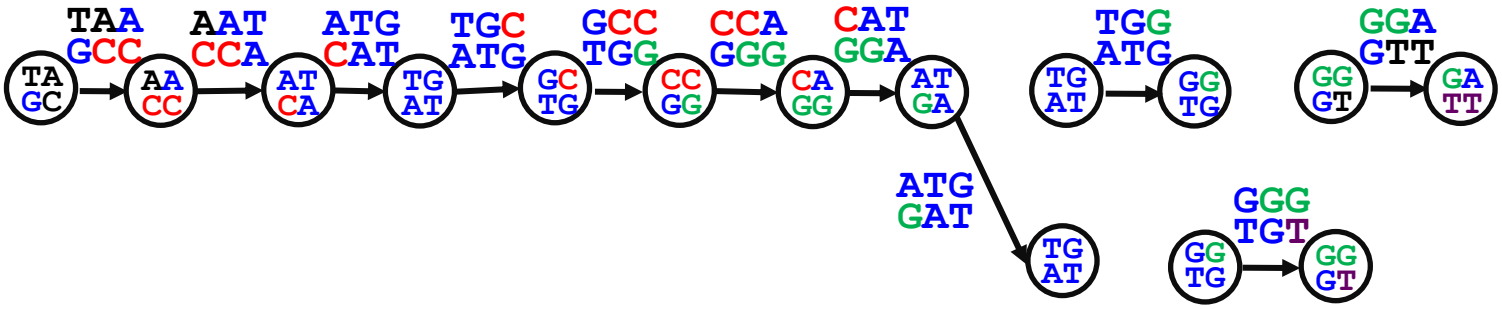
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



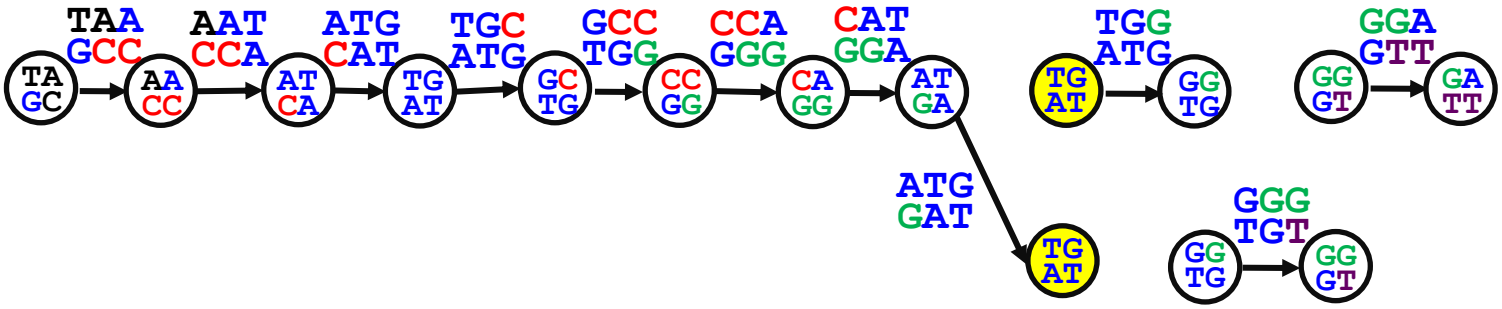
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



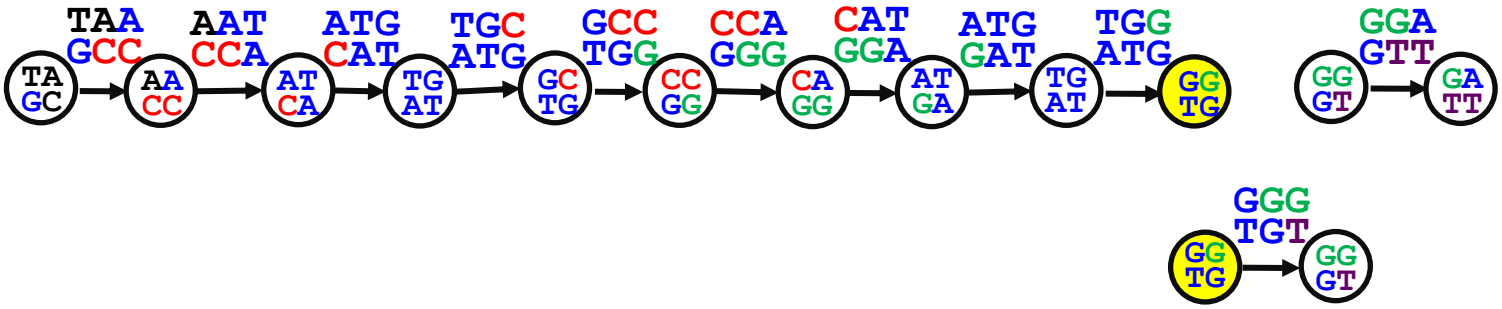
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



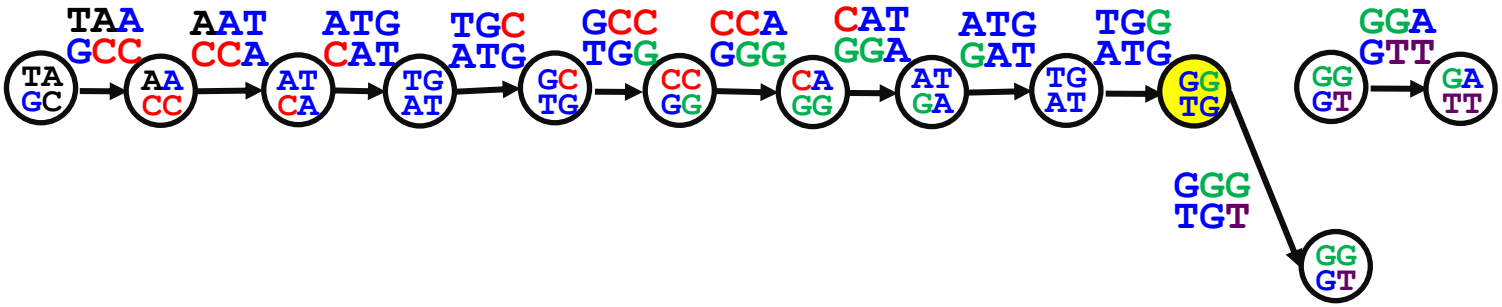
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



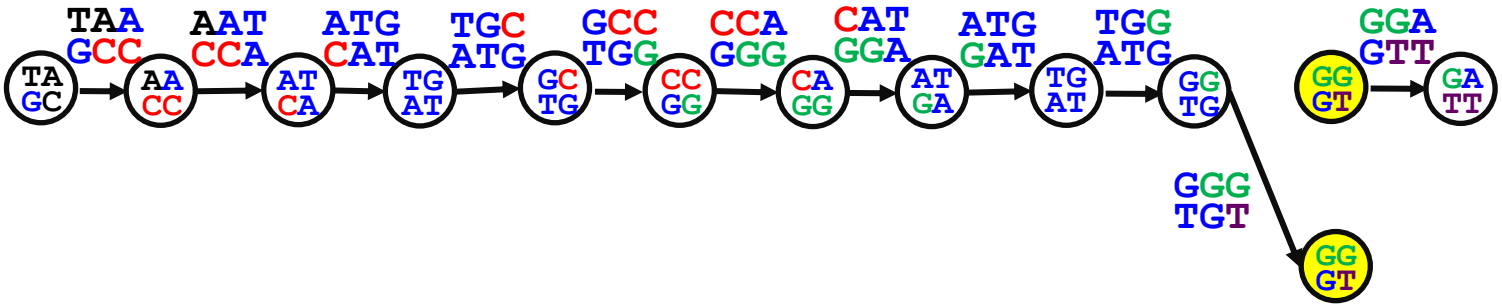
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



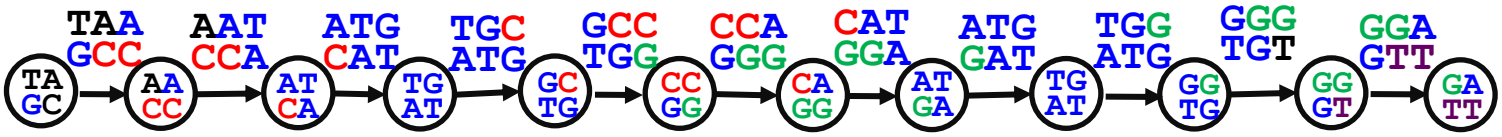
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



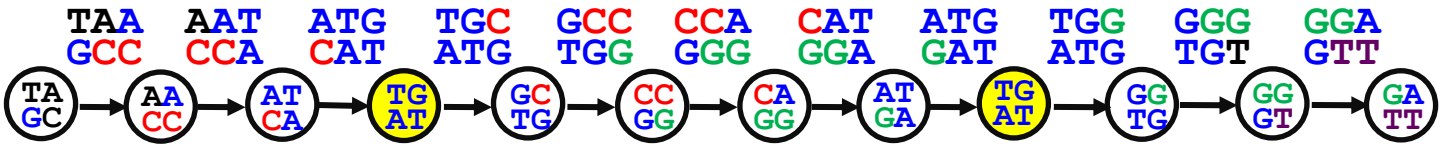
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



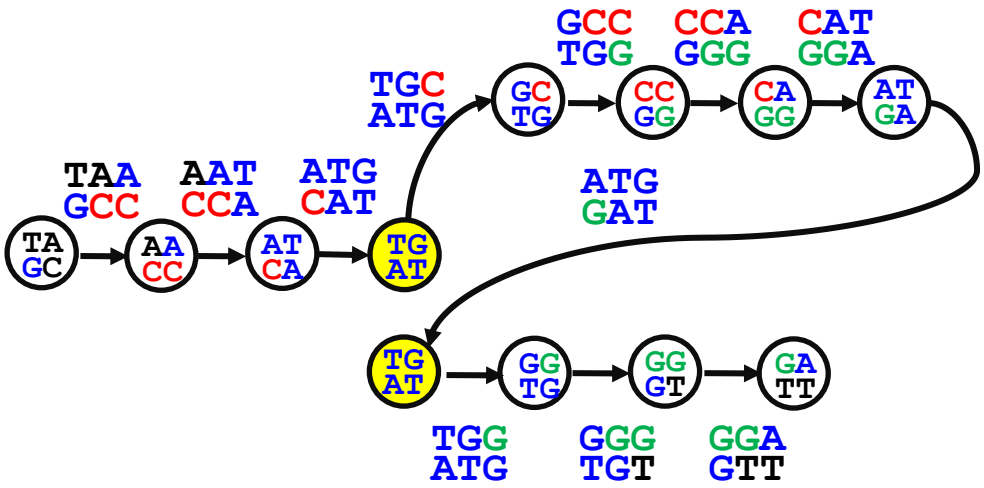
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

We Are Not Done with Gluing Yet



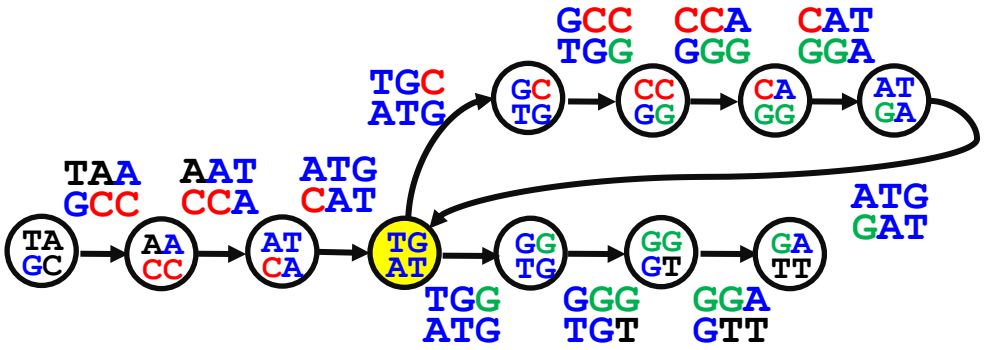
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Constructing Paired de Bruijn Graph



Paired de Bruijn Graph from read-pairs

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Paired de Bruijn Graphs



- **Paired de Bruijn graph for a collection of paired *k*-mers:**
  - Represent every paired *k*-mer as an edge between its paired prefix and paired suffix.
  - Glue **ALL** nodes with identical labels.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Which Graph Represents a Better Assembly?

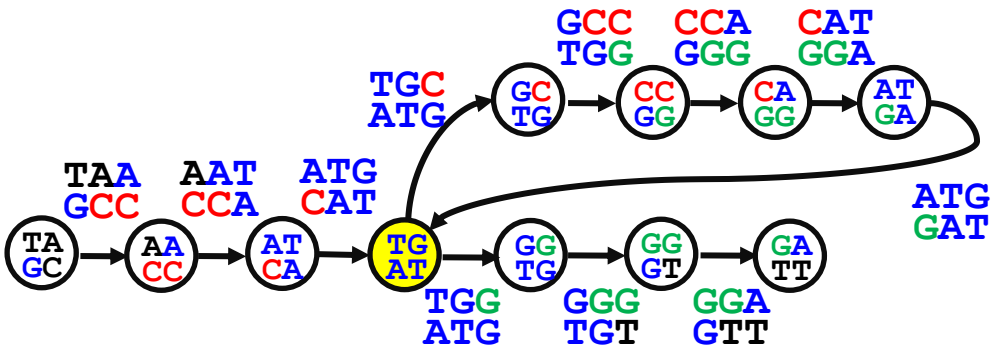
Unique genome reconstruction

TAATGCCATGGGATGTT

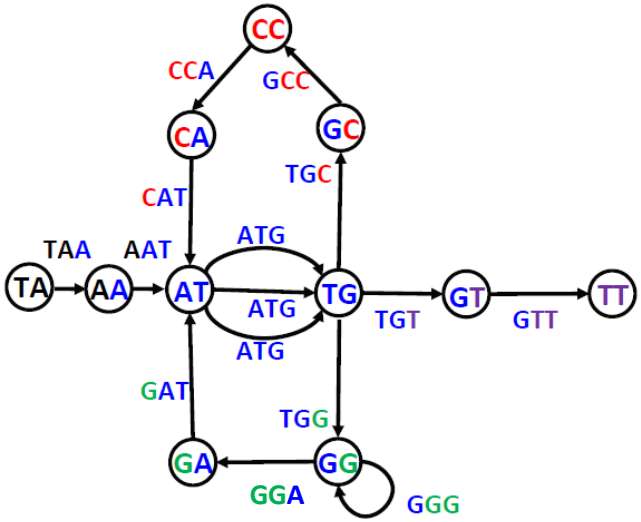
Multiple genome reconstructions

TAATGCCATGGGATGTT

TAATGGGATGCCATGTT



Paired de Bruijn Graph



De Bruijn Graph

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Outline

- What Is Genome Sequencing?
- Exploding Newspapers
- The String Reconstruction Problem
- String Reconstruction as a Hamiltonian Path Problem
- String Reconstruction as an Eulerian Path Problem
- Similar Problems with Different Fates
- De Bruijn Graphs
- Euler's Theorem
- Assembling Read-Pairs
- **De Bruijn Graphs Face Harsh Realities of Assembly**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Some Ridiculously Unrealistic Assumptions

- Perfect coverage of genome by reads (every *k*-mer from the genome is represented by a read)
- Reads are error-free.
- Multiplicities of *k*-mers are known
- Distances between reads within read-pairs are exact.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Some Ridiculously Unrealistic Assumptions

- **Imperfect** coverage of genome by reads (every *k*-mer from the genome is represented by a read)
- Reads are **error-prone**.
- Multiplicities of *k*-mers are **unknown.**
- Distances between reads within read-pairs are **inexact.**
- **Etc., etc., etc.**

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

1st Unrealistic Assumption: Perfect Coverage

```
atgccgtatggacaacgact
atgccgtatg
  gccgtatgga
     gtatggacaa
          gacaacgact
```

250-nucleotide reads generated by Illumina technology capture only a small fraction of 250-mers from the genome, thus violating the key assumption of the de Bruijn graphs.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Breaking Reads into Shorter *k*-mers

```
atgccgtatggacaacgact
atgccgtatg
  gccgtatgga
     gtatggacaa
          gacaacgact
```

```
atgccgtatggacaacgact
atgcc
 tgccg
  gccgt
   ccgta
    cgtat
     gtatg
      tatgg
       atgga
        tggac
         ggaca
          gacaa
           acaac
            caacg
             aacga
              acgac
               cgact
```

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

2nd Unrealistic Assumption: Error-free Reads

```
atgccgtatggacaacgact
atgccgtatg
  gccgtatgga
     gtatggacaa
          gacaacgact
    cgtaCggaca
```

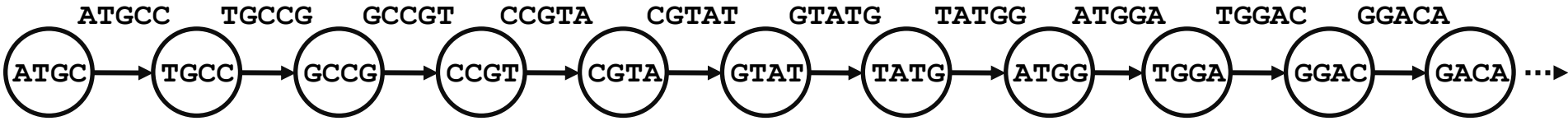
Erroneous read
(change of **t** into **C**)

```
atgccgtatggacaacgact
atgcc
 tgccg
  gccgt
   ccgta
    cgtat
     gtatg
      tatgg
       atgga
        tggac
         ggaca
          gacaa
           acaac
            caacg
             aacga
              acgac
               cgact
cgtaC
 gtaCg
  taCgg
   aCgga
    Cggac
```

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

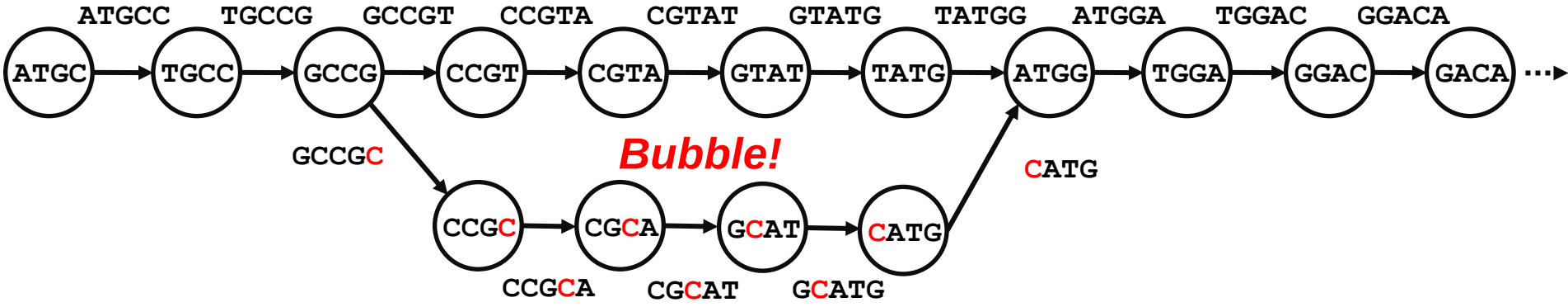
De Bruijn Graph of ATGGCGTGCAATG… Constructed from Error-Free Reads

.



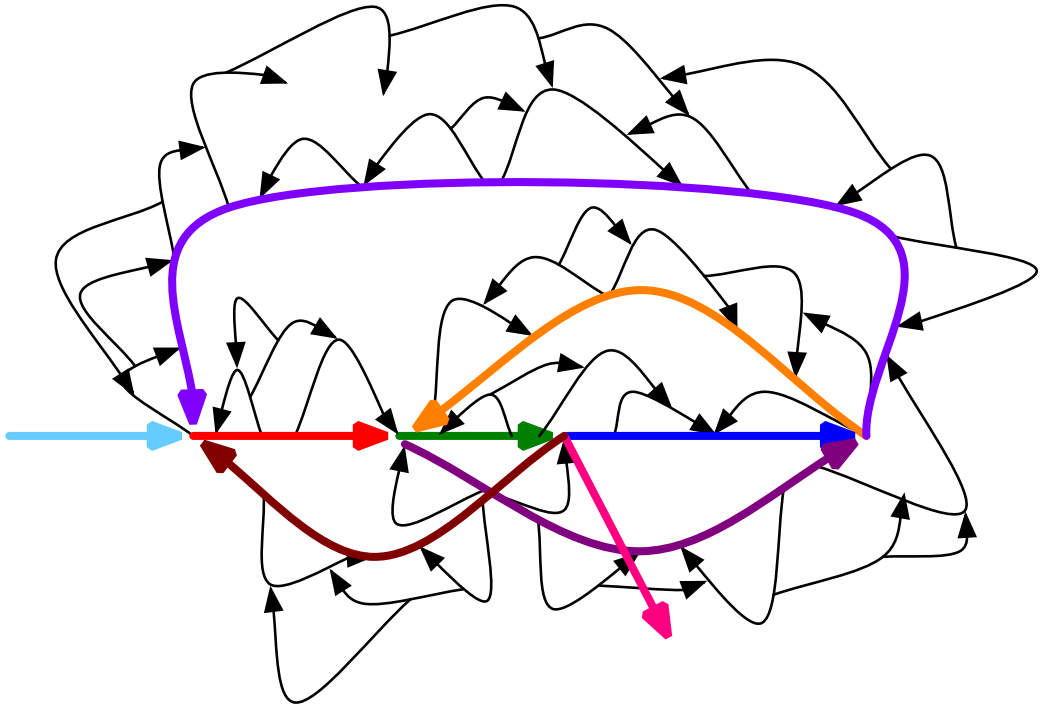
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Errors in Reads Lead to **Bubbles** in the De Bruijn Graph



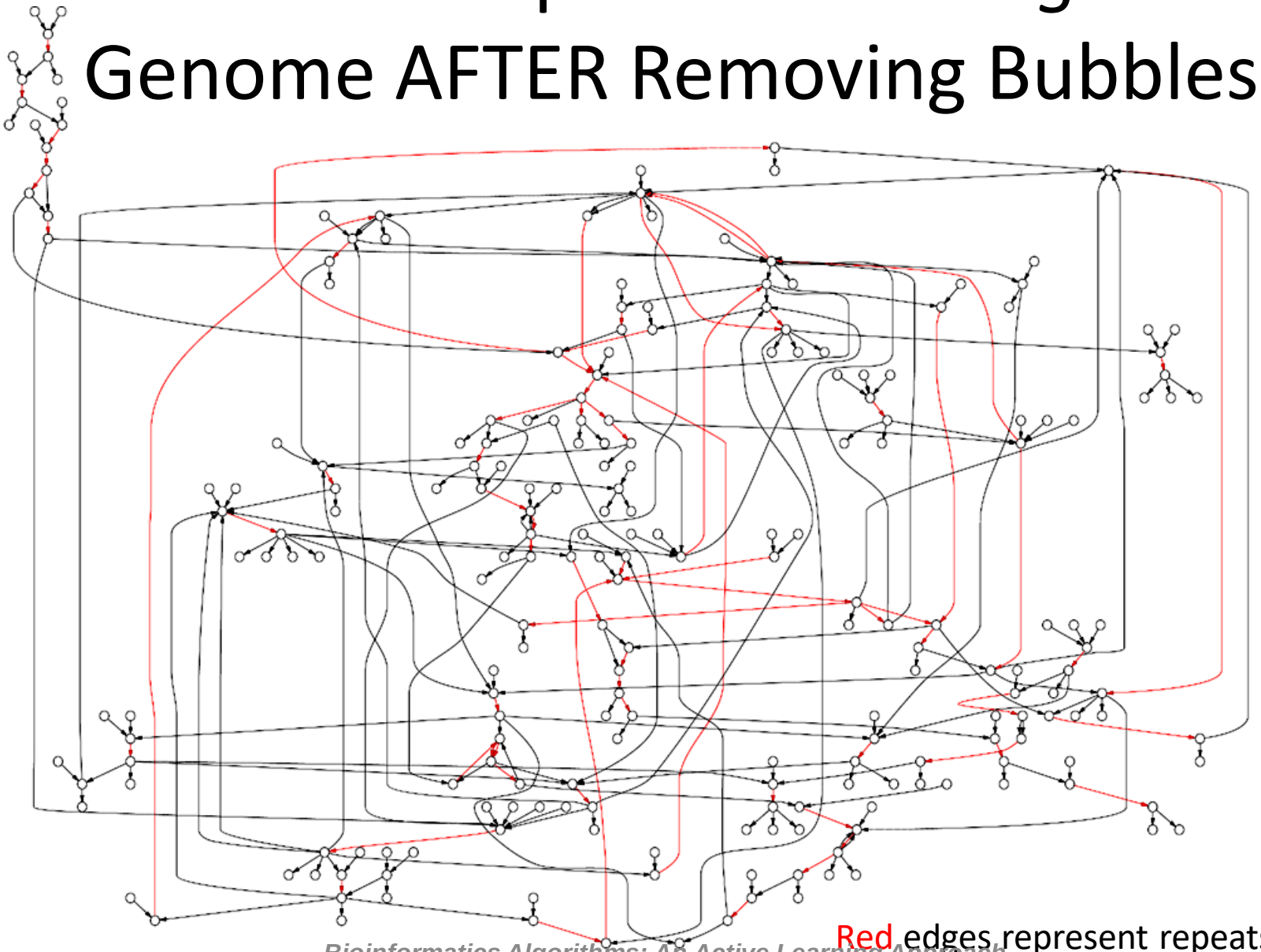
Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Bubble Explosion…Where Are the Correct Edges of the de Bruijn Graph?



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

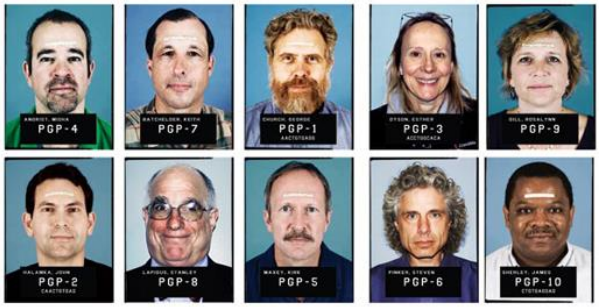
De Bruin Graph of *N. meningitidis* Genome AFTER Removing Bubbles



Red edges represent repeats

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Who Are These People?

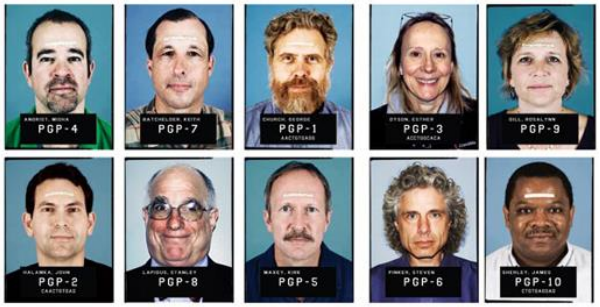


10 scientists and entrepreneurs who made their genomes available in 2009

- Personal genome sequencing will soon expand to millions of individuals.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Who Are These People?



10 scientists and entrepreneurs who made their genomes available in 2009

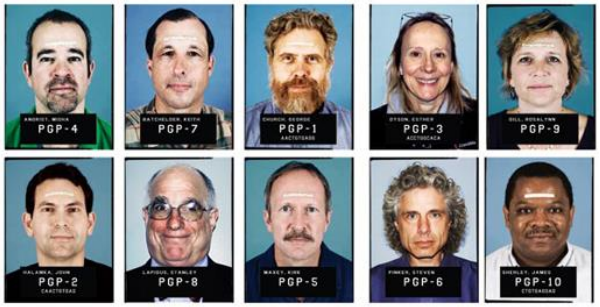
- Personal genome sequencing will soon expand to millions of individuals.

- We will soon sequence genomes of tens of thousands of species.



Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.

Who Are These People?



10 scientists and entrepreneurs who made their genomes available in 2009

- Personal genome sequencing will soon expand to millions of individuals.



- We will soon sequence genomes of tens of thousands of species.

- Thousands of cancer genomes have already been sequenced, and genome sequencing will soon become a routine technique in medicine.

Bioinformatics Algorithms: An Active Learning Approach.
Copyright 2018 Compeau and Pevzner.